

General Algebraic Modeling System
Compilation

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1  * =====
2  * Telecom Staff Scheduling (GAMS) - all 3 notebook parts
3  * =====
4
5  Sets
6      h   "hours"      /h0*h23/
7      j   "companies" /c1*c2/
8      peak(h) "hours 7..17 (inclusive)"
9      off(h)  "hours 0..6 and 18..23"
10     Q(h,h)  "coverage mapping: for each demand hour, staffed start hours in
6-hour window";
11
12     Alias (h,i,k);
13
14     * Subsets for objective weights (ord(h0)=1)
15     peak(h) = yes$(ord(h) >= 8 and ord(h) <= 18);
16     off(h)  = yes$(ord(h) <= 7 or ord(h) >= 19);
17
18     Parameters
19         cost(h)          "cost per operator starting at hour"
20         demand(h,j)      "demand by hour and company"
21         capHour(h)        "RHS for per-hour starting cap"
22         capTotal(h)       "RHS for total active cap per demand hour"
23         minCov(h)         "minimum coverage per demand hour"
24         maxShort(h,j)     "max shortage per hour/company"
25         totShort(j)       "total shortage cap by company";
26
27     * --- Default RHS values (so we can vary them later in part 3)
28     capHour(h) = 100;
29     capTotal(h)= 420;
30     minCov(h)  = 80;
31     maxShort(h,j) = 10;
32     totShort('c1') = 40;
33     totShort('c2') = 30;
34
35     * Costs (hour 0..23)
36     cost('h0') = 189; cost('h1') = 189; cost('h2') = 189; cost('h3') = 180;
37     cost('h4') = 171; cost('h5') = 162; cost('h6') = 153; cost('h7') = 144;
38     cost('h8') = 135; cost('h9') = 126; cost('h10') = 126; cost('h11') = 135;
39     cost('h12') = 144; cost('h13') = 153; cost('h14') = 162; cost('h15') = 171;
40     cost('h16') = 180; cost('h17') = 189; cost('h18') = 189; cost('h19') = 189;
41     cost('h20') = 189; cost('h21') = 189; cost('h22') = 189; cost('h23') = 189;
42
43     * Demand (hour, company)

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44 * Company 1
45 demand('h0','c1') = 120; demand('h1','c1') = 95; demand('h2','c1') = 90;
demand('h3','c1') = 50;
46 demand('h4','c1') = 30; demand('h5','c1') = 30; demand('h6','c1') = 60;
demand('h7','c1') = 110;
47 demand('h8','c1') = 140; demand('h9','c1') = 200; demand('h10','c1') = 180;
demand('h11','c1') = 180;
48 demand('h12','c1') = 240; demand('h13','c1') = 220; demand('h14','c1') = 220;
demand('h15','c1') = 230;
49 demand('h16','c1') = 250; demand('h17','c1') = 260; demand('h18','c1') = 230;
demand('h19','c1') = 250;
50 demand('h20','c1') = 260; demand('h21','c1') = 250; demand('h22','c1') = 220;
demand('h23','c1') = 190;
51
52 * Company 2
53 demand('h0','c2') = 70; demand('h1','c2') = 40; demand('h2','c2') = 20;
demand('h3','c2') = 30;
54 demand('h4','c2') = 50; demand('h5','c2') = 40; demand('h6','c2') = 40;
demand('h7','c2') = 90;
55 demand('h8','c2') = 110; demand('h9','c2') = 110; demand('h10','c2') = 110;
demand('h11','c2') = 130;
56 demand('h12','c2') = 150; demand('h13','c2') = 160; demand('h14','c2') = 120;
demand('h15','c2') = 130;
57 demand('h16','c2') = 160; demand('h17','c2') = 180; demand('h18','c2') = 200;
demand('h19','c2') = 170;
58 demand('h20','c2') = 140; demand('h21','c2') = 110; demand('h22','c2') = 100;
demand('h23','c2') = 90;
59
60 * Coverage mapping Q(i,k): 6-hour cyclic window (wraps around midnight)
61 Q('h0','h18')=yes; Q('h0','h19')=yes; Q('h0','h20')=yes; Q('h0','h22')=yes;
Q('h0','h23')=yes; Q('h0','h0')=yes;
62 Q('h1','h19')=yes; Q('h1','h20')=yes; Q('h1','h21')=yes; Q('h1','h23')=yes;
Q('h1','h0')=yes; Q('h1','h1')=yes;
63 Q('h2','h20')=yes; Q('h2','h21')=yes; Q('h2','h22')=yes; Q('h2','h0')=yes;
Q('h2','h1')=yes; Q('h2','h2')=yes;
64 Q('h3','h21')=yes; Q('h3','h22')=yes; Q('h3','h23')=yes; Q('h3','h1')=yes;
Q('h3','h2')=yes; Q('h3','h3')=yes;
65 Q('h4','h22')=yes; Q('h4','h23')=yes; Q('h4','h0')=yes; Q('h4','h2')=yes;
Q('h4','h3')=yes; Q('h4','h4')=yes;
66 Q('h5','h23')=yes; Q('h5','h0')=yes; Q('h5','h1')=yes; Q('h5','h3')=yes;
Q('h5','h4')=yes; Q('h5','h5')=yes;
67 Q('h6','h0')=yes; Q('h6','h1')=yes; Q('h6','h2')=yes; Q('h6','h4')=yes;
Q('h6','h5')=yes; Q('h6','h6')=yes;
68 Q('h7','h1')=yes; Q('h7','h2')=yes; Q('h7','h3')=yes; Q('h7','h5')=yes;
Q('h7','h6')=yes; Q('h7','h7')=yes;
69 Q('h8','h2')=yes; Q('h8','h3')=yes; Q('h8','h4')=yes; Q('h8','h6')=yes;
Q('h8','h7')=yes; Q('h8','h8')=yes;
70 Q('h9','h3')=yes; Q('h9','h4')=yes; Q('h9','h5')=yes; Q('h9','h7')=yes;
Q('h9','h8')=yes; Q('h9','h9')=yes;
71 Q('h10','h4')=yes; Q('h10','h5')=yes; Q('h10','h6')=yes; Q('h10','h8')=yes;

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Q('h10','h9')=yes; Q('h10','h10')=yes;
72  Q('h11','h5')=yes; Q('h11','h6')=yes; Q('h11','h7')=yes; Q('h11','h9')=yes;
Q('h11','h10')=yes;Q('h11','h11')=yes;
73  Q('h12','h6')=yes; Q('h12','h7')=yes; Q('h12','h8')=yes;
Q('h12','h10')=yes;Q('h12','h11')=yes;Q('h12','h12')=yes;
74  Q('h13','h7')=yes; Q('h13','h8')=yes; Q('h13','h9')=yes;
Q('h13','h11')=yes;Q('h13','h12')=yes;Q('h13','h13')=yes;
75  Q('h14','h8')=yes; Q('h14','h9')=yes;
Q('h14','h10')=yes;Q('h14','h12')=yes;Q('h14','h13')=yes;Q('h14','h14')=yes;
76  Q('h15','h9')=yes;
Q('h15','h10')=yes;Q('h15','h11')=yes;Q('h15','h13')=yes;Q('h15','h14')=yes;Q('h15',
'h15')=yes;
77
Q('h16','h10')=yes;Q('h16','h11')=yes;Q('h16','h12')=yes;Q('h16','h14')=yes;Q('h16',
'h15')=yes;Q('h16','h16')=yes;
78
Q('h17','h11')=yes;Q('h17','h12')=yes;Q('h17','h13')=yes;Q('h17','h15')=yes;Q('h17',
'h16')=yes;Q('h17','h17')=yes;
79
Q('h18','h12')=yes;Q('h18','h13')=yes;Q('h18','h14')=yes;Q('h18','h16')=yes;Q('h18',
'h17')=yes;Q('h18','h18')=yes;
80
Q('h19','h13')=yes;Q('h19','h14')=yes;Q('h19','h15')=yes;Q('h19','h17')=yes;Q('h19',
'h18')=yes;Q('h19','h19')=yes;
81
Q('h20','h14')=yes;Q('h20','h15')=yes;Q('h20','h16')=yes;Q('h20','h18')=yes;Q('h20',
'h19')=yes;Q('h20','h20')=yes;
82
Q('h21','h15')=yes;Q('h21','h16')=yes;Q('h21','h17')=yes;Q('h21','h19')=yes;Q('h21',
'h20')=yes;Q('h21','h21')=yes;
83
Q('h22','h16')=yes;Q('h22','h17')=yes;Q('h22','h18')=yes;Q('h22','h20')=yes;Q('h22',
'h21')=yes;Q('h22','h22')=yes;
84
Q('h23','h17')=yes;Q('h23','h18')=yes;Q('h23','h19')=yes;Q('h23','h21')=yes;Q('h23',
'h22')=yes;Q('h23','h23')=yes;
85
86
87  * =====
88  * PART 1: Base MIP
89  * =====
90
91  Variables
92      z "total cost (objective)";
93
94  Integer Variables
95      x(h,j) "operators starting at hour for company"
96      L(h,j) "shortage at hour for company";
97
98  Equations

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99      obj          "objective definition"
100     cap_per_hour(h)  "sum of operators starting at hour across companies <=
capHour"
101     cap_total(h)     "total operators active in window across companies <=
capTotal"
102     shortage(h,j)    "demand - covered <= L"
103     shortage_cap(h,j) "L <= maxShort"
104     shortage_total(j) "sum_h L(h,j) <= totShort(j)"
105     min_coverage(h)   "minimum coverage in each window across companies >=
minCov";
106
107  obj..
108      z =e=
109          sum((h,j), cost(h)*x(h,j))
110          + 60*sum(i$peak(i), L(i,'c1'))
111          + 30*sum(i$off(i), L(i,'c1'))
112          + 45*sum(i, L(i,'c2'));
113
114  cap_per_hour(h)..
115      sum(j, x(h,j)) =l= capHour(h);
116
117  cap_total(h)..
118      sum((k,j)$Q(h,k), x(k,j)) =l= capTotal(h);
119
120  shortage(h,j)..
121      demand(h,j) - sum(k$Q(h,k), x(k,j)) =l= L(h,j);
122
123  shortage_cap(h,j)..
124      L(h,j) =l= maxShort(h,j);
125
126  shortage_total(j)..
127      sum(h, L(h,j)) =l= totShort(j);
128
129  min_coverage(h)..
130      sum((k,j)$Q(h,k), x(k,j)) =g= minCov(h);
131
132  Model Telecom_MIP /all/;
133  * =====
134  * PHASE 3: Two separate what-if sweeps
135  *   A) Change ONE objective coefficient: cost('h8')
136  *   B) Change ONE RHS value: capHour('h0')
137  * =====
138
139  Set t /t1*t21/;
140
141  Parameter
142      coefVal(t) "trial values for objective coefficient cost('h8')"
143      rhsVal(t)  "trial values for RHS capHour('h0')"
144      objCoef(t) "optimal objective for coefficient sweep"
145      objRhs(t)  "optimal objective for RHS sweep";

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146
147 Scalar origCost8, origCapH0;
148
149 origCost8 = cost('h8');
150 origCapH0 = capHour('h0');
151
152 * --- A) Sweep objective coefficient: from 0.5*orig to 1.5*orig
153 coefVal(t) = (0.5 + (ord(t)-1)*(1.0/(card(t)-1))) * origCost8;
154
155 loop(t,
156     cost('h8') = coefVal(t);
157     Solve Telecom_MIP using MIP minimizing z;
158     objCoef(t) = z.l;
159 );
160
161 * restore
162 cost('h8') = origCost8;
163
164 * --- B) Sweep RHS: from 0.6*orig to 1.4*orig (اگر اثر نداد بازه را بیشتر کن)
165 * --- B) Sweep ONE RHS: totShort('c1') (total shortage cap for company 1)
166 Scalar origTotC1;
167 origTotC1 = totShort('c1');
168
169 rhsVal(t) = 0 + (ord(t)-1)*((80-0)/(card(t)-1));
170
171 loop(t,
172     totShort('c1') = rhsVal(t);
173     Solve Telecom_MIP using MIP minimizing z;
174     objRhs(t) = z.l;
175 );
176
177 totShort('c1') = origTotC1;
178
179
180 * --- Write TWO separate CSV files (easier to plot)
181 file f1 /sweep_coef.csv; put f1;
182 put 't,cost_h8,obj',;
183 loop(t,
184     put t.tl, ',', coefVal(t):0:6, ',', objCoef(t):0:6 /;
185 );
186 putclose f1;
187
188 file f2 /sweep_rhs.csv; put f2;
189 put 't,totShort_c1,obj',;
190 loop(t,
191     put t.tl, ',', rhsVal(t):0:6, ',', objRhs(t):0:6 /;
192 );
193 putclose f2;
194
195 display coefVal, objCoef, rhsVal, objRhs;

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 GAMS 52.5.0 9bc771bf Jan 28, 2026 WEX-WEI x86 64bit/MS Windows - 02/15/26
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 General Algebraic Modeling System
 Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. $z - 189 \times (h0, c1) - 189 \times (h0, c2) - 189 \times (h1, c1) - 189 \times (h1, c2) - 189 \times (h2, c1) - 189 \times (h2, c2) - 180 \times (h3, c1) - 180 \times (h3, c2) - 171 \times (h4, c1) - 171 \times (h4, c2) - 162 \times (h5, c1) - 162 \times (h5, c2) - 153 \times (h6, c1) - 153 \times (h6, c2) - 144 \times (h7, c1) - 144 \times (h7, c2) - 67.5 \times (h8, c1) - 67.5 \times (h8, c2) - 126 \times (h9, c1) - 126 \times (h9, c2) - 126 \times (h10, c1) - 126 \times (h10, c2) - 135 \times (h11, c1) - 135 \times (h11, c2) - 144 \times (h12, c1) - 144 \times (h12, c2) - 153 \times (h13, c1) - 153 \times (h13, c2) - 162 \times (h14, c1) - 162 \times (h14, c2) - 171 \times (h15, c1) - 171 \times (h15, c2) - 180 \times (h16, c1) - 180 \times (h16, c2) - 189 \times (h17, c1) - 189 \times (h17, c2) - 189 \times (h18, c1) - 189 \times (h18, c2) - 189 \times (h19, c1) - 189 \times (h19, c2) - 189 \times (h20, c1) - 189 \times (h20, c2) - 189 \times (h21, c1) - 189 \times (h21, c2) - 189 \times (h22, c1) - 189 \times (h22, c2) - 189 \times (h23, c1) - 189 \times (h23, c2) - 30 \times L(h0, c1) - 45 \times L(h0, c2) - 30 \times L(h1, c1) - 45 \times L(h1, c2) - 30 \times L(h2, c1) - 45 \times L(h2, c2) - 30 \times L(h3, c1) - 45 \times L(h3, c2) - 30 \times L(h4, c1) - 45 \times L(h4, c2) - 30 \times L(h5, c1) - 45 \times L(h5, c2) - 30 \times L(h6, c1) - 45 \times L(h6, c2) - 60 \times L(h7, c1) - 45 \times L(h7, c2) - 60 \times L(h8, c1) - 45 \times L(h8, c2) - 60 \times L(h9, c1) - 45 \times L(h9, c2) - 60 \times L(h10, c1) - 45 \times L(h10, c2) - 60 \times L(h11, c1) - 45 \times L(h11, c2) - 60 \times L(h12, c1) - 45 \times L(h12, c2) - 60 \times L(h13, c1) - 45 \times L(h13, c2) - 60 \times L(h14, c1) - 45 \times L(h14, c2) - 60 \times L(h15, c1) - 45 \times L(h15, c2) - 60 \times L(h16, c1) - 45 \times L(h16, c2) - 60 \times L(h17, c1) - 45 \times L(h17, c2) - 30 \times L(h18, c1) - 45 \times L(h18, c2) - 30 \times L(h19, c1) - 45 \times L(h19, c2) - 30 \times L(h20, c1) - 45 \times L(h20, c2) - 30 \times L(h21, c1) - 45 \times L(h21, c2) - 30 \times L(h22, c1) - 45 \times L(h22, c2) - 30 \times L(h23, c1) - 45 \times L(h23, c2) =E= 0 ; (LHS = 0)$

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0, c1) + x(h0, c2) =L= 100 ; (LHS = 0)$

cap_per_hour(h1).. $x(h1, c1) + x(h1, c2) =L= 100 ; (LHS = 0)$

cap_per_hour(h2).. $x(h2, c1) + x(h2, c2) =L= 100 ; (LHS = 0)$

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0, c1) + x(h0, c2) + x(h18, c1) + x(h18, c2) + x(h19, c1) + x(h19, c2) + x(h20, c1) + x(h20, c2) + x(h22, c1) + x(h22, c2) + x(h23, c1) + x(h23, c2) =L= 420 ; (LHS = 0)$

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420$; (LHS = 0)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420$; (LHS = 0)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) =L= -120$; (LHS = 0, INFES = 120 ****)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) =L= -70$; (LHS = 0, INFES = 70 ****)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) =L= -95$; (LHS = 0, INFES = 95 ****)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. $L(h0,c1) =L= 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) =L= 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) =L= 10$; (LHS = 0)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. $L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) + L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) + L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) + L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40$; (LHS = 0)

shortage_total(c2).. $L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) + L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) + L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) + L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30$; (LHS = 0)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 0, INFES = 80 ****)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 0, INFES = 80 ****)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 0, INFES = 80 ****)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System
Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 0, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
1 (.LO, .L, .UP, .M = 0, 0, +INF, 0)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)


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1      min_coverage(h5)
1      min_coverage(h6)

x(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 0)
-189  obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
      -1     shortage(h0,c2)
      -1     shortage(h1,c2)
      -1     shortage(h2,c2)
      -1     shortage(h4,c2)
      -1     shortage(h5,c2)
      -1     shortage(h6,c2)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

x(h1,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 0)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)
      -1     shortage(h1,c1)
      -1     shortage(h2,c1)
      -1     shortage(h3,c1)
      -1     shortage(h5,c1)
      -1     shortage(h6,c1)
      -1     shortage(h7,c1)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h3)
      1      min_coverage(h5)
      1      min_coverage(h6)
      1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```
L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 0)
-30   obj
-1    shortage(h0,c1)
1     shortage_cap(h0,c1)
1     shortage_total(c1)
```

```
L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 0)
-45   obj
-1    shortage(h0,c2)
1     shortage_cap(h0,c2)
1     shortage_total(c2)
```

```
L(h1,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 0)
-30   obj
-1    shortage(h1,c1)
1     shortage_cap(h1,c1)
1     shortage_total(c1)
```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```
RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [           NA,           NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]
```

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General Algebraic Modeling System

Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t1

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97

NON ZERO ELEMENTS 1,153 DISCRETE VARIABLES 96

GENERATION TIME = 0.016 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
 L O O P S t t1

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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	157

**** SOLVER STATUS 1 Normal Completion
**** MODEL STATUS 8 Integer Solution
**** OBJECTIVE VALUE 178479.0000

RESOURCE USAGE, LIMIT 0.047 10000000000.000
ITERATION COUNT, LIMIT 80 2147483647
*** This solver runs with a demo license. No commercial use.
--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

--- MIP status (102): integer optimal, tolerance.
--- Cplex Time: 0.05sec (det. 7.36 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances
MIP Solution: 178479.000000 (80 iterations, 0 nodes)
Final Solve: 178479.000000 (0 iterations)

Best possible: 178471.000000
Absolute gap: 8.000000
Relative gap: 0.000045

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	8.0000	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	38.0000	100.0000	.
h3	-INF	18.0000	100.0000	.
h4	-INF	12.0000	100.0000	.
h5	-INF	42.0000	100.0000	.
h6	-INF	16.0000	100.0000	.
h7	-INF	76.0000	100.0000	.
h8	-INF	100.0000	100.0000	.
h9	-INF	62.0000	100.0000	.
h10	-INF	58.0000	100.0000	.
h11	-INF	86.0000	100.0000	.
h12	-INF	54.0000	100.0000	.
h13	-INF	39.0000	100.0000	.
h14	-INF	71.0000	100.0000	.
h15	-INF	68.0000	100.0000	.
h16	-INF	73.0000	100.0000	.
h17	-INF	100.0000	100.0000	.
h18	-INF	83.0000	100.0000	.
h19	-INF	59.0000	100.0000	.
h20	-INF	36.0000	100.0000	.
h21	-INF	24.0000	100.0000	.
h22	-INF	4.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	190.0000	420.0000	.
h1	-INF	127.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	84.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	116.0000	420.0000	.
h7	-INF	190.0000	420.0000	.

h8	-INF	260.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	340.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	417.0000	420.0000	.
h14	-INF	384.0000	420.0000	.
h15	-INF	384.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-120.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-54.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-34.0000	-30.0000	.
h5 .c2	-INF	-46.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-56.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-150.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-210.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-257.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.

h14.c2	-INF	-164.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-154.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-251.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	4.0000	10.0000	.
h1 .c2	-INF	4.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	5.0000	10.0000	.
h7 .c2	-INF	5.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.

h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	.	10.0000	.
h18.c2	-INF	10.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	1.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	39.0000	40.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	190.0000	+INF	.
h1	80.0000	127.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	84.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	116.0000	+INF	.
h7	80.0000	190.0000	+INF	.
h8	80.0000	260.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	340.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	417.0000	+INF	.
h14	80.0000	384.0000	+INF	.
h15	80.0000	384.0000	+INF	.

h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	320.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	178479.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	8.0000	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	26.0000	+INF	189.0000
h2 .c2	.	12.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	18.0000	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	12.0000	+INF	171.0000
h5 .c1	.	34.0000	+INF	162.0000
h5 .c2	.	8.0000	+INF	162.0000
h6 .c1	.	.	+INF	153.0000
h6 .c2	.	16.0000	+INF	153.0000
h7 .c1	.	45.0000	+INF	144.0000
h7 .c2	.	31.0000	+INF	144.0000
h8 .c1	.	79.0000	+INF	67.5000
h8 .c2	.	21.0000	+INF	67.5000
h9 .c1	.	42.0000	+INF	126.0000
h9 .c2	.	20.0000	+INF	126.0000
h10.c1	.	25.0000	+INF	126.0000
h10.c2	.	33.0000	+INF	126.0000
h11.c1	.	64.0000	+INF	135.0000
h11.c2	.	22.0000	+INF	135.0000
h12.c1	.	27.0000	+INF	144.0000
h12.c2	.	27.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	39.0000	+INF	153.0000
h14.c1	.	47.0000	+INF	162.0000
h14.c2	.	24.0000	+INF	162.0000

h15.c1	.	52.0000	+INF	171.0000
h15.c2	.	16.0000	+INF	171.0000
h16.c1	.	35.0000	+INF	180.0000
h16.c2	.	38.0000	+INF	180.0000
h17.c1	.	72.0000	+INF	189.0000
h17.c2	.	28.0000	+INF	189.0000
h18.c1	.	49.0000	+INF	189.0000
h18.c2	.	34.0000	+INF	189.0000
h19.c1	.	31.0000	+INF	189.0000
h19.c2	.	28.0000	+INF	189.0000
h20.c1	.	36.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000
h21.c1	.	24.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	4.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	4.0000	+INF	30.0000
h1 .c2	.	4.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	5.0000	+INF	60.0000
h7 .c2	.	5.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000

h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	.	+INF	30.0000
h18.c2	.	10.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	1.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 74.25*x(h8,c1) - 74.25*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -

$45 * L(h17, c2) - 30 * L(h18, c1) - 45 * L(h18, c2) - 30 * L(h19, c1) - 45 * L(h19, c2) -$
 $30 * L(h20, c1) - 45 * L(h20, c2) - 30 * L(h21, c1) - 45 * L(h21, c2) - 30 * L(h22, c1) -$
 $45 * L(h22, c2) - 30 * L(h23, c1) - 45 * L(h23, c2) = E = 0 ; (LHS = -675, INFES = 675 ****)$

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0, c1) + x(h0, c2) = L = 100 ; (LHS = 7.999999999999635)$

cap_per_hour(h1).. $x(h1, c1) + x(h1, c2) = L = 100 ; (LHS = 0)$

cap_per_hour(h2).. $x(h2, c1) + x(h2, c2) = L = 100 ; (LHS = 38.00000000000009)$

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0, c1) + x(h0, c2) + x(h18, c1) + x(h18, c2) + x(h19, c1) + x(h19, c2) +$
 $x(h20, c1) + x(h20, c2) + x(h22, c1) + x(h22, c2) + x(h23, c1) + x(h23, c2) = L = 420 ;$
 (LHS = 190)

cap_total(h1).. $x(h0, c1) + x(h0, c2) + x(h1, c1) + x(h1, c2) + x(h19, c1) + x(h19, c2) +$
 $x(h20, c1) + x(h20, c2) + x(h21, c1) + x(h21, c2) + x(h23, c1) + x(h23, c2) = L = 420 ; (LHS$
 $= 126.9999999999995)$

cap_total(h2).. $x(h0, c1) + x(h0, c2) + x(h1, c1) + x(h1, c2) + x(h2, c1) + x(h2, c2) +$
 $x(h20, c1) + x(h20, c2) + x(h21, c1) + x(h21, c2) + x(h22, c1) + x(h22, c2) = L = 420 ; (LHS$
 $= 110)$

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0, c1).. $- x(h0, c1) - x(h18, c1) - x(h19, c1) - x(h20, c1) - x(h22, c1) -$
 $x(h23, c1) - L(h0, c1) = L = -120 ; (LHS = -120)$

shortage(h0, c2).. $- x(h0, c2) - x(h18, c2) - x(h19, c2) - x(h20, c2) - x(h22, c2) -$
 $x(h23, c2) - L(h0, c2) = L = -70 ; (LHS = -70)$

shortage(h1, c1).. $- x(h0, c1) - x(h1, c1) - x(h19, c1) - x(h20, c1) - x(h21, c1) -$
 $x(h23, c1) - L(h1, c1) = L = -95 ; (LHS = -95)$

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. L(h0,c1) =L= 10 ; (LHS = 0)

shortage_cap(h0,c2).. L(h0,c2) =L= 10 ; (LHS = 0)

shortage_cap(h1,c1).. L(h1,c1) =L= 10 ; (LHS = 4.00000000005381)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40 ; (LHS =
39.0000000000027)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 30)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 190)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 126.999999999995)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z

1 (.LO, .L, .UP, .M = -INF, 178479, +INF, 0)
obj

---- x operators starting at hour for company

```
x(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1  cap_per_hour(h0)
      1  cap_total(h0)
      1  cap_total(h1)
      1  cap_total(h2)
      1  cap_total(h4)
      1  cap_total(h5)
      1  cap_total(h6)
     -1  shortage(h0,c1)
     -1  shortage(h1,c1)
     -1  shortage(h2,c1)
     -1  shortage(h4,c1)
     -1  shortage(h5,c1)
     -1  shortage(h6,c1)
      1  min_coverage(h0)
      1  min_coverage(h1)
      1  min_coverage(h2)
      1  min_coverage(h4)
      1  min_coverage(h5)
      1  min_coverage(h6)

x(h0,c2)
      (.LO, .L, .UP, .M = 0, 7.999999999999635, +INF, 189)
-189  obj
      1  cap_per_hour(h0)
      1  cap_total(h0)
      1  cap_total(h1)
      1  cap_total(h2)
      1  cap_total(h4)
      1  cap_total(h5)
      1  cap_total(h6)
     -1  shortage(h0,c2)
     -1  shortage(h1,c2)
     -1  shortage(h2,c2)
     -1  shortage(h4,c2)
     -1  shortage(h5,c2)
     -1  shortage(h6,c2)
      1  min_coverage(h0)
      1  min_coverage(h1)
      1  min_coverage(h2)
      1  min_coverage(h4)
      1  min_coverage(h5)
      1  min_coverage(h6)
```

```

x(h1,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1  cap_per_hour(h1)
      1  cap_total(h1)
      1  cap_total(h2)
      1  cap_total(h3)
      1  cap_total(h5)
      1  cap_total(h6)
      1  cap_total(h7)
     -1  shortage(h1,c1)
     -1  shortage(h2,c1)
     -1  shortage(h3,c1)
     -1  shortage(h5,c1)
     -1  shortage(h6,c1)
     -1  shortage(h7,c1)
      1  min_coverage(h1)
      1  min_coverage(h2)
      1  min_coverage(h3)
      1  min_coverage(h5)
      1  min_coverage(h6)
      1  min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
    -30  obj
      -1  shortage(h0,c1)
       1  shortage_cap(h0,c1)
       1  shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
    -45  obj
      -1  shortage(h0,c2)
       1  shortage_cap(h0,c2)
       1  shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 4.00000000005381, +INF, 30)
    -30  obj
      -1  shortage(h1,c1)
       1  shortage_cap(h1,c1)
       1  shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

RHS [min, max] : [1.000E+01, 4.200E+02] - Zero values observed as well

Bound [min, max] : [NA, NA] - Zero values observed as well

Matrix [min, max] : [1.000E+00, 1.890E+02]

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General Algebraic Modeling System

Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS

t t2

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
---------------------	---	------------------	-----

BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
---------------------	---	------------------	----

NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96
-------------------	-------	--------------------	----

GENERATION TIME	=	0.000 SECONDS	4 MB	52.5.0 9bc771bf WEX-WEI
LOOPS		t t2		

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General Algebraic Modeling System

Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	157

**** SOLVER STATUS 1 Normal Completion

**** MODEL STATUS 8 Integer Solution

**** OBJECTIVE VALUE 179154.0000

RESOURCE USAGE, LIMIT 0.062 10000000000.000

ITERATION COUNT, LIMIT 89 2147483647

*** This solver runs with a demo license. No commercial use.

--- GMO setup time: 0.00s

```

--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

```

```

--- MIP status (102): integer optimal, tolerance.
--- Cplex Time: 0.06sec (det. 11.48 ticks)

```

```

--- Returning a primal only solution to GAMS (marginals all set to NA).
--- Fixing integer variables and solving final LP...

```

```

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.00sec (det. 0.14 ticks)

```

Solution satisfies tolerances

```

MIP Solution:      179154.000000    (89 iterations, 0 nodes)
Final Solve:      179154.000000    (0 iterations)

```

```

Best possible:      179145.882353
Absolute gap:        8.117647
Relative gap:        0.000045

```

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

```

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

```

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	.	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	9.0000	100.0000	.
h3	-INF	10.0000	100.0000	.
h4	-INF	20.0000	100.0000	.
h5	-INF	50.0000	100.0000	.
h6	-INF	37.0000	100.0000	.
h7	-INF	84.0000	100.0000	.
h8	-INF	100.0000	100.0000	.
h9	-INF	46.0000	100.0000	.
h10	-INF	37.0000	100.0000	.
h11	-INF	86.0000	100.0000	.
h12	-INF	46.0000	100.0000	.

h13	-INF	55.0000	100.0000	.
h14	-INF	100.0000	100.0000	.
h15	-INF	60.0000	100.0000	.
h16	-INF	81.0000	100.0000	.
h17	-INF	92.0000	100.0000	.
h18	-INF	46.0000	100.0000	.
h19	-INF	67.0000	100.0000	.
h20	-INF	36.0000	100.0000	.
h21	-INF	24.0000	100.0000	.
h22	-INF	41.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	190.0000	420.0000	.
h1	-INF	127.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	84.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	116.0000	420.0000	.
h7	-INF	190.0000	420.0000	.
h8	-INF	260.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	340.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	417.0000	420.0000	.
h14	-INF	384.0000	420.0000	.
h15	-INF	384.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-120.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.

h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-54.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-34.0000	-30.0000	.
h5 .c2	-INF	-46.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-56.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-150.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-210.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-257.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-164.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-154.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-251.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.

h1 .c1	-INF	4.0000	10.0000	.
h1 .c2	-INF	4.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	5.0000	10.0000	.
h7 .c2	-INF	5.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	.	10.0000	.
h18.c2	-INF	10.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	1.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

LOWER

LEVEL

UPPER

MARGINAL

c1	-INF	39.0000	40.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	190.0000	+INF	.
h1	80.0000	127.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	84.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	116.0000	+INF	.
h7	80.0000	190.0000	+INF	.
h8	80.0000	260.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	340.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	417.0000	+INF	.
h14	80.0000	384.0000	+INF	.
h15	80.0000	384.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	320.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	179154.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	1.0000	+INF	189.0000
h2 .c2	.	8.0000	+INF	189.0000

h3 .c1	.	.	+INF	180.0000
h3 .c2	.	10.0000	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	20.0000	+INF	171.0000
h5 .c1	.	34.0000	+INF	162.0000
h5 .c2	.	16.0000	+INF	162.0000
h6 .c1	.	25.0000	+INF	153.0000
h6 .c2	.	12.0000	+INF	153.0000
h7 .c1	.	45.0000	+INF	144.0000
h7 .c2	.	39.0000	+INF	144.0000
h8 .c1	.	79.0000	+INF	74.2500
h8 .c2	.	21.0000	+INF	74.2500
h9 .c1	.	42.0000	+INF	126.0000
h9 .c2	.	4.0000	+INF	126.0000
h10.c1	.	.	+INF	126.0000
h10.c2	.	37.0000	+INF	126.0000
h11.c1	.	64.0000	+INF	135.0000
h11.c2	.	22.0000	+INF	135.0000
h12.c1	.	27.0000	+INF	144.0000
h12.c2	.	19.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	55.0000	+INF	153.0000
h14.c1	.	72.0000	+INF	162.0000
h14.c2	.	28.0000	+INF	162.0000
h15.c1	.	52.0000	+INF	171.0000
h15.c2	.	8.0000	+INF	171.0000
h16.c1	.	35.0000	+INF	180.0000
h16.c2	.	46.0000	+INF	180.0000
h17.c1	.	72.0000	+INF	189.0000
h17.c2	.	20.0000	+INF	189.0000
h18.c1	.	24.0000	+INF	189.0000
h18.c2	.	22.0000	+INF	189.0000
h19.c1	.	31.0000	+INF	189.0000
h19.c2	.	36.0000	+INF	189.0000
h20.c1	.	36.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000
h21.c1	.	24.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	29.0000	+INF	189.0000
h22.c2	.	12.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	4.0000	+INF	30.0000

h1 .c2	.	4.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	5.0000	+INF	60.0000
h7 .c2	.	5.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	.	+INF	30.0000
h18.c2	.	10.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	1.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NOOPT
 0 INFEASIBLE
 0 UNBOUNDED

---- obj =E= objective definition

obj.. $z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) - 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) - 144*x(h7,c2) - 81*x(h8,c1) - 81*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) - 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) - 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) - 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) - 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) - 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) - 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) - 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) - 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) - 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) - 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) - 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) - 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) - 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) - 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) - 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -675, INFES = 675 ****)$

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 0)$

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 0)$

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 9)$

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS = 190)$

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS = 127)$

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420$; (LHS = 110)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) =L= -120$; (LHS = -120)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) =L= -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) =L= -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. $L(h0,c1) =L= 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) =L= 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) =L= 10$; (LHS = 4)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. $L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) + L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) + L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) + L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40$; (LHS = 39)

shortage_total(c2).. $L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) + L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) + L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) + L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30$; (LHS = 30)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)$

=G= 80 ; (LHS = 190)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 127)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 179154, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
-189 (.LO, .L, .UP, .M = 0, 0, +INF, 189)
obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

```

x(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1  cap_per_hour(h0)
      1  cap_total(h0)
      1  cap_total(h1)
      1  cap_total(h2)
      1  cap_total(h4)
      1  cap_total(h5)
      1  cap_total(h6)
     -1  shortage(h0,c2)
     -1  shortage(h1,c2)
     -1  shortage(h2,c2)
     -1  shortage(h4,c2)
     -1  shortage(h5,c2)
     -1  shortage(h6,c2)
      1  min_coverage(h0)
      1  min_coverage(h1)
      1  min_coverage(h2)
      1  min_coverage(h4)
      1  min_coverage(h5)
      1  min_coverage(h6)

```

```

x(h1,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1  cap_per_hour(h1)
      1  cap_total(h1)
      1  cap_total(h2)
      1  cap_total(h3)
      1  cap_total(h5)
      1  cap_total(h6)
      1  cap_total(h7)
     -1  shortage(h1,c1)
     -1  shortage(h2,c1)
     -1  shortage(h3,c1)
     -1  shortage(h5,c1)
     -1  shortage(h6,c1)
     -1  shortage(h7,c1)
      1  min_coverage(h1)
      1  min_coverage(h2)
      1  min_coverage(h3)
      1  min_coverage(h5)
      1  min_coverage(h6)
      1  min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1    shortage(h0,c1)
1     shortage_cap(h0,c1)
1     shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1    shortage(h0,c2)
1     shortage_cap(h0,c2)
1     shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 4, +INF, 30)
-30   obj
-1    shortage(h1,c1)
1     shortage_cap(h1,c1)
1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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General Algebraic Modeling System

Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t3

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.000 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t3

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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	Cplex	FROM LINE	157

**** SOLVER STATUS 1 Normal Completion
**** MODEL STATUS 8 Integer Solution
**** OBJECTIVE VALUE 179829.0000

RESOURCE USAGE, LIMIT 0.031 10000000000.000
ITERATION COUNT, LIMIT 94 2147483647
*** This solver runs with a demo license. No commercial use.
--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

--- MIP status (102): integer optimal, tolerance.
--- Cplex Time: 0.03sec (det. 7.22 ticks)
--- Returning a primal only solution to GAMS (marginals all set to NA).
--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances

MIP Solution:	179829.000000	(94 iterations, 0 nodes)
Final Solve:	179829.000000	(0 iterations)

Best possible:	179820.882353
Absolute gap:	8.117647
Relative gap:	0.000045

LOWER	LEVEL	UPPER	MARGINAL
-------	-------	-------	----------

---- EQU obj . . . 1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	.	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	21.0000	100.0000	.
h3	-INF	10.0000	100.0000	.
h4	-INF	20.0000	100.0000	.
h5	-INF	50.0000	100.0000	.
h6	-INF	25.0000	100.0000	.
h7	-INF	84.0000	100.0000	.
h8	-INF	100.0000	100.0000	.
h9	-INF	46.0000	100.0000	.
h10	-INF	49.0000	100.0000	.
h11	-INF	86.0000	100.0000	.
h12	-INF	46.0000	100.0000	.
h13	-INF	55.0000	100.0000	.
h14	-INF	88.0000	100.0000	.
h15	-INF	60.0000	100.0000	.
h16	-INF	81.0000	100.0000	.
h17	-INF	92.0000	100.0000	.
h18	-INF	58.0000	100.0000	.
h19	-INF	67.0000	100.0000	.
h20	-INF	36.0000	100.0000	.
h21	-INF	24.0000	100.0000	.
h22	-INF	29.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	190.0000	420.0000	.
h1	-INF	127.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	84.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	116.0000	420.0000	.
h7	-INF	190.0000	420.0000	.
h8	-INF	260.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.

h11	-INF	340.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	417.0000	420.0000	.
h14	-INF	384.0000	420.0000	.
h15	-INF	384.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-120.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-54.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-34.0000	-30.0000	.
h5 .c2	-INF	-46.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-56.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-150.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-210.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-257.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-164.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-154.0000	-130.0000	.

h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-251.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	4.0000	10.0000	.
h1 .c2	-INF	4.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	5.0000	10.0000	.
h7 .c2	-INF	5.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.

h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	.	10.0000	.
h18.c2	-INF	10.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	1.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	39.0000	40.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	190.0000	+INF	.
h1	80.0000	127.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	84.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	116.0000	+INF	.
h7	80.0000	190.0000	+INF	.
h8	80.0000	260.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	340.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	417.0000	+INF	.
h14	80.0000	384.0000	+INF	.
h15	80.0000	384.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.

h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	320.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	179829.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	1.0000	+INF	189.0000
h2 .c2	.	20.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	10.0000	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	20.0000	+INF	171.0000
h5 .c1	.	34.0000	+INF	162.0000
h5 .c2	.	16.0000	+INF	162.0000
h6 .c1	.	25.0000	+INF	153.0000
h6 .c2	.	.	+INF	153.0000
h7 .c1	.	45.0000	+INF	144.0000
h7 .c2	.	39.0000	+INF	144.0000
h8 .c1	.	79.0000	+INF	81.0000
h8 .c2	.	21.0000	+INF	81.0000
h9 .c1	.	42.0000	+INF	126.0000
h9 .c2	.	4.0000	+INF	126.0000
h10.c1	.	.	+INF	126.0000
h10.c2	.	49.0000	+INF	126.0000
h11.c1	.	64.0000	+INF	135.0000
h11.c2	.	22.0000	+INF	135.0000
h12.c1	.	27.0000	+INF	144.0000
h12.c2	.	19.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	55.0000	+INF	153.0000
h14.c1	.	72.0000	+INF	162.0000
h14.c2	.	16.0000	+INF	162.0000
h15.c1	.	52.0000	+INF	171.0000
h15.c2	.	8.0000	+INF	171.0000
h16.c1	.	35.0000	+INF	180.0000

h16.c2	.	46.0000	+INF	180.0000
h17.c1	.	72.0000	+INF	189.0000
h17.c2	.	20.0000	+INF	189.0000
h18.c1	.	24.0000	+INF	189.0000
h18.c2	.	34.0000	+INF	189.0000
h19.c1	.	31.0000	+INF	189.0000
h19.c2	.	36.0000	+INF	189.0000
h20.c1	.	36.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000
h21.c1	.	24.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	29.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	4.0000	+INF	30.0000
h1 .c2	.	4.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	5.0000	+INF	60.0000
h7 .c2	.	5.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000

h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	.	+INF	30.0000
h18.c2	.	10.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	1.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 87.75*x(h8,c1) - 87.75*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -675, INFES = 675 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0,c1) + x(h0,c2) =L= 100$; (LHS = 0)

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) =L= 100$; (LHS = 0)

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) =L= 100$; (LHS = 21.0000000000728)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420$; (LHS = 190)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420$; (LHS = 127)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420$; (LHS = 110)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) =L= -120$; (LHS = -120)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) =L= -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) =L= -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= $L \leq \text{maxShort}$

shortage_cap(h0,c1).. $L(h0,c1) =L= 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) =L= 10$; (LHS = 0)

shortage_cap(h1,c1).. L(h1,c1) =L= 10 ; (LHS = 3.99999999992719)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40 ; (LHS = 39)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 30)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 190)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 127)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z

1 (.LO, .L, .UP, .M = -INF, 179829, +INF, 0)
obj

---- x operators starting at hour for company

```

x(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1  cap_per_hour(h0)
      1  cap_total(h0)
      1  cap_total(h1)
      1  cap_total(h2)
      1  cap_total(h4)
      1  cap_total(h5)
      1  cap_total(h6)
     -1  shortage(h0,c1)
     -1  shortage(h1,c1)
     -1  shortage(h2,c1)
     -1  shortage(h4,c1)
     -1  shortage(h5,c1)
     -1  shortage(h6,c1)
      1  min_coverage(h0)
      1  min_coverage(h1)
      1  min_coverage(h2)
      1  min_coverage(h4)
      1  min_coverage(h5)
      1  min_coverage(h6)

```

```

x(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1  cap_per_hour(h0)
      1  cap_total(h0)
      1  cap_total(h1)
      1  cap_total(h2)
      1  cap_total(h4)
      1  cap_total(h5)
      1  cap_total(h6)
     -1  shortage(h0,c2)
     -1  shortage(h1,c2)
     -1  shortage(h2,c2)
     -1  shortage(h4,c2)
     -1  shortage(h5,c2)
     -1  shortage(h6,c2)
      1  min_coverage(h0)
      1  min_coverage(h1)
      1  min_coverage(h2)
      1  min_coverage(h4)
      1  min_coverage(h5)
      1  min_coverage(h6)

```

```

x(h1,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1  cap_per_hour(h1)

```

```

1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h3)
1      cap_total(h5)
1      cap_total(h6)
1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h3)
1      min_coverage(h5)
1      min_coverage(h6)
1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.L0, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1    shortage(h0,c1)
1     shortage_cap(h0,c1)
1     shortage_total(c1)

```

```

L(h0,c2)
      (.L0, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1    shortage(h0,c2)
1     shortage_cap(h0,c2)
1     shortage_total(c2)

```

```

L(h1,c1)
      (.L0, .L, .UP, .M = 0, 3.99999999992719, +INF, 30)
-30   obj
-1    shortage(h1,c1)
1     shortage_cap(h1,c1)
1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

RHS [min, max] : [1.000E+01, 4.200E+02] - Zero values observed as well
 Bound [min, max] : [NA, NA] - Zero values observed as well
 Matrix [min, max] : [1.000E+00, 1.890E+02]

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 General Algebraic Modeling System
 Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t4

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.000 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
 L O O P S t t4

GAMS 52.5.0 9bc771bf Jan 28, 2026 WEX-WEI x86 64bit/MS Windows - 02/15/26
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 General Algebraic Modeling System
 Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	157

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 8 Integer Solution
 **** OBJECTIVE VALUE 180504.0000

RESOURCE USAGE, LIMIT 0.110 10000000000.000
 ITERATION COUNT, LIMIT 88 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex


```

--- MIP status (102): integer optimal, tolerance.
--- Cplex Time: 0.11sec (det. 7.23 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.00sec (det. 0.14 ticks)

```

Solution satisfies tolerances

```

MIP Solution:      180504.000000      (88 iterations, 0 nodes)
Final Solve:       180504.000000      (0 iterations)

```

```

Best possible:      180495.882353
Absolute gap:        8.117647
Relative gap:        0.000045

```

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

```

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

```

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	8.0000	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	13.0000	100.0000	.
h3	-INF	.	100.0000	.
h4	-INF	12.0000	100.0000	.
h5	-INF	42.0000	100.0000	.
h6	-INF	41.0000	100.0000	.
h7	-INF	94.0000	100.0000	.
h8	-INF	100.0000	100.0000	.
h9	-INF	62.0000	100.0000	.
h10	-INF	33.0000	100.0000	.
h11	-INF	68.0000	100.0000	.
h12	-INF	54.0000	100.0000	.
h13	-INF	39.0000	100.0000	.
h14	-INF	96.0000	100.0000	.
h15	-INF	86.0000	100.0000	.
h16	-INF	73.0000	100.0000	.

h17	-INF	100.0000	100.0000	.
h18	-INF	58.0000	100.0000	.
h19	-INF	41.0000	100.0000	.
h20	-INF	36.0000	100.0000	.
h21	-INF	24.0000	100.0000	.
h22	-INF	29.0000	100.0000	.
h23	-INF	18.0000	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	190.0000	420.0000	.
h1	-INF	127.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	84.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	116.0000	420.0000	.
h7	-INF	190.0000	420.0000	.
h8	-INF	260.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	340.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	417.0000	420.0000	.
h14	-INF	384.0000	420.0000	.
h15	-INF	384.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-120.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-54.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.

h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-34.0000	-30.0000	.
h5 .c2	-INF	-46.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-56.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-150.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-210.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-257.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-164.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-154.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-251.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	4.0000	10.0000	.
h1 .c2	-INF	4.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.

h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	5.0000	10.0000	.
h7 .c2	-INF	5.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	.	10.0000	.
h18.c2	-INF	10.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	1.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	39.0000	40.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	190.0000	+INF	.
h1	80.0000	127.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	84.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	116.0000	+INF	.
h7	80.0000	190.0000	+INF	.
h8	80.0000	260.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	340.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	417.0000	+INF	.
h14	80.0000	384.0000	+INF	.
h15	80.0000	384.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	320.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	180504.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	8.0000	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	1.0000	+INF	189.0000
h2 .c2	.	12.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	.	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	12.0000	+INF	171.0000

h5 .c1	.	34.0000	+INF	162.0000
h5 .c2	.	8.0000	+INF	162.0000
h6 .c1	.	25.0000	+INF	153.0000
h6 .c2	.	16.0000	+INF	153.0000
h7 .c1	.	45.0000	+INF	144.0000
h7 .c2	.	49.0000	+INF	144.0000
h8 .c1	.	79.0000	+INF	87.7500
h8 .c2	.	21.0000	+INF	87.7500
h9 .c1	.	42.0000	+INF	126.0000
h9 .c2	.	20.0000	+INF	126.0000
h10.c1	.	.	+INF	126.0000
h10.c2	.	33.0000	+INF	126.0000
h11.c1	.	64.0000	+INF	135.0000
h11.c2	.	4.0000	+INF	135.0000
h12.c1	.	27.0000	+INF	144.0000
h12.c2	.	27.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	39.0000	+INF	153.0000
h14.c1	.	72.0000	+INF	162.0000
h14.c2	.	24.0000	+INF	162.0000
h15.c1	.	52.0000	+INF	171.0000
h15.c2	.	34.0000	+INF	171.0000
h16.c1	.	35.0000	+INF	180.0000
h16.c2	.	38.0000	+INF	180.0000
h17.c1	.	72.0000	+INF	189.0000
h17.c2	.	28.0000	+INF	189.0000
h18.c1	.	24.0000	+INF	189.0000
h18.c2	.	34.0000	+INF	189.0000
h19.c1	.	31.0000	+INF	189.0000
h19.c2	.	10.0000	+INF	189.0000
h20.c1	.	36.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000
h21.c1	.	24.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	29.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	18.0000	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	4.0000	+INF	30.0000
h1 .c2	.	4.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000

h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	5.0000	+INF	60.0000
h7 .c2	.	5.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	.	+INF	30.0000
h18.c2	.	10.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	1.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) - 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) - 144*x(h7,c2) - 94.5*x(h8,c1) - 94.5*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) - 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) - 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) - 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) - 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) - 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) - 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) - 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) - 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) - 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) - 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) - 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) - 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) - 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) - 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) - 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -675.0000000000029, INFES = 675.0000000000029 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 8)

cap_per_hour(h1).. x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h2).. x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 13.00000000000266)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS = 190)

cap_total(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS = 127)

cap_total(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS

= 110)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. - x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) -
x(h23,c1) - L(h0,c1) =L= -120 ; (LHS = -120)

shortage(h0,c2).. - x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) -
x(h23,c2) - L(h0,c2) =L= -70 ; (LHS = -70)

shortage(h1,c1).. - x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) -
x(h23,c1) - L(h1,c1) =L= -95 ; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. L(h0,c1) =L= 10 ; (LHS = 0)

shortage_cap(h0,c2).. L(h0,c2) =L= 10 ; (LHS = 0)

shortage_cap(h1,c1).. L(h1,c1) =L= 10 ; (LHS = 3.99999999997342)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40 ; (LHS = 39)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 30)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 190)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +

$x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)$
 $=G= 80 ; (LHS = 127)$

$min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)$
 $+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;$
 $(LHS = 110)$

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z
 1 (.LO, .L, .UP, .M = -INF, 180504, +INF, 0)
 obj

---- x operators starting at hour for company

$x(h0,c1)$
 (.LO, .L, .UP, .M = 0, 0, +INF, 189)
 -189 obj
 1 cap_per_hour(h0)
 1 cap_total(h0)
 1 cap_total(h1)
 1 cap_total(h2)
 1 cap_total(h4)
 1 cap_total(h5)
 1 cap_total(h6)
 -1 shortage(h0,c1)
 -1 shortage(h1,c1)
 -1 shortage(h2,c1)
 -1 shortage(h4,c1)
 -1 shortage(h5,c1)
 -1 shortage(h6,c1)
 1 min_coverage(h0)
 1 min_coverage(h1)
 1 min_coverage(h2)
 1 min_coverage(h4)
 1 min_coverage(h5)
 1 min_coverage(h6)

$x(h0,c2)$
 (.LO, .L, .UP, .M = 0, 8, +INF, 189)
 -189 obj

```

1      cap_per_hour(h0)
1      cap_total(h0)
1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
1      cap_per_hour(h1)
1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h3)
1      cap_total(h5)
1      cap_total(h6)
1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h3)
1      min_coverage(h5)
1      min_coverage(h6)
1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30  obj

```

```

-1 shortage(h0,c1)
1 shortage_cap(h0,c1)
1 shortage_total(c1)

```

L(h0,c2)

```

(.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45 obj
-1 shortage(h0,c2)
1 shortage_cap(h0,c2)
1 shortage_total(c2)

```

L(h1,c1)

```

(.LO, .L, .UP, .M = 0, 3.99999999997342, +INF, 30)
-30 obj
-1 shortage(h1,c1)
1 shortage_cap(h1,c1)
1 shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix    [min, max] : [ 1.000E+00, 1.890E+02]

```

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General Algebraic Modeling System

Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS

t t5

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

```

GENERATION TIME      =      0.000 SECONDS      4 MB  52.5.0 9bc771bf WEX-WEI
      L O O P S      t t5

```

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	157

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 8 Integer Solution
 **** OBJECTIVE VALUE 181179.0000

RESOURCE USAGE, LIMIT 0.187 10000000000.000

ITERATION COUNT, LIMIT 88 2147483647

*** This solver runs with a demo license. No commercial use.

--- GMO setup time: 0.00s

--- GMO memory 0.53 Mb (peak 0.53 Mb)

--- Dictionary memory 0.00 Mb

--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)

--- Starting Cplex

--- MIP status (102): integer optimal, tolerance.

--- Cplex Time: 0.19sec (det. 7.23 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).

--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.

--- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances

MIP Solution: 181179.000000 (88 iterations, 0 nodes)

Final Solve: 181179.000000 (0 iterations)

Best possible: 181170.882353

Absolute gap: 8.117647

Relative gap: 0.000045

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	8.0000	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	13.0000	100.0000	.
h3	-INF	.	100.0000	.
h4	-INF	12.0000	100.0000	.
h5	-INF	42.0000	100.0000	.
h6	-INF	41.0000	100.0000	.
h7	-INF	94.0000	100.0000	.
h8	-INF	100.0000	100.0000	.
h9	-INF	62.0000	100.0000	.
h10	-INF	33.0000	100.0000	.
h11	-INF	68.0000	100.0000	.
h12	-INF	54.0000	100.0000	.
h13	-INF	39.0000	100.0000	.
h14	-INF	96.0000	100.0000	.
h15	-INF	86.0000	100.0000	.
h16	-INF	73.0000	100.0000	.
h17	-INF	100.0000	100.0000	.
h18	-INF	58.0000	100.0000	.
h19	-INF	41.0000	100.0000	.
h20	-INF	36.0000	100.0000	.
h21	-INF	24.0000	100.0000	.
h22	-INF	29.0000	100.0000	.
h23	-INF	18.0000	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	190.0000	420.0000	.
h1	-INF	127.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	84.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	116.0000	420.0000	.
h7	-INF	190.0000	420.0000	.
h8	-INF	260.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	340.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	417.0000	420.0000	.

h14	-INF	384.0000	420.0000	.
h15	-INF	384.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-120.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-54.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-34.0000	-30.0000	.
h5 .c2	-INF	-46.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-56.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-150.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-210.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-257.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-164.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-154.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.

h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-251.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	4.0000	10.0000	.
h1 .c2	-INF	4.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	5.0000	10.0000	.
h7 .c2	-INF	5.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.

h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	.	10.0000	.
h18.c2	-INF	10.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	1.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	39.0000	40.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	190.0000	+INF	.
h1	80.0000	127.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	84.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	116.0000	+INF	.
h7	80.0000	190.0000	+INF	.
h8	80.0000	260.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	340.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	417.0000	+INF	.
h14	80.0000	384.0000	+INF	.
h15	80.0000	384.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.

h22	80.0000	320.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	181179.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	8.0000	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	1.0000	+INF	189.0000
h2 .c2	.	12.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	.	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	12.0000	+INF	171.0000
h5 .c1	.	34.0000	+INF	162.0000
h5 .c2	.	8.0000	+INF	162.0000
h6 .c1	.	25.0000	+INF	153.0000
h6 .c2	.	16.0000	+INF	153.0000
h7 .c1	.	45.0000	+INF	144.0000
h7 .c2	.	49.0000	+INF	144.0000
h8 .c1	.	79.0000	+INF	94.5000
h8 .c2	.	21.0000	+INF	94.5000
h9 .c1	.	42.0000	+INF	126.0000
h9 .c2	.	20.0000	+INF	126.0000
h10.c1	.	.	+INF	126.0000
h10.c2	.	33.0000	+INF	126.0000
h11.c1	.	64.0000	+INF	135.0000
h11.c2	.	4.0000	+INF	135.0000
h12.c1	.	27.0000	+INF	144.0000
h12.c2	.	27.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	39.0000	+INF	153.0000
h14.c1	.	72.0000	+INF	162.0000
h14.c2	.	24.0000	+INF	162.0000
h15.c1	.	52.0000	+INF	171.0000
h15.c2	.	34.0000	+INF	171.0000
h16.c1	.	35.0000	+INF	180.0000
h16.c2	.	38.0000	+INF	180.0000
h17.c1	.	72.0000	+INF	189.0000
h17.c2	.	28.0000	+INF	189.0000

h18.c1	.	24.0000	+INF	189.0000
h18.c2	.	34.0000	+INF	189.0000
h19.c1	.	31.0000	+INF	189.0000
h19.c2	.	10.0000	+INF	189.0000
h20.c1	.	36.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000
h21.c1	.	24.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	29.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	18.0000	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	4.0000	+INF	30.0000
h1 .c2	.	4.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	5.0000	+INF	60.0000
h7 .c2	.	5.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000

h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	.	+INF	30.0000
h18.c2	.	10.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	1.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 101.25*x(h8,c1) - 101.25*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -675.000000000029, INFES =
 675.000000000029 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0,c1) + x(h0,c2) =L= 100$; (LHS = 8)

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) =L= 100$; (LHS = 0)

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) =L= 100$; (LHS = 13.0000000000266)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420$;
(LHS = 190)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420$; (LHS = 127)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420$; (LHS = 110)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) =L= -120$; (LHS = -120)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) =L= -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) =L= -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= $L \leq \text{maxShort}$

shortage_cap(h0,c1).. $L(h0,c1) =L= 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) =L= 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) =L= 10$; (LHS = 3.99999999997342)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40 ; (LHS = 39)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 30)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 190)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 127)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 181179, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
(.LO, .L, .UP, .M = 0, 0, +INF, 189)

```

-189      obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
     -1      shortage(h0,c1)
     -1      shortage(h1,c1)
     -1      shortage(h2,c1)
     -1      shortage(h4,c1)
     -1      shortage(h5,c1)
     -1      shortage(h6,c1)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

```

x(h0,c2)

```
(.LO, .L, .UP, .M = 0, 8, +INF, 189)
```

```

-189      obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
     -1      shortage(h0,c2)
     -1      shortage(h1,c2)
     -1      shortage(h2,c2)
     -1      shortage(h4,c2)
     -1      shortage(h5,c2)
     -1      shortage(h6,c2)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

```

x(h1,c1)

```
(.LO, .L, .UP, .M = 0, 0, +INF, 189)
```

```

-189      obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)

```

```

1      cap_total(h3)
1      cap_total(h5)
1      cap_total(h6)
1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h3)
1      min_coverage(h5)
1      min_coverage(h6)
1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
1      shortage_cap(h0,c1)
1      shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1     shortage(h0,c2)
1      shortage_cap(h0,c2)
1      shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 3.99999999997342, +INF, 30)
-30   obj
-1     shortage(h1,c1)
1      shortage_cap(h1,c1)
1      shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

RHS [min, max] : [1.000E+01, 4.200E+02] - Zero values observed as well
 Bound [min, max] : [NA, NA] - Zero values observed as well
 Matrix [min, max] : [1.000E+00, 1.890E+02]

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 General Algebraic Modeling System
 Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t6

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.016 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
 L O O P S t t6

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 General Algebraic Modeling System
 Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	157

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 8 Integer Solution
 **** OBJECTIVE VALUE 181854.0000

RESOURCE USAGE, LIMIT 0.110 10000000000.000
 ITERATION COUNT, LIMIT 84 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

```

--- MIP status (102): integer optimal, tolerance.
--- Cplex Time: 0.11sec (det. 7.61 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.00sec (det. 0.14 ticks)

```

Solution satisfies tolerances

```

MIP Solution:      181854.000000      (84 iterations, 0 nodes)
Final Solve:      181854.000000      (0 iterations)

```

```

Best possible:      181845.882353
Absolute gap:       8.117647
Relative gap:       0.000045

```

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	8.0000	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	13.0000	100.0000	.
h3	-INF	18.0000	100.0000	.
h4	-INF	12.0000	100.0000	.
h5	-INF	42.0000	100.0000	.
h6	-INF	41.0000	100.0000	.
h7	-INF	76.0000	100.0000	.
h8	-INF	100.0000	100.0000	.
h9	-INF	62.0000	100.0000	.
h10	-INF	33.0000	100.0000	.
h11	-INF	86.0000	100.0000	.
h12	-INF	54.0000	100.0000	.
h13	-INF	39.0000	100.0000	.
h14	-INF	96.0000	100.0000	.
h15	-INF	68.0000	100.0000	.
h16	-INF	73.0000	100.0000	.
h17	-INF	100.0000	100.0000	.
h18	-INF	58.0000	100.0000	.

h19	-INF	59.0000	100.0000	.
h20	-INF	36.0000	100.0000	.
h21	-INF	24.0000	100.0000	.
h22	-INF	29.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	190.0000	420.0000	.
h1	-INF	127.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	84.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	116.0000	420.0000	.
h7	-INF	190.0000	420.0000	.
h8	-INF	260.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	340.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	417.0000	420.0000	.
h14	-INF	384.0000	420.0000	.
h15	-INF	384.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-120.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-54.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-34.0000	-30.0000	.

h5 .c2	-INF	-46.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-56.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-150.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-210.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-257.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-164.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-154.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-251.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	4.0000	10.0000	.
h1 .c2	-INF	4.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.

h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	5.0000	10.0000	.
h7 .c2	-INF	5.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	.	10.0000	.
h18.c2	-INF	10.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	1.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	39.0000	40.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	190.0000	+INF	.
h1	80.0000	127.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	84.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	116.0000	+INF	.
h7	80.0000	190.0000	+INF	.
h8	80.0000	260.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	340.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	417.0000	+INF	.
h14	80.0000	384.0000	+INF	.
h15	80.0000	384.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	320.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	181854.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	8.0000	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	1.0000	+INF	189.0000
h2 .c2	.	12.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	18.0000	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	12.0000	+INF	171.0000
h5 .c1	.	34.0000	+INF	162.0000
h5 .c2	.	8.0000	+INF	162.0000

h6 .c1	.	25.0000	+INF	153.0000
h6 .c2	.	16.0000	+INF	153.0000
h7 .c1	.	45.0000	+INF	144.0000
h7 .c2	.	31.0000	+INF	144.0000
h8 .c1	.	79.0000	+INF	101.2500
h8 .c2	.	21.0000	+INF	101.2500
h9 .c1	.	42.0000	+INF	126.0000
h9 .c2	.	20.0000	+INF	126.0000
h10.c1	.	.	+INF	126.0000
h10.c2	.	33.0000	+INF	126.0000
h11.c1	.	64.0000	+INF	135.0000
h11.c2	.	22.0000	+INF	135.0000
h12.c1	.	27.0000	+INF	144.0000
h12.c2	.	27.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	39.0000	+INF	153.0000
h14.c1	.	72.0000	+INF	162.0000
h14.c2	.	24.0000	+INF	162.0000
h15.c1	.	52.0000	+INF	171.0000
h15.c2	.	16.0000	+INF	171.0000
h16.c1	.	35.0000	+INF	180.0000
h16.c2	.	38.0000	+INF	180.0000
h17.c1	.	72.0000	+INF	189.0000
h17.c2	.	28.0000	+INF	189.0000
h18.c1	.	24.0000	+INF	189.0000
h18.c2	.	34.0000	+INF	189.0000
h19.c1	.	31.0000	+INF	189.0000
h19.c2	.	28.0000	+INF	189.0000
h20.c1	.	36.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000
h21.c1	.	24.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	29.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	4.0000	+INF	30.0000
h1 .c2	.	4.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000

h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	5.0000	+INF	60.0000
h7 .c2	.	5.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	.	+INF	30.0000
h18.c2	.	10.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	1.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
- 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
144*x(h7,c2) - 108*x(h8,c1) - 108*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -675, INFES = 675 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 8)

cap_per_hour(h1).. x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h2).. x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 12.9999999999759)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) +
x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;
(LHS = 190)

cap_total(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +
x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS
= 127)

cap_total(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) +
x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS
= 110)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. - x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) -
x(h23,c1) - L(h0,c1) =L= -120 ; (LHS = -120)

shortage(h0,c2).. - x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) -
x(h23,c2) - L(h0,c2) =L= -70 ; (LHS = -70)

shortage(h1,c1).. - x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) -
x(h23,c1) - L(h1,c1) =L= -95 ; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. L(h0,c1) =L= 10 ; (LHS = 0)

shortage_cap(h0,c2).. L(h0,c2) =L= 10 ; (LHS = 0)

shortage_cap(h1,c1).. L(h1,c1) =L= 10 ; (LHS = 4.00000000002415)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40 ; (LHS = 39)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 30)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 190)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 127)

```

min_coverage(h2)..  x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110)

```

REMAINING 21 ENTRIES SKIPPED

```

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General Algebraic Modeling System
Column Listing      SOLVE Telecom_MIP Using MIP From line 157

```

---- z total cost (objective)

```

z
      (.LO, .L, .UP, .M = -INF, 181854, +INF, 0)
1      obj

```

---- x operators starting at hour for company

```

x(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189      obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
      -1      shortage(h0,c1)
      -1      shortage(h1,c1)
      -1      shortage(h2,c1)
      -1      shortage(h4,c1)
      -1      shortage(h5,c1)
      -1      shortage(h6,c1)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

```

```

x(h0,c2)
      (.LO, .L, .UP, .M = 0, 8, +INF, 189)
-189      obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)

```

```

1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h1,c1)

```

      (.L0, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h3)
      1      min_coverage(h5)
      1      min_coverage(h6)
      1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

      (.L0, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
      1     shortage_cap(h0,c1)
      1     shortage_total(c1)

```

```

L(h0,c2)
      (.L0, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1    shortage(h0,c2)
1     shortage_cap(h0,c2)
1     shortage_total(c2)

```

```

L(h1,c1)
      (.L0, .L, .UP, .M = 0, 4.00000000002415, +INF, 30)
-30   obj
-1    shortage(h1,c1)
1     shortage_cap(h1,c1)
1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System
Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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General Algebraic Modeling System
Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t7

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.016 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t7

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General Algebraic Modeling System

Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	157

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 8 Integer Solution
 **** OBJECTIVE VALUE 182529.0000

RESOURCE USAGE, LIMIT 0.094 10000000000.000
 ITERATION COUNT, LIMIT 77 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (102): integer optimal, tolerance.
 --- Cplex Time: 0.08sec (det. 7.35 ticks)
 --- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...
 --- Fixed MIP status (1): optimal.
 --- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances
 MIP Solution: 182529.000000 (77 iterations, 0 nodes)
 Final Solve: 182529.000000 (0 iterations)
 Best possible: 182520.882353
 Absolute gap: 8.117647
 Relative gap: 0.000044

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	.	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	46.0000	100.0000	.
h3	-INF	10.0000	100.0000	.
h4	-INF	20.0000	100.0000	.
h5	-INF	50.0000	100.0000	.
h6	-INF	.	100.0000	.
h7	-INF	84.0000	100.0000	.
h8	-INF	100.0000	100.0000	.
h9	-INF	46.0000	100.0000	.
h10	-INF	74.0000	100.0000	.
h11	-INF	86.0000	100.0000	.
h12	-INF	46.0000	100.0000	.
h13	-INF	55.0000	100.0000	.
h14	-INF	63.0000	100.0000	.
h15	-INF	60.0000	100.0000	.
h16	-INF	81.0000	100.0000	.
h17	-INF	92.0000	100.0000	.
h18	-INF	83.0000	100.0000	.
h19	-INF	67.0000	100.0000	.
h20	-INF	36.0000	100.0000	.
h21	-INF	24.0000	100.0000	.
h22	-INF	4.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	190.0000	420.0000	.
h1	-INF	127.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	84.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	116.0000	420.0000	.
h7	-INF	190.0000	420.0000	.
h8	-INF	260.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	340.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	417.0000	420.0000	.
h14	-INF	384.0000	420.0000	.
h15	-INF	384.0000	420.0000	.
h16	-INF	410.0000	420.0000	.

h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-120.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-54.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-34.0000	-30.0000	.
h5 .c2	-INF	-46.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-56.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-150.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-210.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-257.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-164.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-154.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.

h19.c1	-INF	-251.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	4.0000	10.0000	.
h1 .c2	-INF	4.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	5.0000	10.0000	.
h7 .c2	-INF	5.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.

h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	.	10.0000	.
h18.c2	-INF	10.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	1.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	39.0000	40.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	190.0000	+INF	.
h1	80.0000	127.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	84.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	116.0000	+INF	.
h7	80.0000	190.0000	+INF	.
h8	80.0000	260.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	340.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	417.0000	+INF	.
h14	80.0000	384.0000	+INF	.
h15	80.0000	384.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	320.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	182529.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	26.0000	+INF	189.0000
h2 .c2	.	20.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	10.0000	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	20.0000	+INF	171.0000
h5 .c1	.	34.0000	+INF	162.0000
h5 .c2	.	16.0000	+INF	162.0000
h6 .c1	.	.	+INF	153.0000
h6 .c2	.	.	+INF	153.0000
h7 .c1	.	45.0000	+INF	144.0000
h7 .c2	.	39.0000	+INF	144.0000
h8 .c1	.	79.0000	+INF	108.0000
h8 .c2	.	21.0000	+INF	108.0000
h9 .c1	.	42.0000	+INF	126.0000
h9 .c2	.	4.0000	+INF	126.0000
h10.c1	.	25.0000	+INF	126.0000
h10.c2	.	49.0000	+INF	126.0000
h11.c1	.	64.0000	+INF	135.0000
h11.c2	.	22.0000	+INF	135.0000
h12.c1	.	27.0000	+INF	144.0000
h12.c2	.	19.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	55.0000	+INF	153.0000
h14.c1	.	47.0000	+INF	162.0000
h14.c2	.	16.0000	+INF	162.0000
h15.c1	.	52.0000	+INF	171.0000
h15.c2	.	8.0000	+INF	171.0000
h16.c1	.	35.0000	+INF	180.0000
h16.c2	.	46.0000	+INF	180.0000
h17.c1	.	72.0000	+INF	189.0000
h17.c2	.	20.0000	+INF	189.0000
h18.c1	.	49.0000	+INF	189.0000
h18.c2	.	34.0000	+INF	189.0000
h19.c1	.	31.0000	+INF	189.0000

h19.c2	.	36.0000	+INF	189.0000
h20.c1	.	36.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000
h21.c1	.	24.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	4.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	4.0000	+INF	30.0000
h1 .c2	.	4.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	5.0000	+INF	60.0000
h7 .c2	.	5.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000

h18.c1	.	.	+INF	30.0000
h18.c2	.	10.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	1.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

GAMS 52.5.0 9bc771bf Jan 28, 2026

WEX-WEI x86 64bit/MS Windows - 02/15/26

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 114.75*x(h8,c1) - 114.75*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -675, INFES = 675 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
 capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) = 100$; (LHS = 0)

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) = 100$; (LHS = 46)

REMAINING 21 ENTRIES SKIPPED

---- cap_total = total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) = 420$;
(LHS = 190)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) = 420$; (LHS = 127)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) = 420$; (LHS = 110)

REMAINING 21 ENTRIES SKIPPED

---- shortage = demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) = -120$; (LHS = -120)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) = -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) = -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap = L <= maxShort

shortage_cap(h0,c1).. $L(h0,c1) = 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) = 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) = 10$; (LHS = 4.0000000002463)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40 ; (LHS = 39)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 30)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 190)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 127)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z
1 (.L0, .L, .UP, .M = -INF, 182529, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
-189 (.L0, .L, .UP, .M = 0, 0, +INF, 189)
obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)

```

1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c1)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h4,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h0,c2)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)

```



```

-1      shortage(h1,c1)
-1      shortage(h2,c1)
-1      shortage(h3,c1)
-1      shortage(h5,c1)
-1      shortage(h6,c1)
-1      shortage(h7,c1)
 1      min_coverage(h1)
 1      min_coverage(h2)
 1      min_coverage(h3)
 1      min_coverage(h5)
 1      min_coverage(h6)
 1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
 1     shortage_cap(h0,c1)
 1     shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1     shortage(h0,c2)
 1     shortage_cap(h0,c2)
 1     shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 4.00000000002463, +INF, 30)
-30   obj
-1     shortage(h1,c1)
 1     shortage_cap(h1,c1)
 1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [           NA,           NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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 General Algebraic Modeling System
 Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t8

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.031 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
 L O O P S t t8

GAMS 52.5.0 9bc771bf Jan 28, 2026 WEX-WEI x86 64bit/MS Windows - 02/15/26
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 General Algebraic Modeling System
 Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	157

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 1 Optimal
 **** OBJECTIVE VALUE 183192.0000

RESOURCE USAGE, LIMIT 0.063 10000000000.000
 ITERATION COUNT, LIMIT 77 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (101): integer optimal solution.
 --- Cplex Time: 0.06sec (det. 6.65 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).

--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.

--- Cplex Time: 0.00sec (det. 0.14 ticks)

Proven optimal solution

MIP Solution: 183192.000000 (77 iterations, 0 nodes)

Final Solve: 183192.000000 (0 iterations)

Best possible: 183192.000000

Absolute gap: -0.000000

Relative gap: -0.000000

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	10.0000	100.0000	.
h1	-INF	16.0000	100.0000	.
h2	-INF	26.0000	100.0000	.
h3	-INF	20.0000	100.0000	.
h4	-INF	10.0000	100.0000	.
h5	-INF	24.0000	100.0000	.
h6	-INF	24.0000	100.0000	.
h7	-INF	88.0000	100.0000	.
h8	-INF	92.0000	100.0000	.
h9	-INF	76.0000	100.0000	.
h10	-INF	64.0000	100.0000	.
h11	-INF	62.0000	100.0000	.
h12	-INF	60.0000	100.0000	.
h13	-INF	35.0000	100.0000	.
h14	-INF	61.0000	100.0000	.
h15	-INF	84.0000	100.0000	.
h16	-INF	79.0000	100.0000	.
h17	-INF	100.0000	100.0000	.
h18	-INF	85.0000	100.0000	.
h19	-INF	55.0000	100.0000	.
h20	-INF	26.0000	100.0000	.
h21	-INF	16.0000	100.0000	.
h22	-INF	14.0000	100.0000	.

h23	-INF	.	100.0000	.
-----	------	---	----------	---

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	190.0000	420.0000	.
h1	-INF	123.0000	420.0000	.
h2	-INF	108.0000	420.0000	.
h3	-INF	92.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	110.0000	420.0000	.
h7	-INF	198.0000	420.0000	.
h8	-INF	260.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	338.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	413.0000	420.0000	.
h14	-INF	388.0000	420.0000	.
h15	-INF	382.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-120.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-62.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-40.0000	-30.0000	.
h5 .c2	-INF	-40.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-54.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.

h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-150.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-208.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-253.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-168.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-152.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	2.0000	10.0000	.
h1 .c2	-INF	10.0000	10.0000	.
h2 .c1	-INF	2.0000	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.

h6 .c1	-INF	4.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	2.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	2.0000	10.0000	.
h18.c2	-INF	8.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	.	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	40.0000	40.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	190.0000	+INF	.
h1	80.0000	123.0000	+INF	.

h2	80.0000	108.0000	+INF	.
h3	80.0000	92.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	110.0000	+INF	.
h7	80.0000	198.0000	+INF	.
h8	80.0000	260.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	338.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	413.0000	+INF	.
h14	80.0000	388.0000	+INF	.
h15	80.0000	382.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	320.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	183192.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	10.0000	+INF	189.0000
h1 .c1	.	16.0000	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	16.0000	+INF	189.0000
h2 .c2	.	10.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	20.0000	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	10.0000	+INF	171.0000
h5 .c1	.	24.0000	+INF	162.0000
h5 .c2	.	.	+INF	162.0000
h6 .c1	.	.	+INF	153.0000
h6 .c2	.	24.0000	+INF	153.0000
h7 .c1	.	54.0000	+INF	144.0000
h7 .c2	.	34.0000	+INF	144.0000

h8 .c1	.	80.0000	+INF	114.7500
h8 .c2	.	12.0000	+INF	114.7500
h9 .c1	.	42.0000	+INF	126.0000
h9 .c2	.	34.0000	+INF	126.0000
h10.c1	.	34.0000	+INF	126.0000
h10.c2	.	30.0000	+INF	126.0000
h11.c1	.	54.0000	+INF	135.0000
h11.c2	.	8.0000	+INF	135.0000
h12.c1	.	18.0000	+INF	144.0000
h12.c2	.	42.0000	+INF	144.0000
h13.c1	.	5.0000	+INF	153.0000
h13.c2	.	30.0000	+INF	153.0000
h14.c1	.	41.0000	+INF	162.0000
h14.c2	.	20.0000	+INF	162.0000
h15.c1	.	54.0000	+INF	171.0000
h15.c2	.	30.0000	+INF	171.0000
h16.c1	.	49.0000	+INF	180.0000
h16.c2	.	30.0000	+INF	180.0000
h17.c1	.	70.0000	+INF	189.0000
h17.c2	.	30.0000	+INF	189.0000
h18.c1	.	45.0000	+INF	189.0000
h18.c2	.	40.0000	+INF	189.0000
h19.c1	.	35.0000	+INF	189.0000
h19.c2	.	20.0000	+INF	189.0000
h20.c1	.	26.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000
h21.c1	.	16.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	14.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	2.0000	+INF	30.0000
h1 .c2	.	10.0000	+INF	45.0000
h2 .c1	.	2.0000	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	4.0000	+INF	30.0000

h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	2.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	2.0000	+INF	30.0000
h18.c2	.	8.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	.	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -

$162 \times (h5, c1) - 162 \times (h5, c2) - 153 \times (h6, c1) - 153 \times (h6, c2) - 144 \times (h7, c1) -$
 $144 \times (h7, c2) - 121.5 \times (h8, c1) - 121.5 \times (h8, c2) - 126 \times (h9, c1) - 126 \times (h9, c2) -$
 $126 \times (h10, c1) - 126 \times (h10, c2) - 135 \times (h11, c1) - 135 \times (h11, c2) - 144 \times (h12, c1) -$
 $144 \times (h12, c2) - 153 \times (h13, c1) - 153 \times (h13, c2) - 162 \times (h14, c1) - 162 \times (h14, c2) -$
 $171 \times (h15, c1) - 171 \times (h15, c2) - 180 \times (h16, c1) - 180 \times (h16, c2) - 189 \times (h17, c1) -$
 $189 \times (h17, c2) - 189 \times (h18, c1) - 189 \times (h18, c2) - 189 \times (h19, c1) - 189 \times (h19, c2) -$
 $189 \times (h20, c1) - 189 \times (h20, c2) - 189 \times (h21, c1) - 189 \times (h21, c2) - 189 \times (h22, c1) -$
 $189 \times (h22, c2) - 189 \times (h23, c1) - 189 \times (h23, c2) - 30 \times L(h0, c1) - 45 \times L(h0, c2) -$
 $30 \times L(h1, c1) - 45 \times L(h1, c2) - 30 \times L(h2, c1) - 45 \times L(h2, c2) - 30 \times L(h3, c1) - 45 \times L(h3, c2) -$
 $30 \times L(h4, c1) - 45 \times L(h4, c2) - 30 \times L(h5, c1) - 45 \times L(h5, c2) - 30 \times L(h6, c1) - 45 \times L(h6, c2) -$
 $60 \times L(h7, c1) - 45 \times L(h7, c2) - 60 \times L(h8, c1) - 45 \times L(h8, c2) - 60 \times L(h9, c1) - 45 \times L(h9, c2) -$
 $60 \times L(h10, c1) - 45 \times L(h10, c2) - 60 \times L(h11, c1) - 45 \times L(h11, c2) - 60 \times L(h12, c1) -$
 $45 \times L(h12, c2) - 60 \times L(h13, c1) - 45 \times L(h13, c2) - 60 \times L(h14, c1) - 45 \times L(h14, c2) -$
 $60 \times L(h15, c1) - 45 \times L(h15, c2) - 60 \times L(h16, c1) - 45 \times L(h16, c2) - 60 \times L(h17, c1) -$
 $45 \times L(h17, c2) - 30 \times L(h18, c1) - 45 \times L(h18, c2) - 30 \times L(h19, c1) - 45 \times L(h19, c2) -$
 $30 \times L(h20, c1) - 45 \times L(h20, c2) - 30 \times L(h21, c1) - 45 \times L(h21, c2) - 30 \times L(h22, c1) -$
 $45 \times L(h22, c2) - 30 \times L(h23, c1) - 45 \times L(h23, c2) = E = 0 ; (LHS = -620.9999999999942, INFES =$
 $620.9999999999942 ****)$

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0, c1) + x(h0, c2) =L= 100 ; (LHS = 10)$

cap_per_hour(h1).. $x(h1, c1) + x(h1, c2) =L= 100 ; (LHS = 16.00000000000838)$

cap_per_hour(h2).. $x(h2, c1) + x(h2, c2) =L= 100 ; (LHS = 25.99999999999835)$

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0, c1) + x(h0, c2) + x(h18, c1) + x(h18, c2) + x(h19, c1) + x(h19, c2) +$
 $x(h20, c1) + x(h20, c2) + x(h22, c1) + x(h22, c2) + x(h23, c1) + x(h23, c2) =L= 420 ;$
 (LHS = 190)

cap_total(h1).. $x(h0, c1) + x(h0, c2) + x(h1, c1) + x(h1, c2) + x(h19, c1) + x(h19, c2) +$
 $x(h20, c1) + x(h20, c2) + x(h21, c1) + x(h21, c2) + x(h23, c1) + x(h23, c2) =L= 420 ; (LHS$
 $= 123.0000000000033)$

cap_total(h2).. $x(h0, c1) + x(h0, c2) + x(h1, c1) + x(h1, c2) + x(h2, c1) + x(h2, c2) +$
 $x(h20, c1) + x(h20, c2) + x(h21, c1) + x(h21, c2) + x(h22, c1) + x(h22, c2) =L= 420 ; (LHS$
 $= 107.9999999999996)$

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) = -120$; (LHS = -120)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) = -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) = -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap $= L \leq \text{maxShort}$

shortage_cap(h0,c1).. $L(h0,c1) = 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) = 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) = 10$; (LHS = 1.99999999996711)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total $= L = \sum_h L(h,j) \leq \text{totShort}(j)$

shortage_total(c1).. $L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) + L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) + L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) + L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) = 40$; (LHS = 40)

shortage_total(c2).. $L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) + L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) + L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) + L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) = 30$; (LHS = 29.9999999999688)

---- min_coverage $= G = \text{minimum coverage in each window across companies} \geq \text{minCov}$

min_coverage(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) = 80$; (LHS = 190)

min_coverage(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) = 80$; (LHS = 123.000000000033)

min_coverage(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) = 80$;

(LHS = 107.9999999999996)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z

1 (.LO, .L, .UP, .M = -INF, 183191.999999999, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)

(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

x(h0,c2)

(.LO, .L, .UP, .M = 0, 10, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)

```

1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 16.0000000000838, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h3)
      1      min_coverage(h5)
      1      min_coverage(h6)
      1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30  obj
-1     shortage(h0,c1)
      1      shortage_cap(h0,c1)
      1      shortage_total(c1)

```

L(h0,c2)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1    shortage(h0,c2)
1     shortage_cap(h0,c2)
1     shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 1.99999999996711, +INF, 30)
-30   obj
-1    shortage(h1,c1)
1     shortage_cap(h1,c1)
1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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 General Algebraic Modeling System
 Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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 General Algebraic Modeling System
 Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t9

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.000 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
 L O O P S t t9

GAMS 52.5.0 9bc771bf Jan 28, 2026 WEX-WEI x86 64bit/MS Windows - 02/15/26
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 General Algebraic Modeling System
 Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL Telecom_MIP OBJECTIVE z
 TYPE MIP DIRECTION MINIMIZE
 SOLVER CPLEX FROM LINE 157

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 1 Optimal
 **** OBJECTIVE VALUE 183804.0000

RESOURCE USAGE, LIMIT 0.125 10000000000.000
 ITERATION COUNT, LIMIT 72 2147483647

*** This solver runs with a demo license. No commercial use.

--- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (101): integer optimal solution.
 --- Cplex Time: 0.12sec (det. 12.65 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
 --- Cplex Time: 0.00sec (det. 0.14 ticks)

Proven optimal solution

MIP Solution: 183804.000000 (72 iterations, 0 nodes)
 Final Solve: 183804.000000 (0 iterations)

Best possible: 183804.000000
 Absolute gap: 0.000000
 Relative gap: 0.000000

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	10.0000	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	10.0000	100.0000	.
h3	-INF	20.0000	100.0000	.
h4	-INF	10.0000	100.0000	.
h5	-INF	40.0000	100.0000	.
h6	-INF	42.0000	100.0000	.
h7	-INF	88.0000	100.0000	.
h8	-INF	90.0000	100.0000	.
h9	-INF	62.0000	100.0000	.
h10	-INF	46.0000	100.0000	.
h11	-INF	58.0000	100.0000	.
h12	-INF	66.0000	100.0000	.
h13	-INF	49.0000	100.0000	.
h14	-INF	75.0000	100.0000	.
h15	-INF	88.0000	100.0000	.
h16	-INF	77.0000	100.0000	.
h17	-INF	82.0000	100.0000	.
h18	-INF	71.0000	100.0000	.
h19	-INF	53.0000	100.0000	.
h20	-INF	26.0000	100.0000	.
h21	-INF	34.0000	100.0000	.
h22	-INF	30.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	190.0000	420.0000	.
h1	-INF	123.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	94.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	112.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	260.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	336.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	413.0000	420.0000	.
h14	-INF	388.0000	420.0000	.
h15	-INF	378.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.

h19	-INF	418.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-120.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-64.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-40.0000	-30.0000	.
h5 .c2	-INF	-40.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-56.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-150.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-206.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-253.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-168.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-148.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.

h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	2.0000	10.0000	.
h1 .c2	-INF	10.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	4.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	1.9982522E-9	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	4.0000	10.0000	.

h18.c2	-INF	6.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	2.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	40.0000	40.0000	.
c2	-INF	28.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	190.0000	+INF	.
h1	80.0000	123.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	94.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	112.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	260.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	336.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	413.0000	+INF	.
h14	80.0000	388.0000	+INF	.
h15	80.0000	378.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	418.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	320.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
--	-------	-------	-------	----------

```

---- VAR z          -INF          183804.0000          +INF          .

```

h20.c2	.	.	+INF	189.0000
h21.c1	.	34.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	30.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	2.0000	+INF	30.0000
h1 .c2	.	10.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	4.0000	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	1.9982522E-9	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	4.0000	+INF	30.0000
h18.c2	.	6.0000	+INF	45.0000

h19.c1	.	.	+INF	30.0000
h19.c2	.	2.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System
 Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 128.25*x(h8,c1) - 128.25*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -607.499999979773, INFES =
 607.499999979773 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
 capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 10)

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) = 100$; (LHS = 0)

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) = 100$; (LHS = 10)

REMAINING 21 ENTRIES SKIPPED

---- cap_total = total operators active in window across companies $\leq$ capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) = 420$;
(LHS = 190)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) = 420$; (LHS = 122.999999997002)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) = 420$; (LHS = 110)

REMAINING 21 ENTRIES SKIPPED

---- shortage = demand - covered $\leq$ L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) = -120$; (LHS = -120)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) = -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) = -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap = $L \leq$ maxShort

shortage_cap(h0,c1).. $L(h0,c1) = 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) = 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) = 10$; (LHS = 2.00000000299757)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total = $\sum_h L(h,j) \leq$ totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40 ; (LHS = 40)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS =
28.0000000004999)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 190)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 122.999999997002)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System
Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z
1 (.L0, .L, .UP, .M = -INF, 183803.9999999991, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
-189 (.L0, .L, .UP, .M = 0, 0, +INF, 189)
obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)


```

1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c1)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h4,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h0,c2)

```

      (.LO, .L, .UP, .M = 0, 10, +INF, 189)
-189  obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)

```

```

-1      shortage(h1,c1)
-1      shortage(h2,c1)
-1      shortage(h3,c1)
-1      shortage(h5,c1)
-1      shortage(h6,c1)
-1      shortage(h7,c1)
 1      min_coverage(h1)
 1      min_coverage(h2)
 1      min_coverage(h3)
 1      min_coverage(h5)
 1      min_coverage(h6)
 1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
 1     shortage_cap(h0,c1)
 1     shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1     shortage(h0,c2)
 1     shortage_cap(h0,c2)
 1     shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 2.00000000299757, +INF, 30)
-30   obj
-1     shortage(h1,c1)
 1     shortage_cap(h1,c1)
 1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [           NA,           NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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 General Algebraic Modeling System
 Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t10

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.000 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
 L O O P S t t10

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 General Algebraic Modeling System
 Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	157

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 8 Integer Solution
 **** OBJECTIVE VALUE 184367.2500

RESOURCE USAGE, LIMIT 1.891 10000000000.000
 ITERATION COUNT, LIMIT 26194 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (102): integer optimal, tolerance.
 --- Cplex Time: 1.89sec (det. 356.05 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).

--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.

--- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances

MIP Solution: 184367.250000 (26194 iterations, 12563 nodes)

Final Solve: 184367.250000 (0 iterations)

Best possible: 184348.833333

Absolute gap: 18.416667

Relative gap: 0.000100

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	13.0000	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	16.0000	100.0000	.
h3	-INF	14.0000	100.0000	.
h4	-INF	16.0000	100.0000	.
h5	-INF	37.0000	100.0000	.
h6	-INF	31.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	73.0000	100.0000	.
h9	-INF	70.0000	100.0000	.
h10	-INF	63.0000	100.0000	.
h11	-INF	45.0000	100.0000	.
h12	-INF	78.0000	100.0000	.
h13	-INF	45.0000	100.0000	.
h14	-INF	59.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	70.0000	100.0000	.
h17	-INF	82.0000	100.0000	.
h18	-INF	86.0000	100.0000	.
h19	-INF	48.0000	100.0000	.
h20	-INF	27.0000	100.0000	.
h21	-INF	33.0000	100.0000	.
h22	-INF	21.0000	100.0000	.

h23	-INF	.	100.0000	.
-----	------	---	----------	---

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	195.0000	420.0000	.
h1	-INF	121.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	84.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	113.0000	420.0000	.
h7	-INF	198.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	346.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	411.0000	420.0000	.
h14	-INF	388.0000	420.0000	.
h15	-INF	382.0000	420.0000	.
h16	-INF	415.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	319.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-125.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-54.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-40.0000	-30.0000	.
h5 .c2	-INF	-40.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-54.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.

h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-216.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-251.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-168.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-152.0000	-130.0000	.
h16.c1	-INF	-255.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	4.0000	10.0000	.
h1 .c2	-INF	10.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.

h6 .c1	-INF	1.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	2.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	2.0000	10.0000	.
h18.c2	-INF	8.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	.	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	1.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	38.0000	40.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	195.0000	+INF	.
h1	80.0000	121.0000	+INF	.

h2	80.0000	110.0000	+INF	.
h3	80.0000	84.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	113.0000	+INF	.
h7	80.0000	198.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	346.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	411.0000	+INF	.
h14	80.0000	388.0000	+INF	.
h15	80.0000	382.0000	+INF	.
h16	80.0000	415.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	319.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	184367.2500	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	9.0000	+INF	189.0000
h0 .c2	.	4.0000	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	.	+INF	189.0000
h2 .c2	.	16.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	14.0000	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	16.0000	+INF	171.0000
h5 .c1	.	31.0000	+INF	162.0000
h5 .c2	.	6.0000	+INF	162.0000
h6 .c1	.	19.0000	+INF	153.0000
h6 .c2	.	12.0000	+INF	153.0000
h7 .c1	.	60.0000	+INF	144.0000
h7 .c2	.	40.0000	+INF	144.0000

h8 .c1	.	61.0000	+INF	128.2500
h8 .c2	.	12.0000	+INF	128.2500
h9 .c1	.	48.0000	+INF	126.0000
h9 .c2	.	22.0000	+INF	126.0000
h10.c1	.	21.0000	+INF	126.0000
h10.c2	.	42.0000	+INF	126.0000
h11.c1	.	37.0000	+INF	135.0000
h11.c2	.	8.0000	+INF	135.0000
h12.c1	.	42.0000	+INF	144.0000
h12.c2	.	36.0000	+INF	144.0000
h13.c1	.	3.0000	+INF	153.0000
h13.c2	.	42.0000	+INF	153.0000
h14.c1	.	45.0000	+INF	162.0000
h14.c2	.	14.0000	+INF	162.0000
h15.c1	.	76.0000	+INF	171.0000
h15.c2	.	24.0000	+INF	171.0000
h16.c1	.	34.0000	+INF	180.0000
h16.c2	.	36.0000	+INF	180.0000
h17.c1	.	58.0000	+INF	189.0000
h17.c2	.	24.0000	+INF	189.0000
h18.c1	.	46.0000	+INF	189.0000
h18.c2	.	40.0000	+INF	189.0000
h19.c1	.	22.0000	+INF	189.0000
h19.c2	.	26.0000	+INF	189.0000
h20.c1	.	27.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000
h21.c1	.	33.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	21.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	4.0000	+INF	30.0000
h1 .c2	.	10.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	1.0000	+INF	30.0000

h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	2.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	2.0000	+INF	30.0000
h18.c2	.	8.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	.	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	1.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -

$162 \times (h5, c1) - 162 \times (h5, c2) - 153 \times (h6, c1) - 153 \times (h6, c2) - 144 \times (h7, c1) -$
 $144 \times (h7, c2) - 135 \times (h8, c1) - 135 \times (h8, c2) - 126 \times (h9, c1) - 126 \times (h9, c2) -$
 $126 \times (h10, c1) - 126 \times (h10, c2) - 135 \times (h11, c1) - 135 \times (h11, c2) - 144 \times (h12, c1) -$
 $144 \times (h12, c2) - 153 \times (h13, c1) - 153 \times (h13, c2) - 162 \times (h14, c1) - 162 \times (h14, c2) -$
 $171 \times (h15, c1) - 171 \times (h15, c2) - 180 \times (h16, c1) - 180 \times (h16, c2) - 189 \times (h17, c1) -$
 $189 \times (h17, c2) - 189 \times (h18, c1) - 189 \times (h18, c2) - 189 \times (h19, c1) - 189 \times (h19, c2) -$
 $189 \times (h20, c1) - 189 \times (h20, c2) - 189 \times (h21, c1) - 189 \times (h21, c2) - 189 \times (h22, c1) -$
 $189 \times (h22, c2) - 189 \times (h23, c1) - 189 \times (h23, c2) - 30 \times L(h0, c1) - 45 \times L(h0, c2) -$
 $30 \times L(h1, c1) - 45 \times L(h1, c2) - 30 \times L(h2, c1) - 45 \times L(h2, c2) - 30 \times L(h3, c1) - 45 \times L(h3, c2) -$
 $30 \times L(h4, c1) - 45 \times L(h4, c2) - 30 \times L(h5, c1) - 45 \times L(h5, c2) - 30 \times L(h6, c1) - 45 \times L(h6, c2) -$
 $60 \times L(h7, c1) - 45 \times L(h7, c2) - 60 \times L(h8, c1) - 45 \times L(h8, c2) - 60 \times L(h9, c1) - 45 \times L(h9, c2) -$
 $60 \times L(h10, c1) - 45 \times L(h10, c2) - 60 \times L(h11, c1) - 45 \times L(h11, c2) - 60 \times L(h12, c1) -$
 $45 \times L(h12, c2) - 60 \times L(h13, c1) - 45 \times L(h13, c2) - 60 \times L(h14, c1) - 45 \times L(h14, c2) -$
 $60 \times L(h15, c1) - 45 \times L(h15, c2) - 60 \times L(h16, c1) - 45 \times L(h16, c2) - 60 \times L(h17, c1) -$
 $45 \times L(h17, c2) - 30 \times L(h18, c1) - 45 \times L(h18, c2) - 30 \times L(h19, c1) - 45 \times L(h19, c2) -$
 $30 \times L(h20, c1) - 45 \times L(h20, c2) - 30 \times L(h21, c1) - 45 \times L(h21, c2) - 30 \times L(h22, c1) -$
 $45 \times L(h22, c2) - 30 \times L(h23, c1) - 45 \times L(h23, c2) = E = 0 ; (LHS = -492.75, INFES = 492.75$
 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0, c1) + x(h0, c2) =L= 100 ; (LHS = 13)$

cap_per_hour(h1).. $x(h1, c1) + x(h1, c2) =L= 100 ; (LHS = 0)$

cap_per_hour(h2).. $x(h2, c1) + x(h2, c2) =L= 100 ; (LHS = 16)$

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0, c1) + x(h0, c2) + x(h18, c1) + x(h18, c2) + x(h19, c1) + x(h19, c2) +$
 $x(h20, c1) + x(h20, c2) + x(h22, c1) + x(h22, c2) + x(h23, c1) + x(h23, c2) =L= 420 ;$
 (LHS = 195)

cap_total(h1).. $x(h0, c1) + x(h0, c2) + x(h1, c1) + x(h1, c2) + x(h19, c1) + x(h19, c2) +$
 $x(h20, c1) + x(h20, c2) + x(h21, c1) + x(h21, c2) + x(h23, c1) + x(h23, c2) =L= 420 ; (LHS$
 = 121)

cap_total(h2).. $x(h0, c1) + x(h0, c2) + x(h1, c1) + x(h1, c2) + x(h2, c1) + x(h2, c2) +$
 $x(h20, c1) + x(h20, c2) + x(h21, c1) + x(h21, c2) + x(h22, c1) + x(h22, c2) =L= 420 ; (LHS$
 = 110)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. - x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) -
x(h23,c1) - L(h0,c1) =L= -120 ; (LHS = -125)

shortage(h0,c2).. - x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) -
x(h23,c2) - L(h0,c2) =L= -70 ; (LHS = -70)

shortage(h1,c1).. - x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) -
x(h23,c1) - L(h1,c1) =L= -95 ; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. L(h0,c1) =L= 10 ; (LHS = 0)

shortage_cap(h0,c2).. L(h0,c2) =L= 10 ; (LHS = 0)

shortage_cap(h1,c1).. L(h1,c1) =L= 10 ; (LHS = 4)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40 ; (LHS = 38)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 30)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 195)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 121)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110)

REMAINING 21 ENTRIES SKIPPED

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Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z

1 (.LO, .L, .UP, .M = -INF, 184367.25, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)

(.LO, .L, .UP, .M = 0, 9, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

x(h0,c2)

(.LO, .L, .UP, .M = 0, 4.000000000000001, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)

```

1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h3)
      1      min_coverage(h5)
      1      min_coverage(h6)
      1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
      1      shortage_cap(h0,c1)
      1      shortage_total(c1)

```

L(h0,c2)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 45)

```

```

-45      obj
-1       shortage(h0,c2)
1        shortage_cap(h0,c2)
1        shortage_total(c2)

```

```

L(h1,c1)
        (.LO, .L, .UP, .M = 0, 4, +INF, 30)
-30     obj
-1      shortage(h1,c1)
1       shortage_cap(h1,c1)
1       shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System
Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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Model Statistics SOLVE Telecom_MIP Using MIP From line 157

```

LOOPS                                t    t11

```

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

```

GENERATION TIME      =      0.016 SECONDS      4 MB  52.5.0 9bc771bf WEX-WEI
                L O O P S                      t t11

```

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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL Telecom_MIP OBJECTIVE z
 TYPE MIP DIRECTION MINIMIZE
 SOLVER CPLEX FROM LINE 157

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 1 Optimal
 **** OBJECTIVE VALUE 184848.0000

RESOURCE USAGE, LIMIT 0.140 10000000000.000
 ITERATION COUNT, LIMIT 82 2147483647

*** This solver runs with a demo license. No commercial use.

--- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (101): integer optimal solution.
 --- Cplex Time: 0.14sec (det. 12.53 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
 --- Cplex Time: 0.00sec (det. 0.14 ticks)

Proven optimal solution

MIP Solution: 184848.000000 (82 iterations, 0 nodes)
 Final Solve: 184848.000000 (0 iterations)

Best possible: 184848.000000
 Absolute gap: 0.000000
 Relative gap: 0.000000

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

LOWER	LEVEL	UPPER	MARGINAL
-------	-------	-------	----------

h0	-INF	.	100.0000	.
h1	-INF	5.0000	100.0000	.
h2	-INF	28.0000	100.0000	.
h3	-INF	13.0000	100.0000	.
h4	-INF	32.0000	100.0000	.
h5	-INF	30.0000	100.0000	.
h6	-INF	24.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	53.0000	100.0000	.
h9	-INF	82.0000	100.0000	.
h10	-INF	70.0000	100.0000	.
h11	-INF	43.0000	100.0000	.
h12	-INF	100.0000	100.0000	.
h13	-INF	37.0000	100.0000	.
h14	-INF	43.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	55.0000	100.0000	.
h17	-INF	85.0000	100.0000	.
h18	-INF	100.0000	100.0000	.
h19	-INF	50.0000	100.0000	.
h20	-INF	42.0000	100.0000	.
h21	-INF	28.0000	100.0000	.
h22	-INF	7.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	199.0000	420.0000	.
h1	-INF	125.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	81.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	119.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	291.0000	420.0000	.
h11	-INF	349.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	415.0000	420.0000	.
h14	-INF	385.0000	420.0000	.
h15	-INF	375.0000	420.0000	.
h16	-INF	411.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	415.0000	420.0000	.

h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	317.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-129.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-51.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-37.0000	-30.0000	.
h5 .c2	-INF	-43.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-59.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-181.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-219.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-255.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-165.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-145.0000	-130.0000	.
h16.c1	-INF	-251.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.

h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	3.0000	10.0000	.
h1 .c2	-INF	7.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	-1.522184E-8	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	4.0000	10.0000	.
h18.c2	-INF	6.0000	10.0000	.

h19.c1	-INF	.	10.0000	.
h19.c2	-INF	5.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	3.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	40.0000	40.0000	.
c2	-INF	28.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	199.0000	+INF	.
h1	80.0000	125.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	81.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	119.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	291.0000	+INF	.
h11	80.0000	349.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	415.0000	+INF	.
h14	80.0000	385.0000	+INF	.
h15	80.0000	375.0000	+INF	.
h16	80.0000	411.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	415.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	317.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	184848.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	5.0000	+INF	189.0000
h2 .c1	.	23.0000	+INF	189.0000
h2 .c2	.	5.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	13.0000	+INF	180.0000
h4 .c1	.	7.0000	+INF	171.0000
h4 .c2	.	25.0000	+INF	171.0000
h5 .c1	.	30.0000	+INF	162.0000
h5 .c2	.	-2.410219E-8	+INF	162.0000
h6 .c1	.	.	+INF	153.0000
h6 .c2	.	24.0000	+INF	153.0000
h7 .c1	.	57.0000	+INF	144.0000
h7 .c2	.	43.0000	+INF	144.0000
h8 .c1	.	53.0000	+INF	135.0000
h8 .c2	.	-3.932403E-8	+INF	135.0000
h9 .c1	.	53.0000	+INF	126.0000
h9 .c2	.	29.0000	+INF	126.0000
h10.c1	.	38.0000	+INF	126.0000
h10.c2	.	32.0000	+INF	126.0000
h11.c1	.	41.0000	+INF	135.0000
h11.c2	.	2.0000	+INF	135.0000
h12.c1	.	51.0000	+INF	144.0000
h12.c2	.	49.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	37.0000	+INF	153.0000
h14.c1	.	25.0000	+INF	162.0000
h14.c2	.	18.0000	+INF	162.0000
h15.c1	.	73.0000	+INF	171.0000
h15.c2	.	27.0000	+INF	171.0000
h16.c1	.	23.0000	+INF	180.0000
h16.c2	.	32.0000	+INF	180.0000
h17.c1	.	62.0000	+INF	189.0000
h17.c2	.	23.0000	+INF	189.0000
h18.c1	.	65.0000	+INF	189.0000
h18.c2	.	35.0000	+INF	189.0000
h19.c1	.	25.0000	+INF	189.0000
h19.c2	.	25.0000	+INF	189.0000
h20.c1	.	39.0000	+INF	189.0000
h20.c2	.	3.0000	+INF	189.0000

h21.c1	.	28.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	-3.180922E-9	+INF	189.0000
h22.c2	.	7.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	3.0000	+INF	30.0000
h1 .c2	.	7.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	-1.522184E-8	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	4.0000	+INF	30.0000
h18.c2	.	6.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000

h19.c2	.	5.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	3.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 141.75*x(h8,c1) - 141.75*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -357.750010662072, INFES =
 357.750010662072 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
 capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h1).. x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 5.0000000374265)

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) = 100$; (LHS = 27.9999999543182)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) = 420$;
(LHS = 199.000000069166)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) = 420$; (LHS = 124.999999979706)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) = 420$; (LHS = 110.000000003181)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) = -120$; (LHS = -129.000000069166)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) = -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) = -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. $L(h0,c1) = 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) = 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) = 10$; (LHS = 3.00000002219792)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40 ; (LHS = 40)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS =
27.9999999999839)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 199.000000069166)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 124.999999979706)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110.000000003181)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System
Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 184848.000000154, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
-189 (.LO, .L, .UP, .M = 0, 0, +INF, 189)
obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)

```

1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c1)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h4,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h0,c2)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)
-1     shortage(h1,c1)

```

```

-1      shortage(h2,c1)
-1      shortage(h3,c1)
-1      shortage(h5,c1)
-1      shortage(h6,c1)
-1      shortage(h7,c1)
 1      min_coverage(h1)
 1      min_coverage(h2)
 1      min_coverage(h3)
 1      min_coverage(h5)
 1      min_coverage(h6)
 1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.L0, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
 1     shortage_cap(h0,c1)
 1     shortage_total(c1)

```

```

L(h0,c2)
      (.L0, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1     shortage(h0,c2)
 1     shortage_cap(h0,c2)
 1     shortage_total(c2)

```

```

L(h1,c1)
      (.L0, .L, .UP, .M = 0, 3.00000002219792, +INF, 30)
-30   obj
-1     shortage(h1,c1)
 1     shortage_cap(h1,c1)
 1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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 General Algebraic Modeling System
 Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t12

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97 4 projected
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.016 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
 L O O P S t t12

GAMS 52.5.0 9bc771bf Jan 28, 2026 WEX-WEI x86 64bit/MS Windows - 02/15/26
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 General Algebraic Modeling System
 Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	157

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 1 Optimal
 **** OBJECTIVE VALUE 185205.0000

RESOURCE USAGE, LIMIT 0.141 10000000000.000
 ITERATION COUNT, LIMIT 76 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (101): integer optimal solution.
 --- Cplex Time: 0.14sec (det. 11.59 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...

```

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.00sec (det. 0.14 ticks)

```

Proven optimal solution

```

MIP Solution:      185205.000000      (76 iterations, 0 nodes)
Final Solve:       185205.000000      (0 iterations)

```

```

Best possible:      185205.000000
Absolute gap:       -0.000000
Relative gap:       -0.000000

```

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

```

---- EQU cap_per_hour  sum of operators starting at hour across companies <= capHour

```

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	1.0000	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	16.0000	100.0000	.
h3	-INF	14.0000	100.0000	.
h4	-INF	32.0000	100.0000	.
h5	-INF	33.0000	100.0000	.
h6	-INF	36.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	52.0000	100.0000	.
h9	-INF	79.0000	100.0000	.
h10	-INF	58.0000	100.0000	.
h11	-INF	44.0000	100.0000	.
h12	-INF	100.0000	100.0000	.
h13	-INF	41.0000	100.0000	.
h14	-INF	54.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	54.0000	100.0000	.
h17	-INF	81.0000	100.0000	.
h18	-INF	90.0000	100.0000	.
h19	-INF	49.0000	100.0000	.
h20	-INF	43.0000	100.0000	.
h21	-INF	33.0000	100.0000	.
h22	-INF	17.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	200.0000	420.0000	.
h1	-INF	126.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	80.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	118.0000	420.0000	.
h7	-INF	199.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	350.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	416.0000	420.0000	.
h14	-INF	384.0000	420.0000	.
h15	-INF	376.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	415.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	318.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-130.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-50.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-36.0000	-30.0000	.
h5 .c2	-INF	-44.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-59.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.

h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-220.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-256.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-164.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-146.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	3.0000	10.0000	.
h1 .c2	-INF	6.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	1.0000	10.0000	.

h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	1.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	3.0000	10.0000	.
h18.c2	-INF	7.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	5.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	2.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	39.0000	40.0000	.
c2	-INF	29.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	200.0000	+INF	.
h1	80.0000	126.0000	+INF	.
h2	80.0000	110.0000	+INF	.

h3	80.0000	80.0000	+INF	.	
h4	80.0000	80.0000	+INF	.	
h5	80.0000	80.0000	+INF	.	
h6	80.0000	118.0000	+INF	.	
h7	80.0000	199.0000	+INF	.	
h8	80.0000	250.0000	+INF	.	
h9	80.0000	310.0000	+INF	.	
h10	80.0000	290.0000	+INF	.	
h11	80.0000	350.0000	+INF	.	
h12	80.0000	390.0000	+INF	.	
h13	80.0000	416.0000	+INF	.	
h14	80.0000	384.0000	+INF	.	
h15	80.0000	376.0000	+INF	.	
h16	80.0000	410.0000	+INF	.	
h17	80.0000	420.0000	+INF	.	
h18	80.0000	420.0000	+INF	.	
h19	80.0000	415.0000	+INF	.	
h20	80.0000	390.0000	+INF	.	
h21	80.0000	360.0000	+INF	.	
h22	80.0000	318.0000	+INF	.	
h23	80.0000	270.0000	+INF	.	
		LOWER	LEVEL	UPPER	MARGINAL
---- VAR z		-INF	185205.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

		LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.		1.0000	+INF	189.0000
h0 .c2	.		.	+INF	189.0000
h1 .c1	.		.	+INF	189.0000
h1 .c2	.		.	+INF	189.0000
h2 .c1	.		.	+INF	189.0000
h2 .c2	.		16.0000	+INF	189.0000
h3 .c1	.		.	+INF	180.0000
h3 .c2	.		14.0000	+INF	180.0000
h4 .c1	.		12.0000	+INF	171.0000
h4 .c2	.		20.0000	+INF	171.0000
h5 .c1	.		23.0000	+INF	162.0000
h5 .c2	.		10.0000	+INF	162.0000
h6 .c1	.		23.0000	+INF	153.0000
h6 .c2	.		13.0000	+INF	153.0000
h7 .c1	.		64.0000	+INF	144.0000
h7 .c2	.		36.0000	+INF	144.0000
h8 .c1	.		41.0000	+INF	141.7500

h8 .c2	.	11.0000	+INF	141.7500
h9 .c1	.	60.0000	+INF	126.0000
h9 .c2	.	19.0000	+INF	126.0000
h10.c1	.	21.0000	+INF	126.0000
h10.c2	.	37.0000	+INF	126.0000
h11.c1	.	29.0000	+INF	135.0000
h11.c2	.	15.0000	+INF	135.0000
h12.c1	.	62.0000	+INF	144.0000
h12.c2	.	38.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	41.0000	+INF	153.0000
h14.c1	.	36.0000	+INF	162.0000
h14.c2	.	18.0000	+INF	162.0000
h15.c1	.	84.0000	+INF	171.0000
h15.c2	.	16.0000	+INF	171.0000
h16.c1	.	18.0000	+INF	180.0000
h16.c2	.	36.0000	+INF	180.0000
h17.c1	.	57.0000	+INF	189.0000
h17.c2	.	24.0000	+INF	189.0000
h18.c1	.	54.0000	+INF	189.0000
h18.c2	.	36.0000	+INF	189.0000
h19.c1	.	19.0000	+INF	189.0000
h19.c2	.	30.0000	+INF	189.0000
h20.c1	.	39.0000	+INF	189.0000
h20.c2	.	4.0000	+INF	189.0000
h21.c1	.	33.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	17.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	3.0000	+INF	30.0000
h1 .c2	.	6.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	1.0000	+INF	30.0000
h6 .c2	.	.	+INF	45.0000

h7 .c1	.	.	+INF	60.0000
h7 .c2	.	1.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	3.0000	+INF	30.0000
h18.c2	.	7.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	5.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	2.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -

$144*x(h7,c2) - 148.5*x(h8,c1) - 148.5*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -$
 $126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -$
 $144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -$
 $171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -$
 $189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -$
 $189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -$
 $189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -$
 $30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -$
 $30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -$
 $60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -$
 $60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -$
 $45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -$
 $60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -$
 $45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -$
 $30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -$
 $45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -351.000000001572, INFES =$
 $351.000000001572 ****)$

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 1.00000000004437)$

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 0)$

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 16.0000000000965)$

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;$
 $(LHS = 199.999999999853)$

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS$
 $= 126.000000000007)$

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS$
 $= 109.999999999959)$

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. - x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) -
x(h23,c1) - L(h0,c1) =L= -120 ; (LHS = -129.999999999853)

shortage(h0,c2).. - x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) -
x(h23,c2) - L(h0,c2) =L= -70 ; (LHS = -70)

shortage(h1,c1).. - x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) -
x(h23,c1) - L(h1,c1) =L= -95 ; (LHS = -94.9999999999589, INFES =
4.11404244005098E-11 ****)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. L(h0,c1) =L= 10 ; (LHS = 0)

shortage_cap(h0,c2).. L(h0,c2) =L= 10 ; (LHS = 0)

shortage_cap(h1,c1).. L(h1,c1) =L= 10 ; (LHS = 2.99999999975897)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40 ; (LHS =
39.0000000002307)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS =
28.9999999996677)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 199.999999999853)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 126.000000000007)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)

+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 109.999999999959)

REMAINING 21 ENTRIES SKIPPED

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Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 185204.9999999994, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
189 (.LO, .L, .UP, .M = 0, 1.00000000004436, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

x(h0,c2)
189 (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)

```

1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h1,c1)

```

      (.L0, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h3)
      1      min_coverage(h5)
      1      min_coverage(h6)
      1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

      (.L0, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
      1     shortage_cap(h0,c1)
      1     shortage_total(c1)

```

L(h0,c2)

	(.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45	obj
-1	shortage(h0,c2)
1	shortage_cap(h0,c2)
1	shortage_total(c2)

L(h1,c1)

	(.LO, .L, .UP, .M = 0, 2.99999999975897, +INF, 30)
-30	obj
-1	shortage(h1,c1)
1	shortage_cap(h1,c1)
1	shortage_total(c1)

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

RHS	[min, max] :	[1.000E+01, 4.200E+02]	- Zero values observed as well
Bound	[min, max] :	NA, NA	- Zero values observed as well
Matrix	[min, max] :	[1.000E+00, 1.890E+02]	

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General Algebraic Modeling System

Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t13

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME	=	0.000 SECONDS	4 MB	52.5.0 9bc771bf WEX-WEI
LOOPS		t t13		

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General Algebraic Modeling System

Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	157

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 8 Integer Solution
 **** OBJECTIVE VALUE 185556.0000

RESOURCE USAGE, LIMIT 0.687 10000000000.000
 ITERATION COUNT, LIMIT 16440 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (102): integer optimal, tolerance.
 --- Cplex Time: 0.69sec (det. 197.60 ticks)
 --- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...
 --- Fixed MIP status (1): optimal.
 --- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances
 MIP Solution: 185556.000000 (16440 iterations, 6688 nodes)
 Final Solve: 185556.000000 (0 iterations)
 Best possible: 185540.502304
 Absolute gap: 15.497696
 Relative gap: 0.000084

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	.	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	26.0000	100.0000	.
h3	-INF	18.0000	100.0000	.
h4	-INF	31.0000	100.0000	.
h5	-INF	31.0000	100.0000	.
h6	-INF	25.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	50.0000	100.0000	.
h9	-INF	80.0000	100.0000	.
h10	-INF	67.0000	100.0000	.
h11	-INF	48.0000	100.0000	.
h12	-INF	100.0000	100.0000	.
h13	-INF	38.0000	100.0000	.
h14	-INF	48.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	47.0000	100.0000	.
h17	-INF	87.0000	100.0000	.
h18	-INF	100.0000	100.0000	.
h19	-INF	47.0000	100.0000	.
h20	-INF	48.0000	100.0000	.
h21	-INF	31.0000	100.0000	.
h22	-INF	5.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	200.0000	420.0000	.
h1	-INF	126.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	80.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	113.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	284.0000	420.0000	.
h11	-INF	351.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	416.0000	420.0000	.
h14	-INF	383.0000	420.0000	.
h15	-INF	381.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.

h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	318.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-130.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-50.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-36.0000	-30.0000	.
h5 .c2	-INF	-44.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-54.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-220.0000	-180.0000	.
h11.c2	-INF	-131.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-256.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-163.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-151.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.

h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	3.0000	10.0000	.
h1 .c2	-INF	6.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	1.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	6.0000	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.

h18.c1	-INF	3.0000	10.0000	.
h18.c2	-INF	7.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	.	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	2.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	39.0000	40.0000	.
c2	-INF	29.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	200.0000	+INF	.
h1	80.0000	126.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	80.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	113.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	284.0000	+INF	.
h11	80.0000	351.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	416.0000	+INF	.
h14	80.0000	383.0000	+INF	.
h15	80.0000	381.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	318.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
--	-------	-------	-------	----------

```

---- VAR z          -INF          185556.0000          +INF

```

z total cost (objective)

```
---- VAR x operators starting at hour for company
```

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	14.0000	+INF	189.0000
h2 .c2	.	12.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	18.0000	+INF	180.0000
h4 .c1	.	11.0000	+INF	171.0000
h4 .c2	.	20.0000	+INF	171.0000
h5 .c1	.	25.0000	+INF	162.0000
h5 .c2	.	6.0000	+INF	162.0000
h6 .c1	.	9.0000	+INF	153.0000
h6 .c2	.	16.0000	+INF	153.0000
h7 .c1	.	62.0000	+INF	144.0000
h7 .c2	.	38.0000	+INF	144.0000
h8 .c1	.	44.0000	+INF	148.5000
h8 .c2	.	6.0000	+INF	148.5000
h9 .c1	.	58.0000	+INF	126.0000
h9 .c2	.	22.0000	+INF	126.0000
h10.c1	.	33.0000	+INF	126.0000
h10.c2	.	34.0000	+INF	126.0000
h11.c1	.	33.0000	+INF	135.0000
h11.c2	.	15.0000	+INF	135.0000
h12.c1	.	59.0000	+INF	144.0000
h12.c2	.	41.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	38.0000	+INF	153.0000
h14.c1	.	26.0000	+INF	162.0000
h14.c2	.	22.0000	+INF	162.0000
h15.c1	.	80.0000	+INF	171.0000
h15.c2	.	20.0000	+INF	171.0000
h16.c1	.	19.0000	+INF	180.0000
h16.c2	.	28.0000	+INF	180.0000
h17.c1	.	59.0000	+INF	189.0000
h17.c2	.	28.0000	+INF	189.0000
h18.c1	.	64.0000	+INF	189.0000
h18.c2	.	36.0000	+INF	189.0000
h19.c1	.	21.0000	+INF	189.0000
h19.c2	.	26.0000	+INF	189.0000

h20.c1	.	40.0000	+INF	189.0000
h20.c2	.	8.0000	+INF	189.0000
h21.c1	.	31.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	5.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	3.0000	+INF	30.0000
h1 .c2	.	6.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	1.0000	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	6.0000	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	3.0000	+INF	30.0000

h18.c2	.	7.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	.	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	2.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System
 Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 155.25*x(h8,c1) - 155.25*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -337.5, INFES = 337.5
 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
 capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) = 100$; (LHS = 0)

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) = 100$; (LHS = 26)

REMAINING 21 ENTRIES SKIPPED

---- cap_total = total operators active in window across companies $\leq$ capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) = 420$;
(LHS = 200)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) = 420$; (LHS = 126)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) = 420$; (LHS = 110)

REMAINING 21 ENTRIES SKIPPED

---- shortage = demand - covered $\leq$ L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) = -120$; (LHS = -130)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) = -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) = -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap = $L \leq$ maxShort

shortage_cap(h0,c1).. $L(h0,c1) = 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) = 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) = 10$; (LHS = 3)

REMAINING 45 ENTRIES SKIPPED

```

---- shortage_total  =L=  sum_h L(h,j) <= totShort(j)

shortage_total(c1)..  L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40 ; (LHS = 39)

shortage_total(c2)..  L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 29)

---- min_coverage  =G=  minimum coverage in each window across companies >= minCov

min_coverage(h0)..  x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 200)

min_coverage(h1)..  x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 126)

min_coverage(h2)..  x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110)

```

REMAINING 21 ENTRIES SKIPPED

```

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General Algebraic Modeling System
Column Listing      SOLVE Telecom_MIP Using MIP From line 157

```

```

---- z  total cost (objective)

```

```

z
      (.LO, .L, .UP, .M = -INF, 185556, +INF, 0)
1      obj

```

```

---- x  operators starting at hour for company

```

```

x(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189      obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)

```

```

1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c1)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h4,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h0,c2)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)

```

```

-1      shortage(h1,c1)
-1      shortage(h2,c1)
-1      shortage(h3,c1)
-1      shortage(h5,c1)
-1      shortage(h6,c1)
-1      shortage(h7,c1)
 1      min_coverage(h1)
 1      min_coverage(h2)
 1      min_coverage(h3)
 1      min_coverage(h5)
 1      min_coverage(h6)
 1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
 1     shortage_cap(h0,c1)
 1     shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1     shortage(h0,c2)
 1     shortage_cap(h0,c2)
 1     shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 3, +INF, 30)
-30   obj
-1     shortage(h1,c1)
 1     shortage_cap(h1,c1)
 1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [           NA,           NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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 General Algebraic Modeling System
 Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t14

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.000 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
 L O O P S t t14

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 General Algebraic Modeling System
 Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	157

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 8 Integer Solution
 **** OBJECTIVE VALUE 185893.5000

RESOURCE USAGE, LIMIT 1.156 10000000000.000
 ITERATION COUNT, LIMIT 12715 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

 --- MIP status (102): integer optimal, tolerance.
 --- Cplex Time: 1.16sec (det. 173.66 ticks)

 --- Returning a primal only solution to GAMS (marginals all set to NA).

--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.

--- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances

MIP Solution: 185893.500000 (12715 iterations, 4083 nodes)

Final Solve: 185893.500000 (0 iterations)

Best possible: 185879.431818

Absolute gap: 14.068182

Relative gap: 0.000076

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	.	100.0000	.
h1	-INF	2.0000	100.0000	.
h2	-INF	2.0000	100.0000	.
h3	-INF	18.0000	100.0000	.
h4	-INF	31.0000	100.0000	.
h5	-INF	29.0000	100.0000	.
h6	-INF	49.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	50.0000	100.0000	.
h9	-INF	82.0000	100.0000	.
h10	-INF	43.0000	100.0000	.
h11	-INF	48.0000	100.0000	.
h12	-INF	100.0000	100.0000	.
h13	-INF	36.0000	100.0000	.
h14	-INF	72.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	47.0000	100.0000	.
h17	-INF	89.0000	100.0000	.
h18	-INF	76.0000	100.0000	.
h19	-INF	47.0000	100.0000	.
h20	-INF	48.0000	100.0000	.
h21	-INF	29.0000	100.0000	.
h22	-INF	29.0000	100.0000	.

h23	-INF	.	100.0000	.
-----	------	---	----------	---

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	200.0000	420.0000	.
h1	-INF	126.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	80.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	113.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	284.0000	420.0000	.
h11	-INF	351.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	416.0000	420.0000	.
h14	-INF	383.0000	420.0000	.
h15	-INF	381.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	318.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-130.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-50.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-36.0000	-30.0000	.
h5 .c2	-INF	-44.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-54.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.

h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-220.0000	-180.0000	.
h11.c2	-INF	-131.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-256.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-163.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-151.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	3.0000	10.0000	.
h1 .c2	-INF	6.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.

h6 .c1	-INF	1.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	6.0000	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	3.0000	10.0000	.
h18.c2	-INF	7.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	.	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	2.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	39.0000	40.0000	.
c2	-INF	29.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	200.0000	+INF	.
h1	80.0000	126.0000	+INF	.

h2	80.0000	110.0000	+INF	.
h3	80.0000	80.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	113.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	284.0000	+INF	.
h11	80.0000	351.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	416.0000	+INF	.
h14	80.0000	383.0000	+INF	.
h15	80.0000	381.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	318.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	185893.5000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	2.0000	+INF	189.0000
h2 .c1	.	.	+INF	189.0000
h2 .c2	.	2.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	18.0000	+INF	180.0000
h4 .c1	.	8.0000	+INF	171.0000
h4 .c2	.	23.0000	+INF	171.0000
h5 .c1	.	28.0000	+INF	162.0000
h5 .c2	.	1.0000	+INF	162.0000
h6 .c1	.	23.0000	+INF	153.0000
h6 .c2	.	26.0000	+INF	153.0000
h7 .c1	.	59.0000	+INF	144.0000
h7 .c2	.	41.0000	+INF	144.0000

h8 .c1	.	50.0000	+INF	155.2500
h8 .c2	.	.	+INF	155.2500
h9 .c1	.	55.0000	+INF	126.0000
h9 .c2	.	27.0000	+INF	126.0000
h10.c1	.	16.0000	+INF	126.0000
h10.c2	.	27.0000	+INF	126.0000
h11.c1	.	39.0000	+INF	135.0000
h11.c2	.	9.0000	+INF	135.0000
h12.c1	.	53.0000	+INF	144.0000
h12.c2	.	47.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	36.0000	+INF	153.0000
h14.c1	.	46.0000	+INF	162.0000
h14.c2	.	26.0000	+INF	162.0000
h15.c1	.	74.0000	+INF	171.0000
h15.c2	.	26.0000	+INF	171.0000
h16.c1	.	22.0000	+INF	180.0000
h16.c2	.	25.0000	+INF	180.0000
h17.c1	.	62.0000	+INF	189.0000
h17.c2	.	27.0000	+INF	189.0000
h18.c1	.	44.0000	+INF	189.0000
h18.c2	.	32.0000	+INF	189.0000
h19.c1	.	24.0000	+INF	189.0000
h19.c2	.	23.0000	+INF	189.0000
h20.c1	.	40.0000	+INF	189.0000
h20.c2	.	8.0000	+INF	189.0000
h21.c1	.	28.0000	+INF	189.0000
h21.c2	.	1.0000	+INF	189.0000
h22.c1	.	22.0000	+INF	189.0000
h22.c2	.	7.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	3.0000	+INF	30.0000
h1 .c2	.	6.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	1.0000	+INF	30.0000

h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	6.0000	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	3.0000	+INF	30.0000
h18.c2	.	7.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	.	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	2.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -

$162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -$
 $144*x(h7,c2) - 162*x(h8,c1) - 162*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -$
 $126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -$
 $144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -$
 $171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -$
 $189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -$
 $189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -$
 $189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -$
 $30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -$
 $30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -$
 $60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -$
 $60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -$
 $45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -$
 $60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -$
 $45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -$
 $30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -$
 $45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -337.5, INFES = 337.5$
 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 0)$

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 2)$

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 2)$

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;$
 (LHS = 200.000000000001)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS$
 $= 125.999999999999)$

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS$
 $= 109.999999999998)$

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) = -120$; (LHS = -130.000000000001)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) = -70$; (LHS = -69.999999999997)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) = -95$; (LHS = -94.999999999994)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= $L \leq \text{maxShort}$

shortage_cap(h0,c1).. $L(h0,c1) = 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) = 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) = 10$; (LHS = 2.9999999999945)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= $\text{sum}_h L(h,j) \leq \text{totShort}(j)$

shortage_total(c1).. $L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) + L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) + L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) + L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) = 40$; (LHS = 39.0000000000014)

shortage_total(c2).. $L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) + L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) + L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) + L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) = 30$; (LHS = 28.9999999999976)

---- min_coverage =G= minimum coverage in each window across companies $\geq \text{minCov}$

min_coverage(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) = 80$; (LHS = 200.000000000001)

min_coverage(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) = 80$; (LHS = 125.999999999999)

min_coverage(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)$

+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 109.999999999998)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System
Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 185893.5, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
1 (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

x(h0,c2)
1 (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)

```

1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h3)
      1      min_coverage(h5)
      1      min_coverage(h6)
      1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
      1     shortage_cap(h0,c1)
      1     shortage_total(c1)

```



```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1    shortage(h0,c2)
1     shortage_cap(h0,c2)
1     shortage_total(c2)

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 2.999999999999945, +INF, 30)
-30   obj
-1    shortage(h1,c1)
1     shortage_cap(h1,c1)
1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System
Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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General Algebraic Modeling System
Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t15

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.016 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t15

GAMS 52.5.0 9bc771bf Jan 28, 2026 WEX-WEI x86 64bit/MS Windows - 02/15/26
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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL Telecom_MIP OBJECTIVE z
 TYPE MIP DIRECTION MINIMIZE
 SOLVER CPLEX FROM LINE 157

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 8 Integer Solution
 **** OBJECTIVE VALUE 186225.0000

RESOURCE USAGE, LIMIT 0.250 10000000000.000
 ITERATION COUNT, LIMIT 1474 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (102): integer optimal, tolerance.
 --- Cplex Time: 0.25sec (det. 42.92 ticks)
 --- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...
 --- Fixed MIP status (1): optimal.
 --- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances
 MIP Solution: 186225.000000 (1474 iterations, 592 nodes)
 Final Solve: 186225.000000 (0 iterations)
 Best possible: 186208.266280
 Absolute gap: 16.733720
 Relative gap: 0.000090

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	.	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	1.0000	100.0000	.
h3	-INF	20.0000	100.0000	.
h4	-INF	31.0000	100.0000	.
h5	-INF	29.0000	100.0000	.
h6	-INF	50.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	48.0000	100.0000	.
h9	-INF	82.0000	100.0000	.
h10	-INF	43.0000	100.0000	.
h11	-INF	49.0000	100.0000	.
h12	-INF	100.0000	100.0000	.
h13	-INF	37.0000	100.0000	.
h14	-INF	72.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	46.0000	100.0000	.
h17	-INF	88.0000	100.0000	.
h18	-INF	77.0000	100.0000	.
h19	-INF	46.0000	100.0000	.
h20	-INF	49.0000	100.0000	.
h21	-INF	31.0000	100.0000	.
h22	-INF	28.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	200.0000	420.0000	.
h1	-INF	126.0000	420.0000	.
h2	-INF	109.0000	420.0000	.
h3	-INF	80.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	111.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	283.0000	420.0000	.
h11	-INF	353.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	416.0000	420.0000	.
h14	-INF	382.0000	420.0000	.
h15	-INF	383.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.

h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	319.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-130.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-50.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-35.0000	-30.0000	.
h5 .c2	-INF	-45.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-53.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-221.0000	-180.0000	.
h11.c2	-INF	-132.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-256.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-162.0000	-120.0000	.
h15.c1	-INF	-231.0000	-230.0000	.
h15.c2	-INF	-152.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.

h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	4.0000	10.0000	.
h1 .c2	-INF	5.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	1.0000	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	2.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	7.0000	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.

h18.c1	-INF	3.0000	10.0000	.
h18.c2	-INF	7.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	.	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	1.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	40.0000	40.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	200.0000	+INF	.
h1	80.0000	126.0000	+INF	.
h2	80.0000	109.0000	+INF	.
h3	80.0000	80.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	111.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	283.0000	+INF	.
h11	80.0000	353.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	416.0000	+INF	.
h14	80.0000	382.0000	+INF	.
h15	80.0000	383.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	319.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
--	-------	-------	-------	----------

---- VAR z -INF 186225.0000 +INF .

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	1.0000	+INF	189.0000
h2 .c2	.	.	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	20.0000	+INF	180.0000
h4 .c1	.	10.0000	+INF	171.0000
h4 .c2	.	21.0000	+INF	171.0000
h5 .c1	.	25.0000	+INF	162.0000
h5 .c2	.	4.0000	+INF	162.0000
h6 .c1	.	22.0000	+INF	153.0000
h6 .c2	.	28.0000	+INF	153.0000
h7 .c1	.	62.0000	+INF	144.0000
h7 .c2	.	38.0000	+INF	144.0000
h8 .c1	.	45.0000	+INF	162.0000
h8 .c2	.	3.0000	+INF	162.0000
h9 .c1	.	58.0000	+INF	126.0000
h9 .c2	.	24.0000	+INF	126.0000
h10.c1	.	20.0000	+INF	126.0000
h10.c2	.	23.0000	+INF	126.0000
h11.c1	.	34.0000	+INF	135.0000
h11.c2	.	15.0000	+INF	135.0000
h12.c1	.	57.0000	+INF	144.0000
h12.c2	.	43.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	37.0000	+INF	153.0000
h14.c1	.	40.0000	+INF	162.0000
h14.c2	.	32.0000	+INF	162.0000
h15.c1	.	79.0000	+INF	171.0000
h15.c2	.	21.0000	+INF	171.0000
h16.c1	.	20.0000	+INF	180.0000
h16.c2	.	26.0000	+INF	180.0000
h17.c1	.	60.0000	+INF	189.0000
h17.c2	.	28.0000	+INF	189.0000
h18.c1	.	50.0000	+INF	189.0000
h18.c2	.	27.0000	+INF	189.0000
h19.c1	.	21.0000	+INF	189.0000
h19.c2	.	25.0000	+INF	189.0000

h20.c1	.	40.0000	+INF	189.0000
h20.c2	.	9.0000	+INF	189.0000
h21.c1	.	30.0000	+INF	189.0000
h21.c2	.	1.0000	+INF	189.0000
h22.c1	.	19.0000	+INF	189.0000
h22.c2	.	9.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	4.0000	+INF	30.0000
h1 .c2	.	5.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	1.0000	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	2.0000	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	7.0000	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	3.0000	+INF	30.0000

h18.c2	.	7.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	.	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	1.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System
 Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 168.75*x(h8,c1) - 168.75*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -324.000000000029, INFES =
 324.000000000029 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
 capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) = 100$; (LHS = 0)

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) = 100$; (LHS = 1)

REMAINING 21 ENTRIES SKIPPED

---- cap_total = total operators active in window across companies $\leq$ capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) = 420$;
(LHS = 200)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) = 420$; (LHS = 126)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) = 420$; (LHS = 109)

REMAINING 21 ENTRIES SKIPPED

---- shortage = demand - covered $\leq$ L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) = -120$; (LHS = -130)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) = -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) = -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap = $L \leq$ maxShort

shortage_cap(h0,c1).. $L(h0,c1) = 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) = 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) = 10$; (LHS = 4)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40 ; (LHS = 40)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS =
29.99999999999993)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 200)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 126)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 109)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 186225, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
-189 (.LO, .L, .UP, .M = 0, 0, +INF, 189)
obj
1 cap_per_hour(h0)
1 cap_total(h0)

```

1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c1)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h4,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h0,c2)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
1      cap_per_hour(h0)
1      cap_total(h0)
1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
1      cap_per_hour(h1)
1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h3)
1      cap_total(h5)
1      cap_total(h6)

```

```

1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h3)
1      min_coverage(h5)
1      min_coverage(h6)
1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
1      shortage_cap(h0,c1)
1      shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1     shortage(h0,c2)
1      shortage_cap(h0,c2)
1      shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 4, +INF, 30)
-30   obj
-1     shortage(h1,c1)
1      shortage_cap(h1,c1)
1      shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well

```

Matrix [min, max] : [1.000E+00, 1.890E+02]

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General Algebraic Modeling System
Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t16

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.015 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t16

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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	Cplex	FROM LINE	157

**** SOLVER STATUS 1 Normal Completion
**** MODEL STATUS 8 Integer Solution
**** OBJECTIVE VALUE 186531.0000

RESOURCE USAGE, LIMIT 0.282 10000000000.000
ITERATION COUNT, LIMIT 1155 2147483647
*** This solver runs with a demo license. No commercial use.
--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

--- MIP status (102): integer optimal, tolerance.
--- Cplex Time: 0.27sec (det. 28.69 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
 --- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances

MIP Solution: 186531.000000 (1155 iterations, 380 nodes)
 Final Solve: 186531.000000 (0 iterations)

Best possible: 186517.545961
 Absolute gap: 13.454039
 Relative gap: 0.000072

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	.	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	6.0000	100.0000	.
h3	-INF	26.0000	100.0000	.
h4	-INF	35.0000	100.0000	.
h5	-INF	19.0000	100.0000	.
h6	-INF	49.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	36.0000	100.0000	.
h9	-INF	91.0000	100.0000	.
h10	-INF	57.0000	100.0000	.
h11	-INF	48.0000	100.0000	.
h12	-INF	100.0000	100.0000	.
h13	-INF	39.0000	100.0000	.
h14	-INF	58.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	47.0000	100.0000	.
h17	-INF	86.0000	100.0000	.
h18	-INF	90.0000	100.0000	.
h19	-INF	46.0000	100.0000	.
h20	-INF	49.0000	100.0000	.
h21	-INF	35.0000	100.0000	.

h22	-INF	13.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	198.0000	420.0000	.
h1	-INF	130.0000	420.0000	.
h2	-INF	103.0000	420.0000	.
h3	-INF	80.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	109.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	252.0000	420.0000	.
h9	-INF	307.0000	420.0000	.
h10	-INF	287.0000	420.0000	.
h11	-INF	364.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	414.0000	420.0000	.
h14	-INF	381.0000	420.0000	.
h15	-INF	393.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	419.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	363.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-128.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-50.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-33.0000	-30.0000	.
h5 .c2	-INF	-47.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-50.0000	-40.0000	.

h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-142.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-220.0000	-180.0000	.
h11.c2	-INF	-144.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-254.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-161.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-163.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-113.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	5.0000	10.0000	.
h1 .c2	-INF	.	10.0000	.
h2 .c1	-INF	1.0000	10.0000	.
h2 .c2	-INF	6.0000	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.

h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	1.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	3.0000	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	3.0000	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	.	10.0000	.
h18.c2	-INF	10.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	1.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	40.0000	40.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	198.0000	+INF	.

h1	80.0000	130.0000	+INF	.
h2	80.0000	103.0000	+INF	.
h3	80.0000	80.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	109.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	252.0000	+INF	.
h9	80.0000	307.0000	+INF	.
h10	80.0000	287.0000	+INF	.
h11	80.0000	364.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	414.0000	+INF	.
h14	80.0000	381.0000	+INF	.
h15	80.0000	393.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	419.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	363.0000	+INF	.
h22	80.0000	320.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	186531.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	2.0000	+INF	189.0000
h2 .c2	.	4.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	26.0000	+INF	180.0000
h4 .c1	.	15.0000	+INF	171.0000
h4 .c2	.	20.0000	+INF	171.0000
h5 .c1	.	18.0000	+INF	162.0000
h5 .c2	.	1.0000	+INF	162.0000
h6 .c1	.	24.0000	+INF	153.0000
h6 .c2	.	25.0000	+INF	153.0000
h7 .c1	.	66.0000	+INF	144.0000

h7 .c2	.	34.0000	+INF	144.0000
h8 .c1	.	35.0000	+INF	168.7500
h8 .c2	.	1.0000	+INF	168.7500
h9 .c1	.	63.0000	+INF	126.0000
h9 .c2	.	28.0000	+INF	126.0000
h10.c1	.	25.0000	+INF	126.0000
h10.c2	.	32.0000	+INF	126.0000
h11.c1	.	24.0000	+INF	135.0000
h11.c2	.	24.0000	+INF	135.0000
h12.c1	.	66.0000	+INF	144.0000
h12.c2	.	34.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	39.0000	+INF	153.0000
h14.c1	.	31.0000	+INF	162.0000
h14.c2	.	27.0000	+INF	162.0000
h15.c1	.	87.0000	+INF	171.0000
h15.c2	.	13.0000	+INF	171.0000
h16.c1	.	17.0000	+INF	180.0000
h16.c2	.	30.0000	+INF	180.0000
h17.c1	.	56.0000	+INF	189.0000
h17.c2	.	30.0000	+INF	189.0000
h18.c1	.	60.0000	+INF	189.0000
h18.c2	.	30.0000	+INF	189.0000
h19.c1	.	16.0000	+INF	189.0000
h19.c2	.	30.0000	+INF	189.0000
h20.c1	.	39.0000	+INF	189.0000
h20.c2	.	10.0000	+INF	189.0000
h21.c1	.	35.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	13.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	5.0000	+INF	30.0000
h1 .c2	.	.	+INF	45.0000
h2 .c1	.	1.0000	+INF	30.0000
h2 .c2	.	6.0000	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000

h6 .c1	.	1.0000	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	3.0000	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	3.0000	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	.	+INF	30.0000
h18.c2	.	10.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	1.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System
 Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)

$- 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -$
 $162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -$
 $144*x(h7,c2) - 175.5*x(h8,c1) - 175.5*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -$
 $126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -$
 $144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -$
 $171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -$
 $189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -$
 $189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -$
 $189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -$
 $30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -$
 $30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -$
 $60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -$
 $60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -$
 $45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -$
 $60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -$
 $45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -$
 $30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -$
 $45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -243, INFES = 243 ****)$

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 0)$

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 0)$

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 6)$

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;$
 (LHS = 198)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS$
 = 130)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS$
 = 103)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) = -120$; (LHS = -128)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) = -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) = -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= $L \leq \text{maxShort}$

shortage_cap(h0,c1).. $L(h0,c1) = 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) = 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) = 10$; (LHS = 5)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= $\sum_h L(h,j) \leq \text{totShort}(j)$

shortage_total(c1).. $L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) + L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) + L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) + L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) = 40$; (LHS = 40)

shortage_total(c2).. $L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) + L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) + L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) + L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) = 30$; (LHS = 30)

---- min_coverage =G= minimum coverage in each window across companies $\geq \text{minCov}$

min_coverage(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) = 80$; (LHS = 198)

min_coverage(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) = 80$; (LHS = 130)

min_coverage(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) = 80$; (LHS = 103)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z

1 (.LO, .L, .UP, .M = -INF, 186531, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)

(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

x(h0,c2)

(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)


```

1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1  cap_per_hour(h1)
      1  cap_total(h1)
      1  cap_total(h2)
      1  cap_total(h3)
      1  cap_total(h5)
      1  cap_total(h6)
      1  cap_total(h7)
     -1  shortage(h1,c1)
     -1  shortage(h2,c1)
     -1  shortage(h3,c1)
     -1  shortage(h5,c1)
     -1  shortage(h6,c1)
     -1  shortage(h7,c1)
      1  min_coverage(h1)
      1  min_coverage(h2)
      1  min_coverage(h3)
      1  min_coverage(h5)
      1  min_coverage(h6)
      1  min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30  obj
     -1  shortage(h0,c1)
      1  shortage_cap(h0,c1)
      1  shortage_total(c1)

```

L(h0,c2)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 45)

```

```

-45      obj
-1       shortage(h0,c2)
1        shortage_cap(h0,c2)
1        shortage_total(c2)

```

```

L(h1,c1)
        (.LO, .L, .UP, .M = 0, 5, +INF, 30)
-30     obj
-1      shortage(h1,c1)
1       shortage_cap(h1,c1)
1       shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System
Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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General Algebraic Modeling System
Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t17

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.016 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t17

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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

```

MODEL   Telecom_MIP          OBJECTIVE  z
TYPE     MIP                  DIRECTION  MINIMIZE
SOLVER   CPLEX                FROM LINE  157

```

```

**** SOLVER STATUS      1 Normal Completion
**** MODEL STATUS      1 Optimal
**** OBJECTIVE VALUE    186729.0000

```

```

RESOURCE USAGE, LIMIT      0.063 10000000000.000
ITERATION COUNT, LIMIT     81   2147483647
*** This solver runs with a demo license. No commercial use.
--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

```

```

--- MIP status (101): integer optimal solution.
--- Cplex Time: 0.06sec (det. 6.53 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.00sec (det. 0.14 ticks)

```

```

Proven optimal solution
MIP Solution:      186729.000000    (81 iterations, 0 nodes)
Final Solve:      186729.000000    (0 iterations)

Best possible:      186729.000000
Absolute gap:       0.000000
Relative gap:       0.000000

```

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

```

---- EQU cap_per_hour  sum of operators starting at hour across companies <= capHour

```

LOWER	LEVEL	UPPER	MARGINAL
-------	-------	-------	----------

h0	-INF	.	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	.	100.0000	.
h3	-INF	30.0000	100.0000	.
h4	-INF	37.0000	100.0000	.
h5	-INF	16.0000	100.0000	.
h6	-INF	54.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	30.0000	100.0000	.
h9	-INF	97.0000	100.0000	.
h10	-INF	56.0000	100.0000	.
h11	-INF	50.0000	100.0000	.
h12	-INF	100.0000	100.0000	.
h13	-INF	40.0000	100.0000	.
h14	-INF	57.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	47.0000	100.0000	.
h17	-INF	83.0000	100.0000	.
h18	-INF	93.0000	100.0000	.
h19	-INF	46.0000	100.0000	.
h20	-INF	47.0000	100.0000	.
h21	-INF	37.0000	100.0000	.
h22	-INF	13.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	199.0000	420.0000	.
h1	-INF	130.0000	420.0000	.
h2	-INF	97.0000	420.0000	.
h3	-INF	80.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	83.0000	420.0000	.
h6	-INF	107.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	251.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	373.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	417.0000	420.0000	.
h14	-INF	380.0000	420.0000	.
h15	-INF	400.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	419.0000	420.0000	.

h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	272.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-129.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-50.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-33.0000	-30.0000	.
h5 .c2	-INF	-50.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-50.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-141.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-223.0000	-180.0000	.
h11.c2	-INF	-150.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-257.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-160.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-170.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.

h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	5.0000	10.0000	.
h1 .c2	-INF	.	10.0000	.
h2 .c1	-INF	3.0000	10.0000	.
h2 .c2	-INF	10.0000	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	3.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	.	10.0000	.
h18.c2	-INF	10.0000	10.0000	.

h19.c1	-INF	1.0000	10.0000	.
h19.c2	-INF	.	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	8.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	40.0000	40.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	199.0000	+INF	.
h1	80.0000	130.0000	+INF	.
h2	80.0000	97.0000	+INF	.
h3	80.0000	80.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	83.0000	+INF	.
h6	80.0000	107.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	251.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	373.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	417.0000	+INF	.
h14	80.0000	380.0000	+INF	.
h15	80.0000	400.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	419.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	320.0000	+INF	.
h23	80.0000	272.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	186729.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	.	+INF	189.0000
h2 .c2	.	.	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	30.0000	+INF	180.0000
h4 .c1	.	17.0000	+INF	171.0000
h4 .c2	.	20.0000	+INF	171.0000
h5 .c1	.	16.0000	+INF	162.0000
h5 .c2	.	.	+INF	162.0000
h6 .c1	.	24.0000	+INF	153.0000
h6 .c2	.	30.0000	+INF	153.0000
h7 .c1	.	70.0000	+INF	144.0000
h7 .c2	.	30.0000	+INF	144.0000
h8 .c1	.	30.0000	+INF	175.5000
h8 .c2	.	.	+INF	175.5000
h9 .c1	.	67.0000	+INF	126.0000
h9 .c2	.	30.0000	+INF	126.0000
h10.c1	.	26.0000	+INF	126.0000
h10.c2	.	30.0000	+INF	126.0000
h11.c1	.	20.0000	+INF	135.0000
h11.c2	.	30.0000	+INF	135.0000
h12.c1	.	70.0000	+INF	144.0000
h12.c2	.	30.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	40.0000	+INF	153.0000
h14.c1	.	27.0000	+INF	162.0000
h14.c2	.	30.0000	+INF	162.0000
h15.c1	.	90.0000	+INF	171.0000
h15.c2	.	10.0000	+INF	171.0000
h16.c1	.	17.0000	+INF	180.0000
h16.c2	.	30.0000	+INF	180.0000
h17.c1	.	53.0000	+INF	189.0000
h17.c2	.	30.0000	+INF	189.0000
h18.c1	.	63.0000	+INF	189.0000
h18.c2	.	30.0000	+INF	189.0000
h19.c1	.	16.0000	+INF	189.0000
h19.c2	.	30.0000	+INF	189.0000
h20.c1	.	37.0000	+INF	189.0000
h20.c2	.	10.0000	+INF	189.0000

h21.c1	.	37.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	13.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	5.0000	+INF	30.0000
h1 .c2	.	.	+INF	45.0000
h2 .c1	.	3.0000	+INF	30.0000
h2 .c2	.	10.0000	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	3.0000	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	.	+INF	30.0000
h18.c2	.	10.0000	+INF	45.0000
h19.c1	.	1.0000	+INF	30.0000

h19.c2	.	.	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	8.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

GAMS 52.5.0 9bc771bf Jan 28, 2026 WEX-WEI x86 64bit/MS Windows - 02/15/26
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General Algebraic Modeling System
 Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 182.25*x(h8,c1) - 182.25*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -202.5000000000349, INFES =
 202.5000000000349 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
 capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h1).. x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) = 100$; (LHS = 0)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) = 420$;
(LHS = 198.99999999988)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) = 420$; (LHS = 130.000000000025)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) = 420$; (LHS = 97.0000000000566)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) = -120$; (LHS = -128.99999999988)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) = -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) = -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. $L(h0,c1) = 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) = 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) = 10$; (LHS = 4.9999999997537)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

```

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40 ; (LHS =
40.0000000000727, INFES = 7.27098381503311E-11 ****)

```

```

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS =
30.0000000000003)

```

```

---- min_coverage =G= minimum coverage in each window across companies >= minCov

```

```

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 198.99999999988)

```

```

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 130.000000000025)

```

```

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 97.0000000000566)

```

```

REMAINING 21 ENTRIES SKIPPED

```

```

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```

```

General Algebraic Modeling System
Column Listing SOLVE Telecom_MIP Using MIP From line 157

```

```

---- z total cost (objective)

```

```

z
1      (.L0, .L, .UP, .M = -INF, 186729.000000004, +INF, 0)
      obj

```

```

---- x operators starting at hour for company

```

```

x(h0,c1)
-189      (.L0, .L, .UP, .M = 0, 0, +INF, 189)
      obj
1      cap_per_hour(h0)
1      cap_total(h0)
1      cap_total(h1)

```

```

1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c1)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h4,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h0,c2)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)

```

```

-1      shortage(h1,c1)
-1      shortage(h2,c1)
-1      shortage(h3,c1)
-1      shortage(h5,c1)
-1      shortage(h6,c1)
-1      shortage(h7,c1)
 1      min_coverage(h1)
 1      min_coverage(h2)
 1      min_coverage(h3)
 1      min_coverage(h5)
 1      min_coverage(h6)
 1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
 1     shortage_cap(h0,c1)
 1     shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1     shortage(h0,c2)
 1     shortage_cap(h0,c2)
 1     shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 4.99999999997537, +INF, 30)
-30   obj
-1     shortage(h1,c1)
 1     shortage_cap(h1,c1)
 1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [           NA,           NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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 General Algebraic Modeling System
 Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t18

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.015 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
 L O O P S t t18

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 General Algebraic Modeling System
 Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	157

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 1 Optimal
 **** OBJECTIVE VALUE 186931.5000

RESOURCE USAGE, LIMIT 0.094 10000000000.000
 ITERATION COUNT, LIMIT 70 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (101): integer optimal solution.
 --- Cplex Time: 0.08sec (det. 5.92 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).

--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.

--- Cplex Time: 0.02sec (det. 0.14 ticks)

Proven optimal solution

MIP Solution: 186931.500000 (70 iterations, 0 nodes)

Final Solve: 186931.500000 (0 iterations)

Best possible: 186931.500000

Absolute gap: 0.000000

Relative gap: 0.000000

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	.	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	.	100.0000	.
h3	-INF	30.0000	100.0000	.
h4	-INF	37.0000	100.0000	.
h5	-INF	16.0000	100.0000	.
h6	-INF	54.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	30.0000	100.0000	.
h9	-INF	97.0000	100.0000	.
h10	-INF	56.0000	100.0000	.
h11	-INF	50.0000	100.0000	.
h12	-INF	100.0000	100.0000	.
h13	-INF	40.0000	100.0000	.
h14	-INF	57.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	47.0000	100.0000	.
h17	-INF	83.0000	100.0000	.
h18	-INF	93.0000	100.0000	.
h19	-INF	46.0000	100.0000	.
h20	-INF	47.0000	100.0000	.
h21	-INF	37.0000	100.0000	.
h22	-INF	13.0000	100.0000	.

h23	-INF	.	100.0000	.
-----	------	---	----------	---

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	199.0000	420.0000	.
h1	-INF	130.0000	420.0000	.
h2	-INF	97.0000	420.0000	.
h3	-INF	80.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	83.0000	420.0000	.
h6	-INF	107.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	251.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	373.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	417.0000	420.0000	.
h14	-INF	380.0000	420.0000	.
h15	-INF	400.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	419.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	272.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-129.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-50.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-33.0000	-30.0000	.
h5 .c2	-INF	-50.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-50.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.

h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-141.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-223.0000	-180.0000	.
h11.c2	-INF	-150.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-257.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-160.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-170.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	5.0000	10.0000	.
h1 .c2	-INF	.	10.0000	.
h2 .c1	-INF	3.0000	10.0000	.
h2 .c2	-INF	10.0000	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.

h6 .c1	-INF	3.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	-1.46471E-10	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	.	10.0000	.
h18.c2	-INF	10.0000	10.0000	.
h19.c1	-INF	1.0000	10.0000	.
h19.c2	-INF	.	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	8.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	40.0000	40.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	199.0000	+INF	.
h1	80.0000	130.0000	+INF	.

h2	80.0000	97.0000	+INF	.
h3	80.0000	80.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	83.0000	+INF	.
h6	80.0000	107.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	251.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	373.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	417.0000	+INF	.
h14	80.0000	380.0000	+INF	.
h15	80.0000	400.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	419.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	320.0000	+INF	.
h23	80.0000	272.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	186931.5000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	.	+INF	189.0000
h2 .c2	.	.	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	30.0000	+INF	180.0000
h4 .c1	.	17.0000	+INF	171.0000
h4 .c2	.	20.0000	+INF	171.0000
h5 .c1	.	16.0000	+INF	162.0000
h5 .c2	.	.	+INF	162.0000
h6 .c1	.	24.0000	+INF	153.0000
h6 .c2	.	30.0000	+INF	153.0000
h7 .c1	.	70.0000	+INF	144.0000
h7 .c2	.	30.0000	+INF	144.0000

h8 .c1	.	30.0000	+INF	182.2500
h8 .c2	.	.	+INF	182.2500
h9 .c1	.	67.0000	+INF	126.0000
h9 .c2	.	30.0000	+INF	126.0000
h10.c1	.	26.0000	+INF	126.0000
h10.c2	.	30.0000	+INF	126.0000
h11.c1	.	20.0000	+INF	135.0000
h11.c2	.	30.0000	+INF	135.0000
h12.c1	.	70.0000	+INF	144.0000
h12.c2	.	30.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	40.0000	+INF	153.0000
h14.c1	.	27.0000	+INF	162.0000
h14.c2	.	30.0000	+INF	162.0000
h15.c1	.	90.0000	+INF	171.0000
h15.c2	.	10.0000	+INF	171.0000
h16.c1	.	17.0000	+INF	180.0000
h16.c2	.	30.0000	+INF	180.0000
h17.c1	.	53.0000	+INF	189.0000
h17.c2	.	30.0000	+INF	189.0000
h18.c1	.	63.0000	+INF	189.0000
h18.c2	.	30.0000	+INF	189.0000
h19.c1	.	16.0000	+INF	189.0000
h19.c2	.	30.0000	+INF	189.0000
h20.c1	.	37.0000	+INF	189.0000
h20.c2	.	10.0000	+INF	189.0000
h21.c1	.	37.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	13.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	5.0000	+INF	30.0000
h1 .c2	.	.	+INF	45.0000
h2 .c1	.	3.0000	+INF	30.0000
h2 .c2	.	10.0000	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	3.0000	+INF	30.0000

h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	-1.46471E-10	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	.	+INF	30.0000
h18.c2	.	10.0000	+INF	45.0000
h19.c1	.	1.0000	+INF	30.0000
h19.c2	.	.	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	8.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -

$162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -$
 $144*x(h7,c2) - 189*x(h8,c1) - 189*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -$
 $126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -$
 $144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -$
 $171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -$
 $189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -$
 $189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -$
 $189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -$
 $30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -$
 $30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -$
 $60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -$
 $60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -$
 $45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -$
 $60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -$
 $45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -$
 $30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -$
 $45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -202.500000009459, INFES =$
 $202.500000009459 ****)$

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 0)$

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 0)$

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 0)$

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;$
 $(LHS = 198.999999999997)$

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS$
 $= 129.9999999999946)$

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS$
 $= 96.99999999999742)$

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) = -120$; (LHS = -128.99999999997)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) = -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) = -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= $L \leq \text{maxShort}$

shortage_cap(h0,c1).. $L(h0,c1) = 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) = 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) = 10$; (LHS = 5.00000000005383)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= $\text{sum}_h L(h,j) \leq \text{totShort}(j)$

shortage_total(c1).. $L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) + L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) + L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) + L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) = 40$; (LHS = 40.0000000001465, INFES = 1.46528122968448E-10 ****)

shortage_total(c2).. $L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) + L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) + L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) + L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) = 30$; (LHS = 29.9999999999488)

---- min_coverage =G= minimum coverage in each window across companies $\geq \text{minCov}$

min_coverage(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) = 80$; (LHS = 198.99999999997)

min_coverage(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) = 80$; (LHS = 129.999999999946)

min_coverage(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)$

+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 96.9999999999742)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z

(.LO, .L, .UP, .M = -INF, 186931.500000015, +INF, 0)
1 obj

---- x operators starting at hour for company

x(h0,c1)

(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

x(h0,c2)

(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)

```

1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h3)
      1      min_coverage(h5)
      1      min_coverage(h6)
      1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
      1     shortage_cap(h0,c1)
      1     shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1    shortage(h0,c2)
1     shortage_cap(h0,c2)
1     shortage_total(c2)

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 5.00000000005383, +INF, 30)
-30   obj
-1    shortage(h1,c1)
1     shortage_cap(h1,c1)
1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System
Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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General Algebraic Modeling System
Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t19

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97 1 projected
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.016 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t19

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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL Telecom_MIP OBJECTIVE z
TYPE MIP DIRECTION MINIMIZE
SOLVER CPLEX FROM LINE 157

**** SOLVER STATUS 1 Normal Completion
**** MODEL STATUS 1 Optimal
**** OBJECTIVE VALUE 187134.0000

RESOURCE USAGE, LIMIT 0.063 10000000000.000
ITERATION COUNT, LIMIT 88 2147483647

*** This solver runs with a demo license. No commercial use.

--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

--- MIP status (101): integer optimal solution.
--- Cplex Time: 0.06sec (det. 13.66 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.00sec (det. 0.14 ticks)

Proven optimal solution

MIP Solution: 187134.000000 (88 iterations, 0 nodes)
Final Solve: 187134.000000 (0 iterations)

Best possible: 187134.000000
Absolute gap: 0.000000
Relative gap: 0.000000

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	.	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	.	100.0000	.
h3	-INF	30.0000	100.0000	.
h4	-INF	37.0000	100.0000	.
h5	-INF	14.0000	100.0000	.
h6	-INF	56.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	28.0000	100.0000	.
h9	-INF	98.0000	100.0000	.
h10	-INF	57.0000	100.0000	.
h11	-INF	49.0000	100.0000	.
h12	-INF	100.0000	100.0000	.
h13	-INF	40.0000	100.0000	.
h14	-INF	57.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	47.0000	100.0000	.
h17	-INF	84.0000	100.0000	.
h18	-INF	92.0000	100.0000	.
h19	-INF	47.0000	100.0000	.
h20	-INF	47.0000	100.0000	.
h21	-INF	37.0000	100.0000	.
h22	-INF	13.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	199.0000	420.0000	.
h1	-INF	131.0000	420.0000	.
h2	-INF	97.0000	420.0000	.
h3	-INF	80.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	81.0000	420.0000	.
h6	-INF	107.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	251.0000	420.0000	.
h9	-INF	307.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	374.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	415.0000	420.0000	.
h14	-INF	380.0000	420.0000	.
h15	-INF	401.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.

h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	362.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	273.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-129.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-50.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-31.0000	-30.0000	.
h5 .c2	-INF	-50.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-50.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-141.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-224.0000	-180.0000	.
h11.c2	-INF	-150.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-255.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-160.0000	-120.0000	.
h15.c1	-INF	-231.0000	-230.0000	.
h15.c2	-INF	-170.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.

h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-252.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	4.0000	10.0000	.
h1 .c2	-INF	.	10.0000	.
h2 .c1	-INF	3.0000	10.0000	.
h2 .c2	-INF	10.0000	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	3.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	3.0000	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.

h18.c1	-INF	.	10.0000	.
h18.c2	-INF	10.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	.	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	7.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	40.0000	40.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	199.0000	+INF	.
h1	80.0000	131.0000	+INF	.
h2	80.0000	97.0000	+INF	.
h3	80.0000	80.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	81.0000	+INF	.
h6	80.0000	107.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	251.0000	+INF	.
h9	80.0000	307.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	374.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	415.0000	+INF	.
h14	80.0000	380.0000	+INF	.
h15	80.0000	401.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	362.0000	+INF	.
h22	80.0000	320.0000	+INF	.
h23	80.0000	273.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
--	-------	-------	-------	----------


```

---- VAR z          -INF          187134.0000          +INF          .

```

z total cost (objective)

```
---- VAR x operators starting at hour for company
```

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	.	+INF	189.0000
h2 .c2	.	.	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	30.0000	+INF	180.0000
h4 .c1	.	17.0000	+INF	171.0000
h4 .c2	.	20.0000	+INF	171.0000
h5 .c1	.	14.0000	+INF	162.0000
h5 .c2	.	.	+INF	162.0000
h6 .c1	.	26.0000	+INF	153.0000
h6 .c2	.	30.0000	+INF	153.0000
h7 .c1	.	70.0000	+INF	144.0000
h7 .c2	.	30.0000	+INF	144.0000
h8 .c1	.	28.0000	+INF	189.0000
h8 .c2	.	.	+INF	189.0000
h9 .c1	.	68.0000	+INF	126.0000
h9 .c2	.	30.0000	+INF	126.0000
h10.c1	.	27.0000	+INF	126.0000
h10.c2	.	30.0000	+INF	126.0000
h11.c1	.	19.0000	+INF	135.0000
h11.c2	.	30.0000	+INF	135.0000
h12.c1	.	70.0000	+INF	144.0000
h12.c2	.	30.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	40.0000	+INF	153.0000
h14.c1	.	27.0000	+INF	162.0000
h14.c2	.	30.0000	+INF	162.0000
h15.c1	.	90.0000	+INF	171.0000
h15.c2	.	10.0000	+INF	171.0000
h16.c1	.	17.0000	+INF	180.0000
h16.c2	.	30.0000	+INF	180.0000
h17.c1	.	54.0000	+INF	189.0000
h17.c2	.	30.0000	+INF	189.0000
h18.c1	.	62.0000	+INF	189.0000
h18.c2	.	30.0000	+INF	189.0000
h19.c1	.	17.0000	+INF	189.0000
h19.c2	.	30.0000	+INF	189.0000

h20.c1	.	37.0000	+INF	189.0000
h20.c2	.	10.0000	+INF	189.0000
h21.c1	.	37.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	13.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	4.0000	+INF	30.0000
h1 .c2	.	.	+INF	45.0000
h2 .c1	.	3.0000	+INF	30.0000
h2 .c2	.	10.0000	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	3.0000	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	3.0000	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	.	+INF	30.0000

h18.c2	.	10.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	.	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	7.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System
 Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 195.75*x(h8,c1) - 195.75*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -189, INFES = 189 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
 capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) = 100$; (LHS = 0)

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) = 100$; (LHS = 0)

REMAINING 21 ENTRIES SKIPPED

---- cap_total = total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) = 420$;
(LHS = 199)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) = 420$; (LHS = 131)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) = 420$; (LHS = 97)

REMAINING 21 ENTRIES SKIPPED

---- shortage = demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) = -120$; (LHS = -129)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) = -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) = -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap = L <= maxShort

shortage_cap(h0,c1).. $L(h0,c1) = 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) = 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) = 10$; (LHS = 4)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total = sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40 ; (LHS = 40)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 30)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 199)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 131)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 97)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System
Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 187134, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
-189 (.LO, .L, .UP, .M = 0, 0, +INF, 189)
obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)

```

1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c1)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h4,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h0,c2)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)
-1     shortage(h1,c1)

```

```

-1      shortage(h2,c1)
-1      shortage(h3,c1)
-1      shortage(h5,c1)
-1      shortage(h6,c1)
-1      shortage(h7,c1)
 1      min_coverage(h1)
 1      min_coverage(h2)
 1      min_coverage(h3)
 1      min_coverage(h5)
 1      min_coverage(h6)
 1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.L0, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
 1     shortage_cap(h0,c1)
 1     shortage_total(c1)

```

```

L(h0,c2)
      (.L0, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1     shortage(h0,c2)
 1     shortage_cap(h0,c2)
 1     shortage_total(c2)

```

```

L(h1,c1)
      (.L0, .L, .UP, .M = 0, 4, +INF, 30)
-30   obj
-1     shortage(h1,c1)
 1     shortage_cap(h1,c1)
 1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.958E+02]

```

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 General Algebraic Modeling System
 Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS t t20

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.016 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
 L O O P S t t20

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 General Algebraic Modeling System
 Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	157

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 8 Integer Solution
 **** OBJECTIVE VALUE 187287.7500

RESOURCE USAGE, LIMIT 0.078 10000000000.000
 ITERATION COUNT, LIMIT 80 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (102): integer optimal, tolerance.
 --- Cplex Time: 0.08sec (det. 11.19 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...


```

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.00sec (det. 0.14 ticks)

```

Solution satisfies tolerances

```

MIP Solution:      187287.750000      (80 iterations, 0 nodes)
Final Solve:      187287.750000      (0 iterations)

```

```

Best possible:      187286.250000
Absolute gap:        1.500000
Relative gap:        0.000008

```

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	.	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	.	100.0000	.
h3	-INF	31.0000	100.0000	.
h4	-INF	39.0000	100.0000	.
h5	-INF	10.0000	100.0000	.
h6	-INF	59.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	21.0000	100.0000	.
h9	-INF	99.0000	100.0000	.
h10	-INF	62.0000	100.0000	.
h11	-INF	48.0000	100.0000	.
h12	-INF	100.0000	100.0000	.
h13	-INF	43.0000	100.0000	.
h14	-INF	54.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	46.0000	100.0000	.
h17	-INF	83.0000	100.0000	.
h18	-INF	94.0000	100.0000	.
h19	-INF	46.0000	100.0000	.
h20	-INF	50.0000	100.0000	.
h21	-INF	39.0000	100.0000	.
h22	-INF	10.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	200.0000	420.0000	.
h1	-INF	135.0000	420.0000	.
h2	-INF	99.0000	420.0000	.
h3	-INF	80.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	108.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	300.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	378.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	411.0000	420.0000	.
h14	-INF	379.0000	420.0000	.
h15	-INF	406.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	364.0000	420.0000	.
h22	-INF	322.0000	420.0000	.
h23	-INF	272.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-130.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-50.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-30.0000	-30.0000	.
h5 .c2	-INF	-50.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-49.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.

h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-225.0000	-180.0000	.
h11.c2	-INF	-153.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-251.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-159.0000	-120.0000	.
h15.c1	-INF	-233.0000	-230.0000	.
h15.c2	-INF	-173.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-254.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-102.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	.	10.0000	.
h1 .c2	-INF	.	10.0000	.
h2 .c1	-INF	1.0000	10.0000	.
h2 .c2	-INF	10.0000	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	1.0000	10.0000	.

h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	10.0000	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	.	10.0000	.
h18.c2	-INF	10.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	.	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	8.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	40.0000	40.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	200.0000	+INF	.
h1	80.0000	135.0000	+INF	.
h2	80.0000	99.0000	+INF	.

h3	80.0000	80.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	108.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	300.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	378.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	411.0000	+INF	.
h14	80.0000	379.0000	+INF	.
h15	80.0000	406.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	364.0000	+INF	.
h22	80.0000	322.0000	+INF	.
h23	80.0000	272.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	187287.7500	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	.	+INF	189.0000
h2 .c2	.	.	+INF	189.0000
h3 .c1	.	1.0000	+INF	180.0000
h3 .c2	.	30.0000	+INF	180.0000
h4 .c1	.	19.0000	+INF	171.0000
h4 .c2	.	20.0000	+INF	171.0000
h5 .c1	.	10.0000	+INF	162.0000
h5 .c2	.	.	+INF	162.0000
h6 .c1	.	30.0000	+INF	153.0000
h6 .c2	.	29.0000	+INF	153.0000
h7 .c1	.	69.0000	+INF	144.0000
h7 .c2	.	31.0000	+INF	144.0000
h8 .c1	.	21.0000	+INF	195.7500

h8 .c2	.	.	+INF	195.7500
h9 .c1	.	70.0000	+INF	126.0000
h9 .c2	.	29.0000	+INF	126.0000
h10.c1	.	30.0000	+INF	126.0000
h10.c2	.	32.0000	+INF	126.0000
h11.c1	.	16.0000	+INF	135.0000
h11.c2	.	32.0000	+INF	135.0000
h12.c1	.	74.0000	+INF	144.0000
h12.c2	.	26.0000	+INF	144.0000
h13.c1	.	1.0000	+INF	153.0000
h13.c2	.	42.0000	+INF	153.0000
h14.c1	.	24.0000	+INF	162.0000
h14.c2	.	30.0000	+INF	162.0000
h15.c1	.	92.0000	+INF	171.0000
h15.c2	.	8.0000	+INF	171.0000
h16.c1	.	14.0000	+INF	180.0000
h16.c2	.	32.0000	+INF	180.0000
h17.c1	.	53.0000	+INF	189.0000
h17.c2	.	30.0000	+INF	189.0000
h18.c1	.	64.0000	+INF	189.0000
h18.c2	.	30.0000	+INF	189.0000
h19.c1	.	16.0000	+INF	189.0000
h19.c2	.	30.0000	+INF	189.0000
h20.c1	.	40.0000	+INF	189.0000
h20.c2	.	10.0000	+INF	189.0000
h21.c1	.	39.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	10.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	.	+INF	30.0000
h1 .c2	.	.	+INF	45.0000
h2 .c1	.	1.0000	+INF	30.0000
h2 .c2	.	10.0000	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	1.0000	+INF	30.0000
h6 .c2	.	.	+INF	45.0000

h7 .c1	.	.	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	10.0000	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	.	+INF	30.0000
h18.c2	.	10.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	.	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	8.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 157

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -

```

144*x(h7,c2) - 202.5*x(h8,c1) - 202.5*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -141.75, INFES = 141.75
****)

```

```

---- cap_per_hour =L= sum of operators starting at hour across companies <=
capHour

```

```

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 0)

```

```

cap_per_hour(h1).. x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 0)

```

```

cap_per_hour(h2).. x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 0)

```

```

REMAINING 21 ENTRIES SKIPPED

```

```

---- cap_total =L= total operators active in window across companies <= capTotal

```

```

cap_total(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2)
+ x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;
(LHS = 200)

```

```

cap_total(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +
x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS
= 135)

```

```

cap_total(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) +
x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS
= 99)

```

```

REMAINING 21 ENTRIES SKIPPED

```

```

---- shortage =L= demand - covered <= L

```


shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) = -120$; (LHS = -130)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) = -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) = -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap $= L \leq \text{maxShort}$

shortage_cap(h0,c1).. $L(h0,c1) = 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) = 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) = 10$; (LHS = 0)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total $= \sum_h L(h,j) \leq \text{totShort}(j)$

shortage_total(c1).. $L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) + L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) + L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) + L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) = 40$; (LHS = 40)

shortage_total(c2).. $L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) + L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) + L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) + L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) = 30$; (LHS = 30)

---- min_coverage $= G = \text{minimum coverage in each window across companies} \geq \text{minCov}$

min_coverage(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) = 80$; (LHS = 200)

min_coverage(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) = 80$; (LHS = 135)

min_coverage(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) = 80$; (LHS = 99)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 157

---- z total cost (objective)

z

(.LO, .L, .UP, .M = -INF, 187287.75, +INF, 0)
1 obj

---- x operators starting at hour for company

x(h0,c1)

(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

x(h0,c2)

(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)

```

-1 shortage(h0,c2)
-1 shortage(h1,c2)
-1 shortage(h2,c2)
-1 shortage(h4,c2)
-1 shortage(h5,c2)
-1 shortage(h6,c2)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

```

x(h1,c1)

```

(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h1)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h3)
1 cap_total(h5)
1 cap_total(h6)
1 cap_total(h7)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h3,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
-1 shortage(h7,c1)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h3)
1 min_coverage(h5)
1 min_coverage(h6)
1 min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

(.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30 obj
-1 shortage(h0,c1)
1 shortage_cap(h0,c1)
1 shortage_total(c1)

```

L(h0,c2)

```

(.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45 obj

```

```

-1      shortage(h0,c2)
 1      shortage_cap(h0,c2)
 1      shortage_total(c2)

```

L(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30    obj
-1      shortage(h1,c1)
 1      shortage_cap(h1,c1)
 1      shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 157

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix    [min, max] : [ 1.000E+00, 2.025E+02]

```

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General Algebraic Modeling System

Model Statistics SOLVE Telecom_MIP Using MIP From line 157

LOOPS

t t21

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.000 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI

L O O P S t t21

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General Algebraic Modeling System

Solution Report SOLVE Telecom_MIP Using MIP From line 157

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	157

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 8 Integer Solution
 **** OBJECTIVE VALUE 187425.0000

RESOURCE USAGE, LIMIT 0.047 10000000000.000
 ITERATION COUNT, LIMIT 74 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (102): integer optimal, tolerance.
 --- Cplex Time: 0.05sec (det. 11.82 ticks)
 --- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
 --- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances
 MIP Solution: 187425.000000 (74 iterations, 0 nodes)
 Final Solve: 187425.000000 (0 iterations)
 Best possible: 187421.250000
 Absolute gap: 3.750000
 Relative gap: 0.000020

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

LOWER	LEVEL	UPPER	MARGINAL
-------	-------	-------	----------

h0	-INF	.	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	.	100.0000	.
h3	-INF	30.0000	100.0000	.
h4	-INF	40.0000	100.0000	.
h5	-INF	10.0000	100.0000	.
h6	-INF	60.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	20.0000	100.0000	.
h9	-INF	100.0000	100.0000	.
h10	-INF	60.0000	100.0000	.
h11	-INF	50.0000	100.0000	.
h12	-INF	100.0000	100.0000	.
h13	-INF	45.0000	100.0000	.
h14	-INF	55.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	45.0000	100.0000	.
h17	-INF	80.0000	100.0000	.
h18	-INF	95.0000	100.0000	.
h19	-INF	45.0000	100.0000	.
h20	-INF	50.0000	100.0000	.
h21	-INF	40.0000	100.0000	.
h22	-INF	10.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	200.0000	420.0000	.
h1	-INF	135.0000	420.0000	.
h2	-INF	100.0000	420.0000	.
h3	-INF	80.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	110.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	300.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	380.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	415.0000	420.0000	.
h14	-INF	380.0000	420.0000	.
h15	-INF	410.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.

h21	-INF	360.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-130.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-50.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-30.0000	-30.0000	.
h5 .c2	-INF	-50.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-50.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-230.0000	-180.0000	.
h11.c2	-INF	-150.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-255.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-160.0000	-120.0000	.
h15.c1	-INF	-240.0000	-230.0000	.
h15.c2	-INF	-170.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.

h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	.	10.0000	.
h1 .c2	-INF	.	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	10.0000	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	10.0000	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	.	10.0000	.
h18.c2	-INF	10.0000	10.0000	.
h19.c1	-INF	.	10.0000	.

h19.c2	-INF	.	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	40.0000	40.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	200.0000	+INF	.
h1	80.0000	135.0000	+INF	.
h2	80.0000	100.0000	+INF	.
h3	80.0000	80.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	110.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	300.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	380.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	415.0000	+INF	.
h14	80.0000	380.0000	+INF	.
h15	80.0000	410.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	320.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	187425.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	.	+INF	189.0000
h2 .c2	.	.	+INF	189.0000
h3 .c1	.	2.0000	+INF	180.0000
h3 .c2	.	28.0000	+INF	180.0000
h4 .c1	.	19.0000	+INF	171.0000
h4 .c2	.	21.0000	+INF	171.0000
h5 .c1	.	9.0000	+INF	162.0000
h5 .c2	.	1.0000	+INF	162.0000
h6 .c1	.	32.0000	+INF	153.0000
h6 .c2	.	28.0000	+INF	153.0000
h7 .c1	.	67.0000	+INF	144.0000
h7 .c2	.	33.0000	+INF	144.0000
h8 .c1	.	20.0000	+INF	202.5000
h8 .c2	.	.	+INF	202.5000
h9 .c1	.	73.0000	+INF	126.0000
h9 .c2	.	27.0000	+INF	126.0000
h10.c1	.	27.0000	+INF	126.0000
h10.c2	.	33.0000	+INF	126.0000
h11.c1	.	22.0000	+INF	135.0000
h11.c2	.	28.0000	+INF	135.0000
h12.c1	.	72.0000	+INF	144.0000
h12.c2	.	28.0000	+INF	144.0000
h13.c1	.	1.0000	+INF	153.0000
h13.c2	.	44.0000	+INF	153.0000
h14.c1	.	27.0000	+INF	162.0000
h14.c2	.	28.0000	+INF	162.0000
h15.c1	.	90.0000	+INF	171.0000
h15.c2	.	10.0000	+INF	171.0000
h16.c1	.	12.0000	+INF	180.0000
h16.c2	.	33.0000	+INF	180.0000
h17.c1	.	53.0000	+INF	189.0000
h17.c2	.	27.0000	+INF	189.0000
h18.c1	.	65.0000	+INF	189.0000
h18.c2	.	30.0000	+INF	189.0000
h19.c1	.	14.0000	+INF	189.0000
h19.c2	.	31.0000	+INF	189.0000
h20.c1	.	42.0000	+INF	189.0000
h20.c2	.	8.0000	+INF	189.0000
h21.c1	.	39.0000	+INF	189.0000

h21.c2	.	1.0000	+INF	189.0000
h22.c1	.	9.0000	+INF	189.0000
h22.c2	.	1.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	.	+INF	30.0000
h1 .c2	.	.	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	10.0000	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	10.0000	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	.	+INF	30.0000
h18.c2	.	10.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	.	+INF	45.0000

h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = 1350, INFES = 1350 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
 capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h1).. x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h2).. x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 0)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420$;
(LHS = 200)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420$; (LHS = 135)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420$; (LHS = 100)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) =L= -120$; (LHS = -130)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) =L= -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) =L= -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. $L(h0,c1) =L= 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) =L= 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) =L= 10$; (LHS = 0)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. $L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) + L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +$

$L(h_{12},c1) + L(h_{13},c1) + L(h_{14},c1) + L(h_{15},c1) + L(h_{16},c1) + L(h_{17},c1) + L(h_{18},c1) + L(h_{19},c1) + L(h_{20},c1) + L(h_{21},c1) + L(h_{22},c1) + L(h_{23},c1) = 0$; (LHS = 40, INFES = 40 ****)

shortage_total(c2).. $L(h_0,c2) + L(h_1,c2) + L(h_2,c2) + L(h_3,c2) + L(h_4,c2) + L(h_5,c2) + L(h_6,c2) + L(h_7,c2) + L(h_8,c2) + L(h_9,c2) + L(h_{10},c2) + L(h_{11},c2) + L(h_{12},c2) + L(h_{13},c2) + L(h_{14},c2) + L(h_{15},c2) + L(h_{16},c2) + L(h_{17},c2) + L(h_{18},c2) + L(h_{19},c2) + L(h_{20},c2) + L(h_{21},c2) + L(h_{22},c2) + L(h_{23},c2) = 30$; (LHS = 30)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. $x(h_0,c1) + x(h_0,c2) + x(h_{18},c1) + x(h_{18},c2) + x(h_{19},c1) + x(h_{19},c2) + x(h_{20},c1) + x(h_{20},c2) + x(h_{22},c1) + x(h_{22},c2) + x(h_{23},c1) + x(h_{23},c2) = 80$; (LHS = 200)

min_coverage(h1).. $x(h_0,c1) + x(h_0,c2) + x(h_1,c1) + x(h_1,c2) + x(h_{19},c1) + x(h_{19},c2) + x(h_{20},c1) + x(h_{20},c2) + x(h_{21},c1) + x(h_{21},c2) + x(h_{23},c1) + x(h_{23},c2) = 80$; (LHS = 135)

min_coverage(h2).. $x(h_0,c1) + x(h_0,c2) + x(h_1,c1) + x(h_1,c2) + x(h_2,c1) + x(h_2,c2) + x(h_{20},c1) + x(h_{20},c2) + x(h_{21},c1) + x(h_{21},c2) + x(h_{22},c1) + x(h_{22},c2) = 80$; (LHS = 100)

REMAINING 21 ENTRIES SKIPPED

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 Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

z
 1 (.LO, .L, .UP, .M = -INF, 187425, +INF, 0)
 obj

---- x operators starting at hour for company

x(h0,c1)
 -189 (.LO, .L, .UP, .M = 0, 0, +INF, 189)
 obj
 1 cap_per_hour(h0)
 1 cap_total(h0)
 1 cap_total(h1)
 1 cap_total(h2)
 1 cap_total(h4)
 1 cap_total(h5)

```

1      cap_total(h6)
-1     shortage(h0,c1)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h4,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h0,c2)

```

      (.L0, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

```

x(h1,c1)

```

      (.L0, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)

```

```

-1      shortage(h5,c1)
-1      shortage(h6,c1)
-1      shortage(h7,c1)
 1      min_coverage(h1)
 1      min_coverage(h2)
 1      min_coverage(h3)
 1      min_coverage(h5)
 1      min_coverage(h6)
 1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
 1     shortage_cap(h0,c1)
 1     shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1     shortage(h0,c2)
 1     shortage_cap(h0,c2)
 1     shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h1,c1)
 1     shortage_cap(h1,c1)
 1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

```

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Range Statistics    SOLVE Telecom_MIP Using MIP From line 173

```

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

```

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```


General Algebraic Modeling System
Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS t t1

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.000 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t1

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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	173

**** SOLVER STATUS 1 Normal Completion
**** MODEL STATUS 10 Integer Infeasible
**** OBJECTIVE VALUE NA

RESOURCE USAGE, LIMIT 0.016 10000000000.000
ITERATION COUNT, LIMIT 0 2147483647
*** This solver runs with a demo license. No commercial use.
--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

--- MIP status (103): integer infeasible.
--- Cplex Time: 0.02sec (det. 2.33 ticks)

--- Problem is integer infeasible

No solution returned

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---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = 1350, INFES = 1350 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
 capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h1).. x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h2).. x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 0)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) +
 + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;
 (LHS = 200)

cap_total(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +
 + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS
 = 135)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420$; (LHS = 100)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) =L= -120$; (LHS = -130)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) =L= -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) =L= -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. $L(h0,c1) =L= 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) =L= 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) =L= 10$; (LHS = 0)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. $L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) + L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) + L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) + L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 4$; (LHS = 40, INFES = 36 ****)

shortage_total(c2).. $L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) + L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) + L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) + L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30$; (LHS = 30)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)$

=G= 80 ; (LHS = 200)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 135)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 100)

REMAINING 21 ENTRIES SKIPPED

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G e n e r a l A l g e b r a i c M o d e l i n g S y s t e m

Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 187425, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
-189 (.LO, .L, .UP, .M = 0, 0, +INF, 189)
obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

```

x(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1  cap_per_hour(h0)
      1  cap_total(h0)
      1  cap_total(h1)
      1  cap_total(h2)
      1  cap_total(h4)
      1  cap_total(h5)
      1  cap_total(h6)
     -1  shortage(h0,c2)
     -1  shortage(h1,c2)
     -1  shortage(h2,c2)
     -1  shortage(h4,c2)
     -1  shortage(h5,c2)
     -1  shortage(h6,c2)
      1  min_coverage(h0)
      1  min_coverage(h1)
      1  min_coverage(h2)
      1  min_coverage(h4)
      1  min_coverage(h5)
      1  min_coverage(h6)

```

```

x(h1,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1  cap_per_hour(h1)
      1  cap_total(h1)
      1  cap_total(h2)
      1  cap_total(h3)
      1  cap_total(h5)
      1  cap_total(h6)
      1  cap_total(h7)
     -1  shortage(h1,c1)
     -1  shortage(h2,c1)
     -1  shortage(h3,c1)
     -1  shortage(h5,c1)
     -1  shortage(h6,c1)
     -1  shortage(h7,c1)
      1  min_coverage(h1)
      1  min_coverage(h2)
      1  min_coverage(h3)
      1  min_coverage(h5)
      1  min_coverage(h6)
      1  min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1    shortage(h0,c1)
1     shortage_cap(h0,c1)
1     shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1    shortage(h0,c2)
1     shortage_cap(h0,c2)
1     shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1    shortage(h1,c1)
1     shortage_cap(h1,c1)
1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 4.000E+00, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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General Algebraic Modeling System

Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS t t2

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.000 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t2

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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	Cplex	FROM LINE	173

**** SOLVER STATUS 1 Normal Completion
**** MODEL STATUS 10 Integer Infeasible
**** OBJECTIVE VALUE NA

RESOURCE USAGE, LIMIT 0.016 10000000000.000

ITERATION COUNT, LIMIT 0 2147483647

*** This solver runs with a demo license. No commercial use.

--- GMO setup time: 0.00s

--- GMO memory 0.53 Mb (peak 0.53 Mb)

--- Dictionary memory 0.00 Mb

--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)

--- Starting Cplex

--- MIP status (103): integer infeasible.

--- Cplex Time: 0.02sec (det. 3.46 ticks)

--- Problem is integer infeasible

No solution returned

GAMS 52.5.0 9bc771bf Jan 28, 2026 WEX-WEI x86 64bit/MS Windows - 02/15/26
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General Algebraic Modeling System
Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)
- 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -

$189 \times (h_{17}, c_2) - 189 \times (h_{18}, c_1) - 189 \times (h_{18}, c_2) - 189 \times (h_{19}, c_1) - 189 \times (h_{19}, c_2) -$
 $189 \times (h_{20}, c_1) - 189 \times (h_{20}, c_2) - 189 \times (h_{21}, c_1) - 189 \times (h_{21}, c_2) - 189 \times (h_{22}, c_1) -$
 $189 \times (h_{22}, c_2) - 189 \times (h_{23}, c_1) - 189 \times (h_{23}, c_2) - 30 \times L(h_0, c_1) - 45 \times L(h_0, c_2) -$
 $30 \times L(h_1, c_1) - 45 \times L(h_1, c_2) - 30 \times L(h_2, c_1) - 45 \times L(h_2, c_2) - 30 \times L(h_3, c_1) - 45 \times L(h_3, c_2) -$
 $30 \times L(h_4, c_1) - 45 \times L(h_4, c_2) - 30 \times L(h_5, c_1) - 45 \times L(h_5, c_2) - 30 \times L(h_6, c_1) - 45 \times L(h_6, c_2) -$
 $60 \times L(h_7, c_1) - 45 \times L(h_7, c_2) - 60 \times L(h_8, c_1) - 45 \times L(h_8, c_2) - 60 \times L(h_9, c_1) - 45 \times L(h_9, c_2) -$
 $60 \times L(h_{10}, c_1) - 45 \times L(h_{10}, c_2) - 60 \times L(h_{11}, c_1) - 45 \times L(h_{11}, c_2) - 60 \times L(h_{12}, c_1) -$
 $45 \times L(h_{12}, c_2) - 60 \times L(h_{13}, c_1) - 45 \times L(h_{13}, c_2) - 60 \times L(h_{14}, c_1) - 45 \times L(h_{14}, c_2) -$
 $60 \times L(h_{15}, c_1) - 45 \times L(h_{15}, c_2) - 60 \times L(h_{16}, c_1) - 45 \times L(h_{16}, c_2) - 60 \times L(h_{17}, c_1) -$
 $45 \times L(h_{17}, c_2) - 30 \times L(h_{18}, c_1) - 45 \times L(h_{18}, c_2) - 30 \times L(h_{19}, c_1) - 45 \times L(h_{19}, c_2) -$
 $30 \times L(h_{20}, c_1) - 45 \times L(h_{20}, c_2) - 30 \times L(h_{21}, c_1) - 45 \times L(h_{21}, c_2) - 30 \times L(h_{22}, c_1) -$
 $45 \times L(h_{22}, c_2) - 30 \times L(h_{23}, c_1) - 45 \times L(h_{23}, c_2) = 0 ; (LHS = 1350, INFES = 1350 ****)$

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h_0, c_1) + x(h_0, c_2) = 100 ; (LHS = 0)$

cap_per_hour(h1).. $x(h_1, c_1) + x(h_1, c_2) = 100 ; (LHS = 0)$

cap_per_hour(h2).. $x(h_2, c_1) + x(h_2, c_2) = 100 ; (LHS = 0)$

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h_0, c_1) + x(h_0, c_2) + x(h_{18}, c_1) + x(h_{18}, c_2) + x(h_{19}, c_1) + x(h_{19}, c_2) +$
 $x(h_{20}, c_1) + x(h_{20}, c_2) + x(h_{22}, c_1) + x(h_{22}, c_2) + x(h_{23}, c_1) + x(h_{23}, c_2) = 420 ;$
 (LHS = 200)

cap_total(h1).. $x(h_0, c_1) + x(h_0, c_2) + x(h_1, c_1) + x(h_1, c_2) + x(h_{19}, c_1) + x(h_{19}, c_2) +$
 $x(h_{20}, c_1) + x(h_{20}, c_2) + x(h_{21}, c_1) + x(h_{21}, c_2) + x(h_{23}, c_1) + x(h_{23}, c_2) = 420 ; (LHS$
 = 135)

cap_total(h2).. $x(h_0, c_1) + x(h_0, c_2) + x(h_1, c_1) + x(h_1, c_2) + x(h_2, c_1) + x(h_2, c_2) +$
 $x(h_{20}, c_1) + x(h_{20}, c_2) + x(h_{21}, c_1) + x(h_{21}, c_2) + x(h_{22}, c_1) + x(h_{22}, c_2) = 420 ; (LHS$
 = 100)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0, c1).. $- x(h_0, c_1) - x(h_{18}, c_1) - x(h_{19}, c_1) - x(h_{20}, c_1) - x(h_{22}, c_1) -$
 $x(h_{23}, c_1) - L(h_0, c_1) = -120 ; (LHS = -130)$

shortage(h0, c2).. $- x(h_0, c_2) - x(h_{18}, c_2) - x(h_{19}, c_2) - x(h_{20}, c_2) - x(h_{22}, c_2) -$
 $x(h_{23}, c_2) - L(h_0, c_2) = -70 ; (LHS = -70)$


```
shortage(h1,c1).. - x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) -
x(h23,c1) - L(h1,c1) =L= -95 ; (LHS = -95)
```

REMAINING 45 ENTRIES SKIPPED

```
---- shortage_cap =L= L <= maxShort
```

```
shortage_cap(h0,c1).. L(h0,c1) =L= 10 ; (LHS = 0)
```

```
shortage_cap(h0,c2).. L(h0,c2) =L= 10 ; (LHS = 0)
```

```
shortage_cap(h1,c1).. L(h1,c1) =L= 10 ; (LHS = 0)
```

REMAINING 45 ENTRIES SKIPPED

```
---- shortage_total =L= sum_h L(h,j) <= totShort(j)
```

```
shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 8 ; (LHS = 40, INFES =
32 ****)
```

```
shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 30)
```

```
---- min_coverage =G= minimum coverage in each window across companies >= minCov
```

```
min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 200)
```

```
min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 135)
```

```
min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 100)
```

REMAINING 21 ENTRIES SKIPPED

General Algebraic Modeling System
Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 187425, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

x(h0,c2)
(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c2)
-1 shortage(h1,c2)
-1 shortage(h2,c2)
-1 shortage(h4,c2)

```

-1 shortage(h5,c2)
-1 shortage(h6,c2)
 1 min_coverage(h0)
 1 min_coverage(h1)
 1 min_coverage(h2)
 1 min_coverage(h4)
 1 min_coverage(h5)
 1 min_coverage(h6)

```

x(h1,c1)

```

      (.L0, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
      1 cap_per_hour(h1)
      1 cap_total(h1)
      1 cap_total(h2)
      1 cap_total(h3)
      1 cap_total(h5)
      1 cap_total(h6)
      1 cap_total(h7)
     -1 shortage(h1,c1)
     -1 shortage(h2,c1)
     -1 shortage(h3,c1)
     -1 shortage(h5,c1)
     -1 shortage(h6,c1)
     -1 shortage(h7,c1)
      1 min_coverage(h1)
      1 min_coverage(h2)
      1 min_coverage(h3)
      1 min_coverage(h5)
      1 min_coverage(h6)
      1 min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

      (.L0, .L, .UP, .M = 0, 0, +INF, 30)
-30 obj
     -1 shortage(h0,c1)
      1 shortage_cap(h0,c1)
      1 shortage_total(c1)

```

L(h0,c2)

```

      (.L0, .L, .UP, .M = 0, 0, +INF, 45)
-45 obj
     -1 shortage(h0,c2)
      1 shortage_cap(h0,c2)
      1 shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30    obj
-1     shortage(h1,c1)
1      shortage_cap(h1,c1)
1      shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 8.000E+00, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix    [min, max] : [ 1.000E+00, 1.890E+02]

```

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General Algebraic Modeling System

Model Statistics SOLVE Telecom_MIP Using MIP From line 173

```

LOOPS                                     t   t3

```

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

```

GENERATION TIME      =      0.015 SECONDS      4 MB  52.5.0 9bc771bf WEX-WEI
                   L O O P S                   t t3

```

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General Algebraic Modeling System

Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	173

```

**** SOLVER STATUS      1 Normal Completion
**** MODEL STATUS       10 Integer Infeasible
**** OBJECTIVE VALUE     NA

```

```

RESOURCE USAGE, LIMIT      0.015 10000000000.000
ITERATION COUNT, LIMIT     0      2147483647

```

*** This solver runs with a demo license. No commercial use.

```

--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

```

```

--- MIP status (103): integer infeasible.
--- Cplex Time: 0.02sec (det. 3.49 ticks)

```

--- Problem is integer infeasible

No solution returned

GAMS 52.5.0 9bc771bf Jan 28, 2026

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

```

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)
- 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = 1350, INFES = 1350 ****)

```

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0,c1) + x(h0,c2) =L= 100$; (LHS = 0)

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) =L= 100$; (LHS = 0)

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) =L= 100$; (LHS = 0)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420$;
(LHS = 200)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420$; (LHS = 135)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420$; (LHS = 100)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) =L= -120$; (LHS = -130)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) =L= -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) =L= -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= $L \leq \text{maxShort}$

shortage_cap(h0,c1).. $L(h0,c1) =L= 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) =L= 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) =L= 10$; (LHS = 0)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 12 ; (LHS = 40, INFES
= 28 ****)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 30)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 200)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 135)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 100)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System
Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 187425, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)

```

        (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189    obj
        1    cap_per_hour(h0)
        1    cap_total(h0)
        1    cap_total(h1)
        1    cap_total(h2)
        1    cap_total(h4)
        1    cap_total(h5)
        1    cap_total(h6)
        -1    shortage(h0,c1)
        -1    shortage(h1,c1)
        -1    shortage(h2,c1)
        -1    shortage(h4,c1)
        -1    shortage(h5,c1)
        -1    shortage(h6,c1)
        1    min_coverage(h0)
        1    min_coverage(h1)
        1    min_coverage(h2)
        1    min_coverage(h4)
        1    min_coverage(h5)
        1    min_coverage(h6)

```

x(h0,c2)

```

        (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189    obj
        1    cap_per_hour(h0)
        1    cap_total(h0)
        1    cap_total(h1)
        1    cap_total(h2)
        1    cap_total(h4)
        1    cap_total(h5)
        1    cap_total(h6)
        -1    shortage(h0,c2)
        -1    shortage(h1,c2)
        -1    shortage(h2,c2)
        -1    shortage(h4,c2)
        -1    shortage(h5,c2)
        -1    shortage(h6,c2)
        1    min_coverage(h0)
        1    min_coverage(h1)
        1    min_coverage(h2)
        1    min_coverage(h4)
        1    min_coverage(h5)
        1    min_coverage(h6)

```

x(h1,c1)

```

        (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189    obj
        1    cap_per_hour(h1)
        1    cap_total(h1)

```



```

1      cap_total(h2)
1      cap_total(h3)
1      cap_total(h5)
1      cap_total(h6)
1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h3)
1      min_coverage(h5)
1      min_coverage(h6)
1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
1      shortage_cap(h0,c1)
1      shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1     shortage(h0,c2)
1      shortage_cap(h0,c2)
1      shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h1,c1)
1      shortage_cap(h1,c1)
1      shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

RHS [min, max] : [1.000E+01, 4.200E+02] - Zero values observed as well
 Bound [min, max] : [NA, NA] - Zero values observed as well
 Matrix [min, max] : [1.000E+00, 1.890E+02]

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 General Algebraic Modeling System
 Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS t t4

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.000 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
 L O O P S t t4

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 General Algebraic Modeling System
 Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	Cplex	FROM LINE	173

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 1 Optimal
 **** OBJECTIVE VALUE 185592.0000

RESOURCE USAGE, LIMIT 0.094 10000000000.000
 ITERATION COUNT, LIMIT 80 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

```

--- MIP status (101): integer optimal solution.
--- Cplex Time: 0.08sec (det. 9.64 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.02sec (det. 0.14 ticks)

```

Proven optimal solution

```

MIP Solution:      185592.000000      (80 iterations, 0 nodes)
Final Solve:       185592.000000      (0 iterations)

```

```

Best possible:      185592.000000
Absolute gap:        -0.000000
Relative gap:        -0.000000

```

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

```

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

```

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	3.0000	100.0000	.
h1	-INF	3.0000	100.0000	.
h2	-INF	.	100.0000	.
h3	-INF	13.0000	100.0000	.
h4	-INF	29.0000	100.0000	.
h5	-INF	32.0000	100.0000	.
h6	-INF	48.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	60.0000	100.0000	.
h9	-INF	76.0000	100.0000	.
h10	-INF	45.0000	100.0000	.
h11	-INF	48.0000	100.0000	.
h12	-INF	90.0000	100.0000	.
h13	-INF	46.0000	100.0000	.
h14	-INF	68.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	66.0000	100.0000	.
h17	-INF	70.0000	100.0000	.

h18	-INF	80.0000	100.0000	.
h19	-INF	55.0000	100.0000	.
h20	-INF	31.0000	100.0000	.
h21	-INF	38.0000	100.0000	.
h22	-INF	35.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	204.0000	420.0000	.
h1	-INF	130.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	89.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	115.0000	420.0000	.
h7	-INF	196.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	349.0000	420.0000	.
h12	-INF	391.0000	420.0000	.
h13	-INF	420.0000	420.0000	.
h14	-INF	385.0000	420.0000	.
h15	-INF	383.0000	420.0000	.
h16	-INF	417.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	419.0000	420.0000	.
h20	-INF	400.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	278.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-133.0000	-120.0000	.
h0 .c2	-INF	-71.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-59.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.

h5 .c1	-INF	-35.0000	-30.0000	.
h5 .c2	-INF	-45.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-55.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-219.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-151.0000	-150.0000	.
h13.c1	-INF	-260.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-165.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-153.0000	-130.0000	.
h16.c1	-INF	-257.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	.	10.0000	.
h1 .c2	-INF	5.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.

h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	4.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	.	10.0000	.
h18.c2	-INF	10.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	1.0000	10.0000	.
h20.c1	-INF	.	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	2.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	12.0000	12.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	204.0000	+INF	.
h1	80.0000	130.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	89.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	115.0000	+INF	.
h7	80.0000	196.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	349.0000	+INF	.
h12	80.0000	391.0000	+INF	.
h13	80.0000	420.0000	+INF	.
h14	80.0000	385.0000	+INF	.
h15	80.0000	383.0000	+INF	.
h16	80.0000	417.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	419.0000	+INF	.
h20	80.0000	400.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	320.0000	+INF	.
h23	80.0000	278.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	185592.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	3.0000	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	3.0000	+INF	189.0000
h2 .c1	.	.	+INF	189.0000
h2 .c2	.	.	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	13.0000	+INF	180.0000
h4 .c1	.	3.0000	+INF	171.0000
h4 .c2	.	26.0000	+INF	171.0000
h5 .c1	.	32.0000	+INF	162.0000

h5 .c2	.	.	+INF	162.0000
h6 .c1	.	25.0000	+INF	153.0000
h6 .c2	.	23.0000	+INF	153.0000
h7 .c1	.	53.0000	+INF	144.0000
h7 .c2	.	47.0000	+INF	144.0000
h8 .c1	.	59.0000	+INF	135.0000
h8 .c2	.	1.0000	+INF	135.0000
h9 .c1	.	53.0000	+INF	126.0000
h9 .c2	.	23.0000	+INF	126.0000
h10.c1	.	8.0000	+INF	126.0000
h10.c2	.	37.0000	+INF	126.0000
h11.c1	.	48.0000	+INF	135.0000
h11.c2	.	.	+INF	135.0000
h12.c1	.	47.0000	+INF	144.0000
h12.c2	.	43.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	46.0000	+INF	153.0000
h14.c1	.	53.0000	+INF	162.0000
h14.c2	.	15.0000	+INF	162.0000
h15.c1	.	68.0000	+INF	171.0000
h15.c2	.	32.0000	+INF	171.0000
h16.c1	.	33.0000	+INF	180.0000
h16.c2	.	33.0000	+INF	180.0000
h17.c1	.	54.0000	+INF	189.0000
h17.c2	.	16.0000	+INF	189.0000
h18.c1	.	43.0000	+INF	189.0000
h18.c2	.	37.0000	+INF	189.0000
h19.c1	.	32.0000	+INF	189.0000
h19.c2	.	23.0000	+INF	189.0000
h20.c1	.	31.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000
h21.c1	.	32.0000	+INF	189.0000
h21.c2	.	6.0000	+INF	189.0000
h22.c1	.	27.0000	+INF	189.0000
h22.c2	.	8.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	.	+INF	30.0000
h1 .c2	.	5.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000

h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	4.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	.	+INF	30.0000
h18.c2	.	10.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	1.0000	+INF	45.0000
h20.c1	.	.	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	2.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
- 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -2.91038304567337E-11,
INFES = 2.91038304567337E-11 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 2.99999999996011)

cap_per_hour(h1).. x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 3.00000000004448)

cap_per_hour(h2).. x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 0)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) +
x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;
(LHS = 203.999999999921)

cap_total(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +
x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS
= 129.999999999919)

cap_total(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) +
x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS
= 110)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. - x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) -
x(h23,c1) - L(h0,c1) =L= -120 ; (LHS = -132.9999999999919)

shortage(h0,c2).. - x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) -
x(h23,c2) - L(h0,c2) =L= -70 ; (LHS = -71.0000000000019)

shortage(h1,c1).. - x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) -
x(h23,c1) - L(h1,c1) =L= -95 ; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. L(h0,c1) =L= 10 ; (LHS = 0)

shortage_cap(h0,c2).. L(h0,c2) =L= 10 ; (LHS = 0)

shortage_cap(h1,c1).. L(h1,c1) =L= 10 ; (LHS = 0)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 16 ; (LHS = 12)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 30)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 203.9999999999921)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)

=G= 80 ; (LHS = 129.999999999919)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 185591.9999999992, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
1 (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

x(h0,c2)
1 (.LO, .L, .UP, .M = 0, 2.99999999996012, +INF, 189)
-189 obj
1 cap_per_hour(h0)

```

1      cap_total(h0)
1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
1      cap_per_hour(h1)
1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h3)
1      cap_total(h5)
1      cap_total(h6)
1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h3)
1      min_coverage(h5)
1      min_coverage(h6)
1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)

```

```

1 shortage_cap(h0,c1)
1 shortage_total(c1)

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45 obj
-1 shortage(h0,c2)
1 shortage_cap(h0,c2)
1 shortage_total(c2)

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30 obj
-1 shortage(h1,c1)
1 shortage_cap(h1,c1)
1 shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System
Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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General Algebraic Modeling System
Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS t t5

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.016 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t5

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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	173

**** SOLVER STATUS 1 Normal Completion
**** MODEL STATUS 8 Integer Solution
**** OBJECTIVE VALUE 185487.0000

RESOURCE USAGE, LIMIT 0.625 10000000000.000
ITERATION COUNT, LIMIT 8058 2147483647

*** This solver runs with a demo license. No commercial use.

--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

--- MIP status (102): integer optimal, tolerance.
--- Cplex Time: 0.62sec (det. 100.50 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances

MIP Solution: 185487.000000 (8058 iterations, 3859 nodes)
Final Solve: 185487.000000 (0 iterations)

Best possible: 185470.666667
Absolute gap: 16.333333
Relative gap: 0.000088

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	7.0000	100.0000	.
h1	-INF	6.0000	100.0000	.
h2	-INF	-1.35969E-10	100.0000	.
h3	-INF	13.0000	100.0000	.
h4	-INF	26.0000	100.0000	.
h5	-INF	28.0000	100.0000	.
h6	-INF	49.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	62.0000	100.0000	.
h9	-INF	81.0000	100.0000	.
h10	-INF	45.0000	100.0000	.
h11	-INF	47.0000	100.0000	.
h12	-INF	87.0000	100.0000	.
h13	-INF	43.0000	100.0000	.
h14	-INF	67.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	68.0000	100.0000	.
h17	-INF	75.0000	100.0000	.
h18	-INF	80.0000	100.0000	.
h19	-INF	54.0000	100.0000	.
h20	-INF	26.0000	100.0000	.
h21	-INF	37.0000	100.0000	.
h22	-INF	34.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	201.0000	420.0000	.
h1	-INF	130.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	90.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	116.0000	420.0000	.
h7	-INF	196.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	291.0000	420.0000	.
h11	-INF	350.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	420.0000	420.0000	.
h14	-INF	385.0000	420.0000	.

h15	-INF	383.0000	420.0000	.
h16	-INF	414.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	419.0000	420.0000	.
h20	-INF	395.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	280.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-131.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-60.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-35.0000	-30.0000	.
h5 .c2	-INF	-45.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-56.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-181.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-220.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-260.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-165.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-153.0000	-130.0000	.
h16.c1	-INF	-254.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.

h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	.	10.0000	.
h1 .c2	-INF	5.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	4.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.

h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	1.0000	10.0000	.
h18.c2	-INF	9.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	1.0000	10.0000	.
h20.c1	-INF	5.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	.	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	16.0000	16.0000	.
c2	-INF	29.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	201.0000	+INF	.
h1	80.0000	130.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	90.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	116.0000	+INF	.
h7	80.0000	196.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	291.0000	+INF	.
h11	80.0000	350.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	420.0000	+INF	.
h14	80.0000	385.0000	+INF	.
h15	80.0000	383.0000	+INF	.
h16	80.0000	414.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	419.0000	+INF	.
h20	80.0000	395.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	320.0000	+INF	.

h23	80.0000	280.0000	+INF	.	
		LOWER	LEVEL	UPPER	MARGINAL
---- VAR z		-INF	185487.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	7.0000	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	1.359708E-10	+INF	189.0000
h1 .c2	.	6.0000	+INF	189.0000
h2 .c1	.	.	+INF	189.0000
h2 .c2	.	-1.35971E-10	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	13.0000	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	26.0000	+INF	171.0000
h5 .c1	.	28.0000	+INF	162.0000
h5 .c2	.	-1.35971E-10	+INF	162.0000
h6 .c1	.	25.0000	+INF	153.0000
h6 .c2	.	24.0000	+INF	153.0000
h7 .c1	.	57.0000	+INF	144.0000
h7 .c2	.	43.0000	+INF	144.0000
h8 .c1	.	58.0000	+INF	135.0000
h8 .c2	.	4.0000	+INF	135.0000
h9 .c1	.	57.0000	+INF	126.0000
h9 .c2	.	24.0000	+INF	126.0000
h10.c1	.	13.0000	+INF	126.0000
h10.c2	.	32.0000	+INF	126.0000
h11.c1	.	40.0000	+INF	135.0000
h11.c2	.	7.0000	+INF	135.0000
h12.c1	.	47.0000	+INF	144.0000
h12.c2	.	40.0000	+INF	144.0000
h13.c1	.	1.0000	+INF	153.0000
h13.c2	.	42.0000	+INF	153.0000
h14.c1	.	44.0000	+INF	162.0000
h14.c2	.	23.0000	+INF	162.0000
h15.c1	.	75.0000	+INF	171.0000
h15.c2	.	25.0000	+INF	171.0000
h16.c1	.	35.0000	+INF	180.0000
h16.c2	.	33.0000	+INF	180.0000
h17.c1	.	52.0000	+INF	189.0000
h17.c2	.	23.0000	+INF	189.0000
h18.c1	.	50.0000	+INF	189.0000

h18.c2	.	30.0000	+INF	189.0000
h19.c1	.	28.0000	+INF	189.0000
h19.c2	.	26.0000	+INF	189.0000
h20.c1	.	23.0000	+INF	189.0000
h20.c2	.	3.0000	+INF	189.0000
h21.c1	.	37.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	23.0000	+INF	189.0000
h22.c2	.	11.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	.	+INF	30.0000
h1 .c2	.	5.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	4.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000

h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	1.0000	+INF	30.0000
h18.c2	.	9.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	1.0000	+INF	45.0000
h20.c1	.	5.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	.	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -4.77593857795E-8, INFES =
 4.77593857795E-8 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=

capHour

cap_per_hour(h0).. $x(h0,c1) + x(h0,c2) =L= 100$; (LHS = 7)

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) =L= 100$; (LHS = 6.00000000020395)

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) =L= 100$; (LHS = 0)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420$;
(LHS = 200.999999999745)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420$; (LHS = 129.999999999932)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420$; (LHS = 110.000000000204)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) =L= -120$; (LHS = -130.999999999745)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) =L= -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) =L= -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. $L(h0,c1) =L= 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) =L= 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) =L= 10$; (LHS = 0)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 20 ; (LHS =
16.0000000003569)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 29)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 200.999999999745)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 129.999999999932)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110.000000000204)

REMAINING 21 ENTRIES SKIPPED

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Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

z

1 (.LO, .L, .UP, .M = -INF, 185486.9999999998, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)

(.LO, .L, .UP, .M = 0, 7, +INF, 189)


```

-189      obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
     -1      shortage(h0,c1)
     -1      shortage(h1,c1)
     -1      shortage(h2,c1)
     -1      shortage(h4,c1)
     -1      shortage(h5,c1)
     -1      shortage(h6,c1)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

```

x(h0,c2)

(.LO, .L, .UP, .M = 0, 0, +INF, 189)

```

-189      obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
     -1      shortage(h0,c2)
     -1      shortage(h1,c2)
     -1      shortage(h2,c2)
     -1      shortage(h4,c2)
     -1      shortage(h5,c2)
     -1      shortage(h6,c2)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

```

x(h1,c1)

(.LO, .L, .UP, .M = 0, 1.35970758462039E-10, +INF, 189)

```

-189      obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)

```

```

1      cap_total(h3)
1      cap_total(h5)
1      cap_total(h6)
1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h3)
1      min_coverage(h5)
1      min_coverage(h6)
1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
1      shortage_cap(h0,c1)
1      shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1     shortage(h0,c2)
1      shortage_cap(h0,c2)
1      shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h1,c1)
1      shortage_cap(h1,c1)
1      shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

RHS [min, max] : [1.000E+01, 4.200E+02] - Zero values observed as well
 Bound [min, max] : [NA, NA] - Zero values observed as well
 Matrix [min, max] : [1.000E+00, 1.890E+02]

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 General Algebraic Modeling System
 Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS t t6

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97 2 projected
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.000 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
 L O O P S t t6

GAMS 52.5.0 9bc771bf Jan 28, 2026 WEX-WEI x86 64bit/MS Windows - 02/15/26
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 General Algebraic Modeling System
 Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	173

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 1 Optimal
 **** OBJECTIVE VALUE 185355.0000

RESOURCE USAGE, LIMIT 0.172 10000000000.000
 ITERATION COUNT, LIMIT 86 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

```

--- MIP status (101): integer optimal solution.
--- Cplex Time: 0.17sec (det. 12.65 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
--- Fixing integer variables and solving final LP...

```

```

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.00sec (det. 0.14 ticks)

```

Proven optimal solution

```

MIP Solution:      185355.000000      (86 iterations, 0 nodes)
Final Solve:      185355.000000      (0 iterations)

```

```

Best possible:      185355.000000
Absolute gap:        0.000000
Relative gap:        0.000000

```

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

```

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

```

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	6.0000	100.0000	.
h1	-INF	6.0000	100.0000	.
h2	-INF	.	100.0000	.
h3	-INF	13.0000	100.0000	.
h4	-INF	25.0000	100.0000	.
h5	-INF	30.0000	100.0000	.
h6	-INF	50.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	62.0000	100.0000	.
h9	-INF	80.0000	100.0000	.
h10	-INF	45.0000	100.0000	.
h11	-INF	44.0000	100.0000	.
h12	-INF	90.0000	100.0000	.
h13	-INF	42.0000	100.0000	.
h14	-INF	66.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	68.0000	100.0000	.
h17	-INF	76.0000	100.0000	.
h18	-INF	78.0000	100.0000	.

h19	-INF	54.0000	100.0000	.
h20	-INF	26.0000	100.0000	.
h21	-INF	36.0000	100.0000	.
h22	-INF	36.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	200.0000	420.0000	.
h1	-INF	128.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	91.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	117.0000	420.0000	.
h7	-INF	199.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	292.0000	420.0000	.
h11	-INF	349.0000	420.0000	.
h12	-INF	391.0000	420.0000	.
h13	-INF	418.0000	420.0000	.
h14	-INF	385.0000	420.0000	.
h15	-INF	377.0000	420.0000	.
h16	-INF	413.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	416.0000	420.0000	.
h20	-INF	392.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	280.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-129.0000	-120.0000	.
h0 .c2	-INF	-71.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-61.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-37.0000	-30.0000	.

h5 .c2	-INF	-43.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-57.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-182.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-219.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-151.0000	-150.0000	.
h13.c1	-INF	-258.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-165.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-147.0000	-130.0000	.
h16.c1	-INF	-253.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	.	10.0000	.
h1 .c2	-INF	7.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.

h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	1.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	2.0000	10.0000	.
h18.c2	-INF	8.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	4.0000	10.0000	.
h20.c1	-INF	8.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	.	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	20.0000	20.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	200.0000	+INF	.
h1	80.0000	128.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	91.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	117.0000	+INF	.
h7	80.0000	199.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	292.0000	+INF	.
h11	80.0000	349.0000	+INF	.
h12	80.0000	391.0000	+INF	.
h13	80.0000	418.0000	+INF	.
h14	80.0000	385.0000	+INF	.
h15	80.0000	377.0000	+INF	.
h16	80.0000	413.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	416.0000	+INF	.
h20	80.0000	392.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	320.0000	+INF	.
h23	80.0000	280.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	185355.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	5.0000	+INF	189.0000
h0 .c2	.	1.0000	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	6.0000	+INF	189.0000
h2 .c1	.	.	+INF	189.0000
h2 .c2	.	.	+INF	189.0000
h3 .c1	.	2.0000	+INF	180.0000
h3 .c2	.	11.0000	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	25.0000	+INF	171.0000
h5 .c1	.	30.0000	+INF	162.0000
h5 .c2	.	.	+INF	162.0000

h6 .c1	.	25.0000	+INF	153.0000
h6 .c2	.	25.0000	+INF	153.0000
h7 .c1	.	53.0000	+INF	144.0000
h7 .c2	.	47.0000	+INF	144.0000
h8 .c1	.	60.0000	+INF	135.0000
h8 .c2	.	2.0000	+INF	135.0000
h9 .c1	.	55.0000	+INF	126.0000
h9 .c2	.	25.0000	+INF	126.0000
h10.c1	.	12.0000	+INF	126.0000
h10.c2	.	33.0000	+INF	126.0000
h11.c1	.	44.0000	+INF	135.0000
h11.c2	.	3.9423966E-9	+INF	135.0000
h12.c1	.	46.0000	+INF	144.0000
h12.c2	.	44.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	42.0000	+INF	153.0000
h14.c1	.	47.0000	+INF	162.0000
h14.c2	.	19.0000	+INF	162.0000
h15.c1	.	72.0000	+INF	171.0000
h15.c2	.	28.0000	+INF	171.0000
h16.c1	.	32.0000	+INF	180.0000
h16.c2	.	36.0000	+INF	180.0000
h17.c1	.	56.0000	+INF	189.0000
h17.c2	.	20.0000	+INF	189.0000
h18.c1	.	47.0000	+INF	189.0000
h18.c2	.	31.0000	+INF	189.0000
h19.c1	.	28.0000	+INF	189.0000
h19.c2	.	26.0000	+INF	189.0000
h20.c1	.	26.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000
h21.c1	.	36.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	23.0000	+INF	189.0000
h22.c2	.	13.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	.	+INF	30.0000
h1 .c2	.	7.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000

h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	1.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	2.0000	+INF	30.0000
h18.c2	.	8.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	4.0000	+INF	45.0000
h20.c1	.	8.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	.	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
- 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -2.91038304567337E-11,
INFES = 2.91038304567337E-11 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 6.00000000328728)

cap_per_hour(h1).. x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 6.0000000065651)

cap_per_hour(h2).. x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 0)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) +
x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;
(LHS = 200.000000002629)

cap_total(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +
x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS
= 128.000000002629)

cap_total(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) +
x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS
= 110)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. - x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) -
x(h23,c1) - L(h0,c1) =L= -120 ; (LHS = -129.000000002628)

shortage(h0,c2).. - x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) -
x(h23,c2) - L(h0,c2) =L= -70 ; (LHS = -71.000000000014)

shortage(h1,c1).. - x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) -
x(h23,c1) - L(h1,c1) =L= -95 ; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. L(h0,c1) =L= 10 ; (LHS = 0)

shortage_cap(h0,c2).. L(h0,c2) =L= 10 ; (LHS = 0)

shortage_cap(h1,c1).. L(h1,c1) =L= 10 ; (LHS = 0)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 24 ; (LHS = 20)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 30)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 200.000000002629)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 128.000000002629)

```

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110)

```

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

```

z
      (.LO, .L, .UP, .M = -INF, 185355, +INF, 0)
1      obj

```

---- x operators starting at hour for company

```

x(h0,c1)
      (.LO, .L, .UP, .M = 0, 5.00000000197156, +INF, 189)
-189  obj
      1  cap_per_hour(h0)
      1  cap_total(h0)
      1  cap_total(h1)
      1  cap_total(h2)
      1  cap_total(h4)
      1  cap_total(h5)
      1  cap_total(h6)
      -1 shortage(h0,c1)
      -1 shortage(h1,c1)
      -1 shortage(h2,c1)
      -1 shortage(h4,c1)
      -1 shortage(h5,c1)
      -1 shortage(h6,c1)
      1  min_coverage(h0)
      1  min_coverage(h1)
      1  min_coverage(h2)
      1  min_coverage(h4)
      1  min_coverage(h5)
      1  min_coverage(h6)

```

```

x(h0,c2)
      (.LO, .L, .UP, .M = 0, 1.00000000131573, +INF, 189)
-189  obj
      1  cap_per_hour(h0)
      1  cap_total(h0)

```

```

1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
1      cap_per_hour(h1)
1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h3)
1      cap_total(h5)
1      cap_total(h6)
1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h3)
1      min_coverage(h5)
1      min_coverage(h6)
1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
1      shortage_cap(h0,c1)

```

```

1      shortage_total(c1)

L(h0,c2)
      (.L0, .L, .UP, .M = 0, 0, +INF, 45)
-45    obj
-1      shortage(h0,c2)
1      shortage_cap(h0,c2)
1      shortage_total(c2)

L(h1,c1)
      (.L0, .L, .UP, .M = 0, 0, +INF, 30)
-30    obj
-1      shortage(h1,c1)
1      shortage_cap(h1,c1)
1      shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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General Algebraic Modeling System

Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS t t7

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

```

GENERATION TIME      =      0.016 SECONDS      4 MB  52.5.0 9bc771bf WEX-WEI
      L O O P S      t t7

```

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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	173

**** SOLVER STATUS 1 Normal Completion
**** MODEL STATUS 1 Optimal
**** OBJECTIVE VALUE 185214.0000

RESOURCE USAGE, LIMIT 0.157 10000000000.000
ITERATION COUNT, LIMIT 94 2147483647
*** This solver runs with a demo license. No commercial use.
--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

--- MIP status (101): integer optimal solution.
--- Cplex Time: 0.16sec (det. 10.39 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.00sec (det. 0.14 ticks)

Proven optimal solution
MIP Solution: 185214.000000 (94 iterations, 0 nodes)
Final Solve: 185214.000000 (0 iterations)

Best possible: 185214.000000
Absolute gap: -0.000000
Relative gap: -0.000000

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	7.0000	100.0000	.
h1	-INF	2.0000	100.0000	.
h2	-INF	.	100.0000	.
h3	-INF	16.0000	100.0000	.
h4	-INF	25.0000	100.0000	.
h5	-INF	30.0000	100.0000	.
h6	-INF	49.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	60.0000	100.0000	.
h9	-INF	79.0000	100.0000	.
h10	-INF	47.0000	100.0000	.
h11	-INF	46.0000	100.0000	.
h12	-INF	90.0000	100.0000	.
h13	-INF	43.0000	100.0000	.
h14	-INF	68.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	60.0000	100.0000	.
h17	-INF	81.0000	100.0000	.
h18	-INF	78.0000	100.0000	.
h19	-INF	50.0000	100.0000	.
h20	-INF	39.0000	100.0000	.
h21	-INF	30.0000	100.0000	.
h22	-INF	32.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	206.0000	420.0000	.
h1	-INF	128.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	80.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	113.0000	420.0000	.
h7	-INF	197.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	351.0000	420.0000	.
h12	-INF	392.0000	420.0000	.
h13	-INF	418.0000	420.0000	.
h14	-INF	387.0000	420.0000	.
h15	-INF	383.0000	420.0000	.

h16	-INF	411.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	395.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	320.0000	420.0000	.
h23	-INF	271.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-134.0000	-120.0000	.
h0 .c2	-INF	-72.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-50.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-37.0000	-30.0000	.
h5 .c2	-INF	-43.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-53.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-221.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-152.0000	-150.0000	.
h13.c1	-INF	-258.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-167.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-153.0000	-130.0000	.
h16.c1	-INF	-251.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.

h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	.	10.0000	.
h1 .c2	-INF	7.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	3.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.

h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	.	10.0000	.
h18.c2	-INF	10.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	.	10.0000	.
h20.c1	-INF	5.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	.	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	9.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	24.0000	24.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	206.0000	+INF	.
h1	80.0000	128.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	80.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	113.0000	+INF	.
h7	80.0000	197.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	351.0000	+INF	.
h12	80.0000	392.0000	+INF	.
h13	80.0000	418.0000	+INF	.
h14	80.0000	387.0000	+INF	.
h15	80.0000	383.0000	+INF	.
h16	80.0000	411.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	395.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	320.0000	+INF	.
h23	80.0000	271.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	185214.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	7.0000	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	2.0000	+INF	189.0000
h2 .c1	.	.	+INF	189.0000
h2 .c2	.	.	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	16.0000	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	25.0000	+INF	171.0000
h5 .c1	.	30.0000	+INF	162.0000
h5 .c2	.	.	+INF	162.0000
h6 .c1	.	23.0000	+INF	153.0000
h6 .c2	.	26.0000	+INF	153.0000
h7 .c1	.	57.0000	+INF	144.0000
h7 .c2	.	43.0000	+INF	144.0000
h8 .c1	.	60.0000	+INF	135.0000
h8 .c2	.	.	+INF	135.0000
h9 .c1	.	53.0000	+INF	126.0000
h9 .c2	.	26.0000	+INF	126.0000
h10.c1	.	14.0000	+INF	126.0000
h10.c2	.	33.0000	+INF	126.0000
h11.c1	.	44.0000	+INF	135.0000
h11.c2	.	2.0000	+INF	135.0000
h12.c1	.	42.0000	+INF	144.0000
h12.c2	.	48.0000	+INF	144.0000
h13.c1	.	2.0000	+INF	153.0000
h13.c2	.	41.0000	+INF	153.0000
h14.c1	.	49.0000	+INF	162.0000
h14.c2	.	19.0000	+INF	162.0000
h15.c1	.	68.0000	+INF	171.0000
h15.c2	.	32.0000	+INF	171.0000
h16.c1	.	34.0000	+INF	180.0000
h16.c2	.	26.0000	+INF	180.0000
h17.c1	.	60.0000	+INF	189.0000
h17.c2	.	21.0000	+INF	189.0000
h18.c1	.	43.0000	+INF	189.0000
h18.c2	.	35.0000	+INF	189.0000

h19.c1	.	28.0000	+INF	189.0000
h19.c2	.	22.0000	+INF	189.0000
h20.c1	.	33.0000	+INF	189.0000
h20.c2	.	6.0000	+INF	189.0000
h21.c1	.	27.0000	+INF	189.0000
h21.c2	.	3.0000	+INF	189.0000
h22.c1	.	23.0000	+INF	189.0000
h22.c2	.	9.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	.	+INF	30.0000
h1 .c2	.	7.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	3.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000

h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	.	+INF	30.0000
h18.c2	.	10.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	.	+INF	45.0000
h20.c1	.	5.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	.	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	9.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -5.82076609134674E-11,
 INFES = 5.82076609134674E-11 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
 capHour

cap_per_hour(h0).. $x(h0,c1) + x(h0,c2) =L= 100$; (LHS = 6.999999999999542)

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) =L= 100$; (LHS = 1.999999999999567)

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) =L= 100$; (LHS = 0)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420$;
(LHS = 205.9999999999992)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420$; (LHS = 127.9999999999991)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420$; (LHS = 109.9999999999992)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) =L= -120$; (LHS = -134)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) =L= -70$; (LHS = -71.99999999999923)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) =L= -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. $L(h0,c1) =L= 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) =L= 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) =L= 10$; (LHS = 0)

REMAINING 45 ENTRIES SKIPPED


```
---- shortage_total  =L=  sum_h L(h,j) <= totShort(j)
```

```
shortage_total(c1)..  L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 28 ; (LHS = 24)
```

```
shortage_total(c2)..  L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 30)
```

```
---- min_coverage  =G=  minimum coverage in each window across companies >= minCov
```

```
min_coverage(h0)..  x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 205.999999999992)
```

```
min_coverage(h1)..  x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 127.999999999991)
```

```
min_coverage(h2)..  x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 109.999999999992)
```

REMAINING 21 ENTRIES SKIPPED

```
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General Algebraic Modeling System
Column Listing      SOLVE Telecom_MIP Using MIP From line 173
```

```
---- z  total cost (objective)
```

```
z
      (.LO, .L, .UP, .M = -INF, 185213.9999999998, +INF, 0)
1      obj
```

```
---- x  operators starting at hour for company
```

```
x(h0,c1)
      (.LO, .L, .UP, .M = 0, 6.99999999999543, +INF, 189)
-189      obj
      1      cap_per_hour(h0)
```

```

1      cap_total(h0)
1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c1)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h4,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h0,c2)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
1      cap_per_hour(h0)
1      cap_total(h0)
1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
1      cap_per_hour(h1)
1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h3)
1      cap_total(h5)

```

```

1      cap_total(h6)
1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h3)
1      min_coverage(h5)
1      min_coverage(h6)
1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
1      shortage_cap(h0,c1)
1      shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1     shortage(h0,c2)
1      shortage_cap(h0,c2)
1      shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h1,c1)
1      shortage_cap(h1,c1)
1      shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

RHS [min, max] : [1.000E+01, 4.200E+02] - Zero values observed as well

Bound [min, max] : [NA, NA] - Zero values observed as well
Matrix [min, max] : [1.000E+00, 1.890E+02]

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General Algebraic Modeling System
Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS t t8

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.047 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t8

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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	Cplex	FROM LINE	173

**** SOLVER STATUS 1 Normal Completion
**** MODEL STATUS 8 Integer Solution
**** OBJECTIVE VALUE 185127.0000

RESOURCE USAGE, LIMIT 1.203 10000000000.000
ITERATION COUNT, LIMIT 8840 2147483647
*** This solver runs with a demo license. No commercial use.
--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

--- MIP status (102): integer optimal, tolerance.
--- Cplex Time: 1.19sec (det. 96.77 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
 --- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances

MIP Solution: 185127.000000 (8840 iterations, 3577 nodes)
 Final Solve: 185127.000000 (0 iterations)

Best possible: 185110.666667
 Absolute gap: 16.333333
 Relative gap: 0.000088

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	13.0000	100.0000	.
h1	-INF	4.0000	100.0000	.
h2	-INF	2.0000	100.0000	.
h3	-INF	16.0000	100.0000	.
h4	-INF	19.0000	100.0000	.
h5	-INF	28.0000	100.0000	.
h6	-INF	48.0000	100.0000	.
h7	-INF	99.0000	100.0000	.
h8	-INF	66.0000	100.0000	.
h9	-INF	82.0000	100.0000	.
h10	-INF	47.0000	100.0000	.
h11	-INF	47.0000	100.0000	.
h12	-INF	84.0000	100.0000	.
h13	-INF	40.0000	100.0000	.
h14	-INF	68.0000	100.0000	.
h15	-INF	99.0000	100.0000	.
h16	-INF	65.0000	100.0000	.
h17	-INF	85.0000	100.0000	.
h18	-INF	78.0000	100.0000	.
h19	-INF	50.0000	100.0000	.
h20	-INF	30.0000	100.0000	.

h21	-INF	31.0000	100.0000	.
h22	-INF	30.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	201.0000	420.0000	.
h1	-INF	128.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	83.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	114.0000	420.0000	.
h7	-INF	197.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	351.0000	420.0000	.
h12	-INF	391.0000	420.0000	.
h13	-INF	418.0000	420.0000	.
h14	-INF	387.0000	420.0000	.
h15	-INF	383.0000	420.0000	.
h16	-INF	410.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	319.0000	420.0000	.
h23	-INF	274.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-130.0000	-120.0000	.
h0 .c2	-INF	-71.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-53.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-37.0000	-30.0000	.
h5 .c2	-INF	-43.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.

h6 .c2	-INF	-54.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-221.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-151.0000	-150.0000	.
h13.c1	-INF	-258.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-167.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-153.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	.	10.0000	.
h1 .c2	-INF	7.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.

h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	3.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	1.0000	10.0000	.
h18.c2	-INF	9.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	.	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	1.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	6.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	28.0000	28.0000	.
c2	-INF	29.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
--	-------	-------	-------	----------

h0	80.0000	201.0000	+INF	.
h1	80.0000	128.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	83.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	114.0000	+INF	.
h7	80.0000	197.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	351.0000	+INF	.
h12	80.0000	391.0000	+INF	.
h13	80.0000	418.0000	+INF	.
h14	80.0000	387.0000	+INF	.
h15	80.0000	383.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	319.0000	+INF	.
h23	80.0000	274.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	185127.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	8.0000	+INF	189.0000
h0 .c2	.	5.0000	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	4.0000	+INF	189.0000
h2 .c1	.	2.0000	+INF	189.0000
h2 .c2	.	.	+INF	189.0000
h3 .c1	.	1.0000	+INF	180.0000
h3 .c2	.	15.0000	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	19.0000	+INF	171.0000
h5 .c1	.	28.0000	+INF	162.0000
h5 .c2	.	.	+INF	162.0000
h6 .c1	.	22.0000	+INF	153.0000
h6 .c2	.	26.0000	+INF	153.0000

h7 .c1	.	57.0000	+INF	144.0000
h7 .c2	.	42.0000	+INF	144.0000
h8 .c1	.	58.0000	+INF	135.0000
h8 .c2	.	8.0000	+INF	135.0000
h9 .c1	.	56.0000	+INF	126.0000
h9 .c2	.	26.0000	+INF	126.0000
h10.c1	.	16.0000	+INF	126.0000
h10.c2	.	31.0000	+INF	126.0000
h11.c1	.	42.0000	+INF	135.0000
h11.c2	.	5.0000	+INF	135.0000
h12.c1	.	45.0000	+INF	144.0000
h12.c2	.	39.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	40.0000	+INF	153.0000
h14.c1	.	45.0000	+INF	162.0000
h14.c2	.	23.0000	+INF	162.0000
h15.c1	.	71.0000	+INF	171.0000
h15.c2	.	28.0000	+INF	171.0000
h16.c1	.	31.0000	+INF	180.0000
h16.c2	.	34.0000	+INF	180.0000
h17.c1	.	61.0000	+INF	189.0000
h17.c2	.	24.0000	+INF	189.0000
h18.c1	.	47.0000	+INF	189.0000
h18.c2	.	31.0000	+INF	189.0000
h19.c1	.	26.0000	+INF	189.0000
h19.c2	.	24.0000	+INF	189.0000
h20.c1	.	30.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000
h21.c1	.	31.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	19.0000	+INF	189.0000
h22.c2	.	11.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	.	+INF	30.0000
h1 .c2	.	7.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000

h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	3.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	1.0000	+INF	30.0000
h18.c2	.	9.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	.	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	1.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	6.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

```

obj..  z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)
- 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -2.91038304567337E-11,
INFES = 2.91038304567337E-11 ****)

```

```

---- cap_per_hour  =L=  sum of operators starting at hour across companies <=
capHour

```

```

cap_per_hour(h0)..  x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 12.99999999999998)

```

```

cap_per_hour(h1)..  x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 4)

```

```

cap_per_hour(h2)..  x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 2)

```

REMAINING 21 ENTRIES SKIPPED

```

---- cap_total  =L=  total operators active in window across companies <= capTotal

```

```

cap_total(h0)..  x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2)
+ x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;
(LHS = 201)

```

```

cap_total(h1)..  x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +
x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS
= 128)

```

```

cap_total(h2)..  x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) +
x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS
= 110)

```

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. - x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) -
x(h23,c1) - L(h0,c1) =L= -120 ; (LHS = -130)

shortage(h0,c2).. - x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) -
x(h23,c2) - L(h0,c2) =L= -70 ; (LHS = -71)

shortage(h1,c1).. - x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) -
x(h23,c1) - L(h1,c1) =L= -95 ; (LHS = -94.99999999999999)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. L(h0,c1) =L= 10 ; (LHS = 0)

shortage_cap(h0,c2).. L(h0,c2) =L= 10 ; (LHS = 0)

shortage_cap(h1,c1).. L(h1,c1) =L= 10 ; (LHS = 0)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 32 ; (LHS =
27.999999999999997)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 29)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 201)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 128)

```

min_coverage(h2)..  x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110)

```

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

```

z
      (.LO, .L, .UP, .M = -INF, 185127, +INF, 0)
1      obj

```

---- x operators starting at hour for company

```

x(h0,c1)
      (.LO, .L, .UP, .M = 0, 7.999999999999978, +INF, 189)
-189  obj
      1  cap_per_hour(h0)
      1  cap_total(h0)
      1  cap_total(h1)
      1  cap_total(h2)
      1  cap_total(h4)
      1  cap_total(h5)
      1  cap_total(h6)
      -1  shortage(h0,c1)
      -1  shortage(h1,c1)
      -1  shortage(h2,c1)
      -1  shortage(h4,c1)
      -1  shortage(h5,c1)
      -1  shortage(h6,c1)
      1  min_coverage(h0)
      1  min_coverage(h1)
      1  min_coverage(h2)
      1  min_coverage(h4)
      1  min_coverage(h5)
      1  min_coverage(h6)

```

```

x(h0,c2)
      (.LO, .L, .UP, .M = 0, 5, +INF, 189)
-189  obj
      1  cap_per_hour(h0)
      1  cap_total(h0)
      1  cap_total(h1)

```

```

1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h1,c1)

```

      (.L0, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h3)
      1      min_coverage(h5)
      1      min_coverage(h6)
      1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

      (.L0, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
      1     shortage_cap(h0,c1)
      1     shortage_total(c1)

```

L(h0,c2)

	(.L0, .L, .UP, .M = 0, 0, +INF, 45)
-45	obj
-1	shortage(h0,c2)
1	shortage_cap(h0,c2)
1	shortage_total(c2)

L(h1,c1)

	(.L0, .L, .UP, .M = 0, 0, +INF, 30)
-30	obj
-1	shortage(h1,c1)
1	shortage_cap(h1,c1)
1	shortage_total(c1)

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System
Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

RHS	[min, max] :	[1.000E+01, 4.200E+02]	- Zero values observed as well
Bound	[min, max] :	[NA, NA]	- Zero values observed as well
Matrix	[min, max] :	[1.000E+00, 1.890E+02]	

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General Algebraic Modeling System
Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS t t9

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME	=	0.000 SECONDS	4 MB	52.5.0 9bc771bf WEX-WEI
L O O P S		t t9		

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General Algebraic Modeling System

Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	173

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 1 Optimal
 **** OBJECTIVE VALUE 184977.0000

RESOURCE USAGE, LIMIT 0.172 10000000000.000
 ITERATION COUNT, LIMIT 74 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (101): integer optimal solution.
 --- Cplex Time: 0.17sec (det. 6.21 ticks)
 --- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
 --- Cplex Time: 0.00sec (det. 0.14 ticks)

Proven optimal solution
 MIP Solution: 184977.000000 (74 iterations, 0 nodes)
 Final Solve: 184977.000000 (0 iterations)
 Best possible: 184977.000000
 Absolute gap: 0.000000
 Relative gap: 0.000000

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	.	100.0000	.
h1	-INF	8.0000	100.0000	.
h2	-INF	26.0000	100.0000	.
h3	-INF	15.0000	100.0000	.
h4	-INF	31.0000	100.0000	.
h5	-INF	26.0000	100.0000	.
h6	-INF	23.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	55.0000	100.0000	.
h9	-INF	83.0000	100.0000	.
h10	-INF	73.0000	100.0000	.
h11	-INF	44.0000	100.0000	.
h12	-INF	95.0000	100.0000	.
h13	-INF	38.0000	100.0000	.
h14	-INF	44.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	54.0000	100.0000	.
h17	-INF	89.0000	100.0000	.
h18	-INF	100.0000	100.0000	.
h19	-INF	49.0000	100.0000	.
h20	-INF	44.0000	100.0000	.
h21	-INF	24.0000	100.0000	.
h22	-INF	8.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	201.0000	420.0000	.
h1	-INF	125.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	81.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	114.0000	420.0000	.
h7	-INF	198.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	291.0000	420.0000	.
h11	-INF	349.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	415.0000	420.0000	.
h14	-INF	388.0000	420.0000	.
h15	-INF	382.0000	420.0000	.
h16	-INF	410.0000	420.0000	.

h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	391.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	319.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-131.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-51.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-40.0000	-30.0000	.
h5 .c2	-INF	-40.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-54.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-181.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-219.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-255.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-168.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-152.0000	-130.0000	.
h16.c1	-INF	-250.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.

h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	.	10.0000	.
h1 .c2	-INF	10.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	2.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.

h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	2.0000	10.0000	.
h18.c2	-INF	8.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	.	10.0000	.
h20.c1	-INF	9.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	1.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	32.0000	32.0000	.
c2	-INF	30.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	201.0000	+INF	.
h1	80.0000	125.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	81.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	114.0000	+INF	.
h7	80.0000	198.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	291.0000	+INF	.
h11	80.0000	349.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	415.0000	+INF	.
h14	80.0000	388.0000	+INF	.
h15	80.0000	382.0000	+INF	.
h16	80.0000	410.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	391.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	319.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	184977.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	6.0000	+INF	189.0000
h1 .c2	.	2.0000	+INF	189.0000
h2 .c1	.	20.0000	+INF	189.0000
h2 .c2	.	6.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	15.0000	+INF	180.0000
h4 .c1	.	9.0000	+INF	171.0000
h4 .c2	.	22.0000	+INF	171.0000
h5 .c1	.	25.0000	+INF	162.0000
h5 .c2	.	1.0000	+INF	162.0000
h6 .c1	.	.	+INF	153.0000
h6 .c2	.	23.0000	+INF	153.0000
h7 .c1	.	59.0000	+INF	144.0000
h7 .c2	.	41.0000	+INF	144.0000
h8 .c1	.	52.0000	+INF	135.0000
h8 .c2	.	3.0000	+INF	135.0000
h9 .c1	.	55.0000	+INF	126.0000
h9 .c2	.	28.0000	+INF	126.0000
h10.c1	.	40.0000	+INF	126.0000
h10.c2	.	33.0000	+INF	126.0000
h11.c1	.	40.0000	+INF	135.0000
h11.c2	.	4.0000	+INF	135.0000
h12.c1	.	49.0000	+INF	144.0000
h12.c2	.	46.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	38.0000	+INF	153.0000
h14.c1	.	24.0000	+INF	162.0000
h14.c2	.	20.0000	+INF	162.0000
h15.c1	.	71.0000	+INF	171.0000
h15.c2	.	29.0000	+INF	171.0000
h16.c1	.	26.0000	+INF	180.0000
h16.c2	.	28.0000	+INF	180.0000
h17.c1	.	64.0000	+INF	189.0000
h17.c2	.	25.0000	+INF	189.0000
h18.c1	.	65.0000	+INF	189.0000
h18.c2	.	35.0000	+INF	189.0000
h19.c1	.	26.0000	+INF	189.0000

h19.c2	.	23.0000	+INF	189.0000
h20.c1	.	39.0000	+INF	189.0000
h20.c2	.	5.0000	+INF	189.0000
h21.c1	.	24.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	1.0000	+INF	189.0000
h22.c2	.	7.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	.	+INF	30.0000
h1 .c2	.	10.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	2.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000

h18.c1	.	2.0000	+INF	30.0000
h18.c2	.	8.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	.	+INF	45.0000
h20.c1	.	9.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	1.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = 5.82076609134674E-11,
 INFES = 5.82076609134674E-11 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
 capHour

cap_per_hour(h0).. $x(h0,c1) + x(h0,c2) = 100$; (LHS = 0)

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) = 100$; (LHS = 8.00000000011266)

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) = 100$; (LHS = 26.0000000000791)

REMAINING 21 ENTRIES SKIPPED

---- cap_total = total operators active in window across companies $\leq$ capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) = 420$;
(LHS = 200.999999999921)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) = 420$; (LHS = 125)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) = 420$; (LHS = 110)

REMAINING 21 ENTRIES SKIPPED

---- shortage = demand - covered $\leq$ L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) = -120$; (LHS = -130.999999999921)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) = -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) = -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap = $L \leq$ maxShort

shortage_cap(h0,c1).. $L(h0,c1) = 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) = 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) = 10$; (LHS = 0)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 36 ; (LHS = 32)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 30)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 200.999999999921)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 125)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 184977, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
-189 (.LO, .L, .UP, .M = 0, 0, +INF, 189)
obj
1 cap_per_hour(h0)
1 cap_total(h0)

```

1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c1)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h4,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h0,c2)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
1      cap_per_hour(h0)
1      cap_total(h0)
1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 6.00000000023722, +INF, 189)
-189  obj
1      cap_per_hour(h1)
1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h3)
1      cap_total(h5)
1      cap_total(h6)

```

```

1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h3)
1      min_coverage(h5)
1      min_coverage(h6)
1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1    shortage(h0,c1)
1     shortage_cap(h0,c1)
1     shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1    shortage(h0,c2)
1     shortage_cap(h0,c2)
1     shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1    shortage(h1,c1)
1     shortage_cap(h1,c1)
1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well

```

Matrix [min, max] : [1.000E+00, 1.890E+02]

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General Algebraic Modeling System
Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS t t10

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.031 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t10

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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	Cplex	FROM LINE	173

**** SOLVER STATUS 1 Normal Completion
**** MODEL STATUS 1 Optimal
**** OBJECTIVE VALUE 184908.0000

RESOURCE USAGE, LIMIT 0.516 10000000000.000
ITERATION COUNT, LIMIT 232 2147483647
*** This solver runs with a demo license. No commercial use.
--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

--- MIP status (101): integer optimal solution.
--- Cplex Time: 0.52sec (det. 20.93 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
 --- Cplex Time: 0.00sec (det. 0.14 ticks)

Proven optimal solution

MIP Solution: 184908.000000 (232 iterations, 44 nodes)
 Final Solve: 184908.000000 (0 iterations)

Best possible: 184908.000000
 Absolute gap: 0.000000
 Relative gap: 0.000000

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	1.0000	100.0000	.
h1	-INF	4.0000	100.0000	.
h2	-INF	29.0000	100.0000	.
h3	-INF	15.0000	100.0000	.
h4	-INF	29.0000	100.0000	.
h5	-INF	31.0000	100.0000	.
h6	-INF	20.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	57.0000	100.0000	.
h9	-INF	78.0000	100.0000	.
h10	-INF	74.0000	100.0000	.
h11	-INF	45.0000	100.0000	.
h12	-INF	94.0000	100.0000	.
h13	-INF	41.0000	100.0000	.
h14	-INF	44.0000	100.0000	.
h15	-INF	99.0000	100.0000	.
h16	-INF	55.0000	100.0000	.
h17	-INF	86.0000	100.0000	.
h18	-INF	100.0000	100.0000	.
h19	-INF	50.0000	100.0000	.
h20	-INF	42.0000	100.0000	.
h21	-INF	28.0000	100.0000	.

h22	-INF	6.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	199.0000	420.0000	.
h1	-INF	125.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	82.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	114.0000	420.0000	.
h7	-INF	199.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	289.0000	420.0000	.
h11	-INF	348.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	415.0000	420.0000	.
h14	-INF	388.0000	420.0000	.
h15	-INF	381.0000	420.0000	.
h16	-INF	411.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	420.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	317.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-129.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-52.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-40.0000	-30.0000	.
h5 .c2	-INF	-40.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-54.0000	-40.0000	.

h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-218.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-255.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-168.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-151.0000	-130.0000	.
h16.c1	-INF	-251.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	3.500844E-10	10.0000	.
h1 .c2	-INF	10.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.

h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	1.0000	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	1.0000	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	3.0000	10.0000	.
h18.c2	-INF	7.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	.	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	3.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	36.0000	36.0000	.
c2	-INF	29.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	199.0000	+INF	.

h1	80.0000	125.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	82.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	114.0000	+INF	.
h7	80.0000	199.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	289.0000	+INF	.
h11	80.0000	348.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	415.0000	+INF	.
h14	80.0000	388.0000	+INF	.
h15	80.0000	381.0000	+INF	.
h16	80.0000	411.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	420.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	317.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	184908.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	1.0000	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	1.0000	+INF	189.0000
h1 .c2	.	3.0000	+INF	189.0000
h2 .c1	.	21.0000	+INF	189.0000
h2 .c2	.	8.0000	+INF	189.0000
h3 .c1	.	2.0000	+INF	180.0000
h3 .c2	.	13.0000	+INF	180.0000
h4 .c1	.	6.0000	+INF	171.0000
h4 .c2	.	23.0000	+INF	171.0000
h5 .c1	.	30.0000	+INF	162.0000
h5 .c2	.	1.0000	+INF	162.0000
h6 .c1	.	1.0000	+INF	153.0000
h6 .c2	.	19.0000	+INF	153.0000
h7 .c1	.	55.0000	+INF	144.0000

h7 .c2	.	45.0000	+INF	144.0000
h8 .c1	.	55.0000	+INF	135.0000
h8 .c2	.	2.0000	+INF	135.0000
h9 .c1	.	52.0000	+INF	126.0000
h9 .c2	.	26.0000	+INF	126.0000
h10.c1	.	36.0000	+INF	126.0000
h10.c2	.	38.0000	+INF	126.0000
h11.c1	.	44.0000	+INF	135.0000
h11.c2	.	1.0000	+INF	135.0000
h12.c1	.	49.0000	+INF	144.0000
h12.c2	.	45.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	41.0000	+INF	153.0000
h14.c1	.	28.0000	+INF	162.0000
h14.c2	.	16.0000	+INF	162.0000
h15.c1	.	70.0000	+INF	171.0000
h15.c2	.	29.0000	+INF	171.0000
h16.c1	.	24.0000	+INF	180.0000
h16.c2	.	31.0000	+INF	180.0000
h17.c1	.	63.0000	+INF	189.0000
h17.c2	.	23.0000	+INF	189.0000
h18.c1	.	63.0000	+INF	189.0000
h18.c2	.	37.0000	+INF	189.0000
h19.c1	.	26.0000	+INF	189.0000
h19.c2	.	24.0000	+INF	189.0000
h20.c1	.	39.0000	+INF	189.0000
h20.c2	.	3.0000	+INF	189.0000
h21.c1	.	28.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	.	+INF	189.0000
h22.c2	.	6.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	3.500844E-10	+INF	30.0000
h1 .c2	.	10.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000

h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	1.0000	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	1.0000	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	3.0000	+INF	30.0000
h18.c2	.	7.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	.	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	3.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System
 Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)

- 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -5.82076609134674E-11,
 INFES = 5.82076609134674E-11 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 0.99999999978084)

cap_per_hour(h1).. x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 4.00000000002989)

cap_per_hour(h2).. x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 29.0000000000142)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2)
 + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;
 (LHS = 198.999999999592)

cap_total(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +
 x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS
 = 124.999999999825)

cap_total(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) +
 x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS
 = 110)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. - x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) -
x(h23,c1) - L(h0,c1) =L= -120 ; (LHS = -128.999999999592)

shortage(h0,c2).. - x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) -
x(h23,c2) - L(h0,c2) =L= -70 ; (LHS = -70)

shortage(h1,c1).. - x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) -
x(h23,c1) - L(h1,c1) =L= -95 ; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. L(h0,c1) =L= 10 ; (LHS = 0)

shortage_cap(h0,c2).. L(h0,c2) =L= 10 ; (LHS = 0)

shortage_cap(h1,c1).. L(h1,c1) =L= 10 ; (LHS = 3.50084405909001E-10)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 40 ; (LHS = 36)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS =
28.999999999701)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 198.999999999592)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 124.999999999825)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)

+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

z

(.LO, .L, .UP, .M = -INF, 184908.000000002, +INF, 0)
1 obj

---- x operators starting at hour for company

x(h0,c1)

(.LO, .L, .UP, .M = 0, 0.999999999780847, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

x(h0,c2)

(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)

```

1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h1,c1)

```

      (.L0, .L, .UP, .M = 0, 0.999999999854836, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h3)
      1      min_coverage(h5)
      1      min_coverage(h6)
      1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

      (.L0, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
      1     shortage_cap(h0,c1)
      1     shortage_total(c1)

```



```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1    shortage(h0,c2)
1     shortage_cap(h0,c2)
1     shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 3.50084405909001E-10, +INF, 30)
-30   obj
-1    shortage(h1,c1)
1     shortage_cap(h1,c1)
1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System
Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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General Algebraic Modeling System
Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS t t11

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.015 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t11

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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL Telecom_MIP OBJECTIVE z
 TYPE MIP DIRECTION MINIMIZE
 SOLVER CPLEX FROM LINE 173

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 1 Optimal
 **** OBJECTIVE VALUE 184848.0000

RESOURCE USAGE, LIMIT 0.343 10000000000.000
 ITERATION COUNT, LIMIT 82 2147483647

*** This solver runs with a demo license. No commercial use.

--- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (101): integer optimal solution.
 --- Cplex Time: 0.34sec (det. 12.53 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
 --- Cplex Time: 0.00sec (det. 0.14 ticks)

Proven optimal solution

MIP Solution: 184848.000000 (82 iterations, 0 nodes)
 Final Solve: 184848.000000 (0 iterations)

Best possible: 184848.000000
 Absolute gap: 0.000000
 Relative gap: 0.000000

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	.	100.0000	.
h1	-INF	5.0000	100.0000	.
h2	-INF	28.0000	100.0000	.
h3	-INF	13.0000	100.0000	.
h4	-INF	32.0000	100.0000	.
h5	-INF	30.0000	100.0000	.
h6	-INF	24.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	53.0000	100.0000	.
h9	-INF	82.0000	100.0000	.
h10	-INF	70.0000	100.0000	.
h11	-INF	43.0000	100.0000	.
h12	-INF	100.0000	100.0000	.
h13	-INF	37.0000	100.0000	.
h14	-INF	43.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	55.0000	100.0000	.
h17	-INF	85.0000	100.0000	.
h18	-INF	100.0000	100.0000	.
h19	-INF	50.0000	100.0000	.
h20	-INF	42.0000	100.0000	.
h21	-INF	28.0000	100.0000	.
h22	-INF	7.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	199.0000	420.0000	.
h1	-INF	125.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	81.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	119.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	291.0000	420.0000	.
h11	-INF	349.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	415.0000	420.0000	.
h14	-INF	385.0000	420.0000	.
h15	-INF	375.0000	420.0000	.
h16	-INF	411.0000	420.0000	.
h17	-INF	420.0000	420.0000	.

h18	-INF	420.0000	420.0000	.
h19	-INF	415.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	317.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-129.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-51.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-37.0000	-30.0000	.
h5 .c2	-INF	-43.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-59.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-181.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-219.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-255.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-165.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-145.0000	-130.0000	.
h16.c1	-INF	-251.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.

h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	3.0000	10.0000	.
h1 .c2	-INF	7.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	-1.522184E-8	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.

h18.c1	-INF	4.0000	10.0000	.
h18.c2	-INF	6.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	5.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	3.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	40.0000	40.0000	.
c2	-INF	28.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	199.0000	+INF	.
h1	80.0000	125.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	81.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	119.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	291.0000	+INF	.
h11	80.0000	349.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	415.0000	+INF	.
h14	80.0000	385.0000	+INF	.
h15	80.0000	375.0000	+INF	.
h16	80.0000	411.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	415.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	317.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
--	-------	-------	-------	----------

---- VAR z -INF 184848.0000 +INF .

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	5.0000	+INF	189.0000
h2 .c1	.	23.0000	+INF	189.0000
h2 .c2	.	5.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	13.0000	+INF	180.0000
h4 .c1	.	7.0000	+INF	171.0000
h4 .c2	.	25.0000	+INF	171.0000
h5 .c1	.	30.0000	+INF	162.0000
h5 .c2	.	-2.410219E-8	+INF	162.0000
h6 .c1	.	.	+INF	153.0000
h6 .c2	.	24.0000	+INF	153.0000
h7 .c1	.	57.0000	+INF	144.0000
h7 .c2	.	43.0000	+INF	144.0000
h8 .c1	.	53.0000	+INF	135.0000
h8 .c2	.	-3.932403E-8	+INF	135.0000
h9 .c1	.	53.0000	+INF	126.0000
h9 .c2	.	29.0000	+INF	126.0000
h10.c1	.	38.0000	+INF	126.0000
h10.c2	.	32.0000	+INF	126.0000
h11.c1	.	41.0000	+INF	135.0000
h11.c2	.	2.0000	+INF	135.0000
h12.c1	.	51.0000	+INF	144.0000
h12.c2	.	49.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	37.0000	+INF	153.0000
h14.c1	.	25.0000	+INF	162.0000
h14.c2	.	18.0000	+INF	162.0000
h15.c1	.	73.0000	+INF	171.0000
h15.c2	.	27.0000	+INF	171.0000
h16.c1	.	23.0000	+INF	180.0000
h16.c2	.	32.0000	+INF	180.0000
h17.c1	.	62.0000	+INF	189.0000
h17.c2	.	23.0000	+INF	189.0000
h18.c1	.	65.0000	+INF	189.0000
h18.c2	.	35.0000	+INF	189.0000
h19.c1	.	25.0000	+INF	189.0000
h19.c2	.	25.0000	+INF	189.0000

h20.c1	.	39.0000	+INF	189.0000
h20.c2	.	3.0000	+INF	189.0000
h21.c1	.	28.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	-3.180922E-9	+INF	189.0000
h22.c2	.	7.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	3.0000	+INF	30.0000
h1 .c2	.	7.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	-1.522184E-8	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	4.0000	+INF	30.0000

h18.c2	.	6.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	5.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	3.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -1.04994396679103E-5,
 INFES = 1.04994396679103E-5 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
 capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) =L= 100$; (LHS = 5.0000000374265)

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) =L= 100$; (LHS = 27.9999999543182)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420$;
(LHS = 199.000000069166)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420$; (LHS = 124.999999979706)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420$; (LHS = 110.00000003181)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) =L= -120$; (LHS = -129.000000069166)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) =L= -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) =L= -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= $L \leq \text{maxShort}$

shortage_cap(h0,c1).. $L(h0,c1) =L= 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) =L= 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) =L= 10$; (LHS = 3.00000002219792)

REMAINING 45 ENTRIES SKIPPED

```

---- shortage_total  =L=  sum_h L(h,j) <= totShort(j)

shortage_total(c1)..  L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 44 ; (LHS = 40)

shortage_total(c2)..  L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS =
27.9999999999839)

---- min_coverage  =G=  minimum coverage in each window across companies >= minCov

min_coverage(h0)..  x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 199.000000069166)

min_coverage(h1)..  x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 124.999999979706)

min_coverage(h2)..  x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110.000000003181)

REMAINING 21 ENTRIES SKIPPED

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Column Listing      SOLVE Telecom_MIP Using MIP From line 173

---- z  total cost (objective)

z
1      (.LO, .L, .UP, .M = -INF, 184848.000000154, +INF, 0)
      obj

---- x  operators starting at hour for company

x(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189   obj
1      cap_per_hour(h0)
1      cap_total(h0)

```

```

1      cap_total(h1)
1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c1)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h4,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h0,c2)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)

```

```

1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
-1     shortage(h7,c1)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h3)
1      min_coverage(h5)
1      min_coverage(h6)
1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
1      shortage_cap(h0,c1)
1      shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1     shortage(h0,c2)
1      shortage_cap(h0,c2)
1      shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 3.00000002219792, +INF, 30)
-30   obj
-1     shortage(h1,c1)
1      shortage_cap(h1,c1)
1      shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well

```

Matrix [min, max] : [1.000E+00, 1.890E+02]

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General Algebraic Modeling System
Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS t t12

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97 4 projected
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.031 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t12

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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	Cplex	FROM LINE	173

**** SOLVER STATUS 1 Normal Completion
**** MODEL STATUS 8 Integer Solution
**** OBJECTIVE VALUE 184794.0000

RESOURCE USAGE, LIMIT 0.359 10000000000.000
ITERATION COUNT, LIMIT 86 2147483647
*** This solver runs with a demo license. No commercial use.
--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

--- MIP status (102): integer optimal, tolerance.
--- Cplex Time: 0.34sec (det. 15.23 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
 --- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances

MIP Solution: 184794.000000 (86 iterations, 0 nodes)
 Final Solve: 184794.000000 (0 iterations)

Best possible: 184787.000000
 Absolute gap: 7.000000
 Relative gap: 0.000038

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	1.0000	100.0000	.
h1	-INF	3.0000	100.0000	.
h2	-INF	34.0000	100.0000	.
h3	-INF	12.0000	100.0000	.
h4	-INF	27.0000	100.0000	.
h5	-INF	37.0000	100.0000	.
h6	-INF	14.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	63.0000	100.0000	.
h9	-INF	71.0000	100.0000	.
h10	-INF	78.0000	100.0000	.
h11	-INF	44.0000	100.0000	.
h12	-INF	91.0000	100.0000	.
h13	-INF	41.0000	100.0000	.
h14	-INF	44.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	61.0000	100.0000	.
h17	-INF	83.0000	100.0000	.
h18	-INF	100.0000	100.0000	.
h19	-INF	50.0000	100.0000	.
h20	-INF	35.0000	100.0000	.
h21	-INF	31.0000	100.0000	.

h22	-INF	6.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	192.0000	420.0000	.
h1	-INF	120.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	86.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	116.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	344.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	410.0000	420.0000	.
h14	-INF	388.0000	420.0000	.
h15	-INF	378.0000	420.0000	.
h16	-INF	418.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	418.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	316.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-122.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-56.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-40.0000	-30.0000	.
h5 .c2	-INF	-40.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-56.0000	-40.0000	.

h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-214.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-250.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-168.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-148.0000	-130.0000	.
h16.c1	-INF	-258.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	5.0000	10.0000	.
h1 .c2	-INF	10.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.

h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	.	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	4.0000	10.0000	.
h18.c2	-INF	6.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	2.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	4.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	43.0000	44.0000	.
c2	-INF	28.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	192.0000	+INF	.

h1	80.0000	120.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	86.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	116.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	344.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	410.0000	+INF	.
h14	80.0000	388.0000	+INF	.
h15	80.0000	378.0000	+INF	.
h16	80.0000	418.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	418.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	316.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	184794.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	1.0000	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	3.0000	+INF	189.0000
h2 .c1	.	20.0000	+INF	189.0000
h2 .c2	.	14.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	12.0000	+INF	180.0000
h4 .c1	.	4.0000	+INF	171.0000
h4 .c2	.	23.0000	+INF	171.0000
h5 .c1	.	35.0000	+INF	162.0000
h5 .c2	.	2.0000	+INF	162.0000
h6 .c1	.	.	+INF	153.0000
h6 .c2	.	14.0000	+INF	153.0000
h7 .c1	.	55.0000	+INF	144.0000

h7 .c2	.	45.0000	+INF	144.0000
h8 .c1	.	61.0000	+INF	135.0000
h8 .c2	.	2.0000	+INF	135.0000
h9 .c1	.	45.0000	+INF	126.0000
h9 .c2	.	26.0000	+INF	126.0000
h10.c1	.	35.0000	+INF	126.0000
h10.c2	.	43.0000	+INF	126.0000
h11.c1	.	44.0000	+INF	135.0000
h11.c2	.	.	+INF	135.0000
h12.c1	.	45.0000	+INF	144.0000
h12.c2	.	46.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	41.0000	+INF	153.0000
h14.c1	.	34.0000	+INF	162.0000
h14.c2	.	10.0000	+INF	162.0000
h15.c1	.	72.0000	+INF	171.0000
h15.c2	.	28.0000	+INF	171.0000
h16.c1	.	28.0000	+INF	180.0000
h16.c2	.	33.0000	+INF	180.0000
h17.c1	.	61.0000	+INF	189.0000
h17.c2	.	22.0000	+INF	189.0000
h18.c1	.	58.0000	+INF	189.0000
h18.c2	.	42.0000	+INF	189.0000
h19.c1	.	25.0000	+INF	189.0000
h19.c2	.	25.0000	+INF	189.0000
h20.c1	.	33.0000	+INF	189.0000
h20.c2	.	2.0000	+INF	189.0000
h21.c1	.	31.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	5.0000	+INF	189.0000
h22.c2	.	1.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	5.0000	+INF	30.0000
h1 .c2	.	10.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000

h6 .c1	.	.	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	4.0000	+INF	30.0000
h18.c2	.	6.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	2.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	4.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

GAMS 52.5.0 9bc771bf Jan 28, 2026 WEX-WEI x86 64bit/MS Windows - 02/15/26
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General Algebraic Modeling System
 Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)

$- 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -$
 $162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -$
 $144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -$
 $126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -$
 $144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -$
 $171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -$
 $189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -$
 $189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -$
 $189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -$
 $30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -$
 $30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -$
 $60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -$
 $60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -$
 $45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -$
 $60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -$
 $45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -$
 $30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -$
 $45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = 0)$

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 1)$

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 3)$

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 34)$

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;$
 (LHS = 192)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS$
 = 120)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS$
 = 110)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) = -120$; (LHS = -122)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) = -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) = -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= $L \leq \text{maxShort}$

shortage_cap(h0,c1).. $L(h0,c1) = 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) = 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) = 10$; (LHS = 5)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= $\text{sum}_h L(h,j) \leq \text{totShort}(j)$

shortage_total(c1).. $L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) + L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) + L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) + L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) = 48$; (LHS = 43)

shortage_total(c2).. $L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) + L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) + L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) + L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) = 30$; (LHS = 28)

---- min_coverage =G= minimum coverage in each window across companies $\geq \text{minCov}$

min_coverage(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) = 80$; (LHS = 192)

min_coverage(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) = 80$; (LHS = 120)

min_coverage(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) = 80$; (LHS = 110)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

z

1 (.LO, .L, .UP, .M = -INF, 184794, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)

(.LO, .L, .UP, .M = 0, 1, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

x(h0,c2)

(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)


```

1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1  cap_per_hour(h1)
      1  cap_total(h1)
      1  cap_total(h2)
      1  cap_total(h3)
      1  cap_total(h5)
      1  cap_total(h6)
      1  cap_total(h7)
     -1  shortage(h1,c1)
     -1  shortage(h2,c1)
     -1  shortage(h3,c1)
     -1  shortage(h5,c1)
     -1  shortage(h6,c1)
     -1  shortage(h7,c1)
      1  min_coverage(h1)
      1  min_coverage(h2)
      1  min_coverage(h3)
      1  min_coverage(h5)
      1  min_coverage(h6)
      1  min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30  obj
     -1  shortage(h0,c1)
      1  shortage_cap(h0,c1)
      1  shortage_total(c1)

```

L(h0,c2)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 45)

```

```

-45      obj
-1       shortage(h0,c2)
1        shortage_cap(h0,c2)
1        shortage_total(c2)

```

```

L(h1,c1)
        (.LO, .L, .UP, .M = 0, 5, +INF, 30)
-30     obj
-1      shortage(h1,c1)
1       shortage_cap(h1,c1)
1       shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System
Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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General Algebraic Modeling System
Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS t t13

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.015 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t13

GAMS 52.5.0 9bc771bf Jan 28, 2026 WEX-WEI x86 64bit/MS Windows - 02/15/26
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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL Telecom_MIP OBJECTIVE z
 TYPE MIP DIRECTION MINIMIZE
 SOLVER CPLEX FROM LINE 173

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 8 Integer Solution
 **** OBJECTIVE VALUE 184749.0000

RESOURCE USAGE, LIMIT 1.312 10000000000.000
 ITERATION COUNT, LIMIT 12061 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (102): integer optimal, tolerance.
 --- Cplex Time: 1.31sec (det. 152.84 ticks)
 --- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...
 --- Fixed MIP status (1): optimal.
 --- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances
 MIP Solution: 184749.000000 (12061 iterations, 4520 nodes)
 Final Solve: 184749.000000 (0 iterations)
 Best possible: 184733.188776
 Absolute gap: 15.811224
 Relative gap: 0.000086

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

LOWER	LEVEL	UPPER	MARGINAL
-------	-------	-------	----------

h0	-INF	4.0000	100.0000	.
h1	-INF	3.0000	100.0000	.
h2	-INF	19.0000	100.0000	.
h3	-INF	12.0000	100.0000	.
h4	-INF	23.0000	100.0000	.
h5	-INF	38.0000	100.0000	.
h6	-INF	28.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	68.0000	100.0000	.
h9	-INF	69.0000	100.0000	.
h10	-INF	64.0000	100.0000	.
h11	-INF	44.0000	100.0000	.
h12	-INF	86.0000	100.0000	.
h13	-INF	41.0000	100.0000	.
h14	-INF	59.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	66.0000	100.0000	.
h17	-INF	83.0000	100.0000	.
h18	-INF	85.0000	100.0000	.
h19	-INF	49.0000	100.0000	.
h20	-INF	31.0000	100.0000	.
h21	-INF	31.0000	100.0000	.
h22	-INF	22.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	191.0000	420.0000	.
h1	-INF	118.0000	420.0000	.
h2	-INF	110.0000	420.0000	.
h3	-INF	87.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	115.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	343.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	408.0000	420.0000	.
h14	-INF	387.0000	420.0000	.
h15	-INF	377.0000	420.0000	.
h16	-INF	419.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	417.0000	420.0000	.

h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	318.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-121.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-57.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-39.0000	-30.0000	.
h5 .c2	-INF	-41.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-57.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-213.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-248.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-167.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-147.0000	-130.0000	.
h16.c1	-INF	-259.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.

h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	8.0000	10.0000	.
h1 .c2	-INF	9.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	.	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	2.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	4.0000	10.0000	.
h18.c2	-INF	6.0000	10.0000	.

h19.c1	-INF	.	10.0000	.
h19.c2	-INF	3.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	2.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	46.0000	48.0000	.
c2	-INF	28.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	191.0000	+INF	.
h1	80.0000	118.0000	+INF	.
h2	80.0000	110.0000	+INF	.
h3	80.0000	87.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	115.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	343.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	408.0000	+INF	.
h14	80.0000	387.0000	+INF	.
h15	80.0000	377.0000	+INF	.
h16	80.0000	419.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	417.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	318.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	184749.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	2.0000	+INF	189.0000
h0 .c2	.	2.0000	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	3.0000	+INF	189.0000
h2 .c1	.	19.0000	+INF	189.0000
h2 .c2	.	.	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	12.0000	+INF	180.0000
h4 .c1	.	2.0000	+INF	171.0000
h4 .c2	.	21.0000	+INF	171.0000
h5 .c1	.	35.0000	+INF	162.0000
h5 .c2	.	3.0000	+INF	162.0000
h6 .c1	.	.	+INF	153.0000
h6 .c2	.	28.0000	+INF	153.0000
h7 .c1	.	56.0000	+INF	144.0000
h7 .c2	.	44.0000	+INF	144.0000
h8 .c1	.	63.0000	+INF	135.0000
h8 .c2	.	5.0000	+INF	135.0000
h9 .c1	.	44.0000	+INF	126.0000
h9 .c2	.	25.0000	+INF	126.0000
h10.c1	.	36.0000	+INF	126.0000
h10.c2	.	28.0000	+INF	126.0000
h11.c1	.	42.0000	+INF	135.0000
h11.c2	.	2.0000	+INF	135.0000
h12.c1	.	43.0000	+INF	144.0000
h12.c2	.	43.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	41.0000	+INF	153.0000
h14.c1	.	34.0000	+INF	162.0000
h14.c2	.	25.0000	+INF	162.0000
h15.c1	.	74.0000	+INF	171.0000
h15.c2	.	26.0000	+INF	171.0000
h16.c1	.	30.0000	+INF	180.0000
h16.c2	.	36.0000	+INF	180.0000
h17.c1	.	61.0000	+INF	189.0000
h17.c2	.	22.0000	+INF	189.0000
h18.c1	.	58.0000	+INF	189.0000
h18.c2	.	27.0000	+INF	189.0000
h19.c1	.	23.0000	+INF	189.0000
h19.c2	.	26.0000	+INF	189.0000
h20.c1	.	31.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000

h21.c1	.	31.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	7.0000	+INF	189.0000
h22.c2	.	15.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	8.0000	+INF	30.0000
h1 .c2	.	9.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	.	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	2.0000	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	4.0000	+INF	30.0000
h18.c2	.	6.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000

h19.c2	.	3.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	2.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = 5.82076609134674E-11,
 INFES = 5.82076609134674E-11 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
 capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 4)

cap_per_hour(h1).. x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 2.99999999998114)

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) = 100$; (LHS = 19)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) = 420$;
(LHS = 191.000000000025)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) = 420$; (LHS = 117.999999999981)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) = 420$; (LHS = 110.000000000006)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) = -120$; (LHS = -121.000000000019)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) = -70$; (LHS = -70.000000000063)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) = -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. $L(h0,c1) = 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) = 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) = 10$; (LHS = 8)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

```

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 52 ; (LHS =
45.9999999999623)

```

```

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS =
28.000000000044)

```

```

---- min_coverage =G= minimum coverage in each window across companies >= minCov

```

```

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 191.000000000025)

```

```

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 117.999999999981)

```

```

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 110.000000000006)

```

```

REMAINING 21 ENTRIES SKIPPED

```

```

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```

```

General Algebraic Modeling System
Column Listing SOLVE Telecom_MIP Using MIP From line 173

```

```

---- z total cost (objective)

```

```

z
1      (.L0, .L, .UP, .M = -INF, 184749.000000009, +INF, 0)
      obj

```

```

---- x operators starting at hour for company

```

```

x(h0,c1)
-189      (.L0, .L, .UP, .M = 0, 2, +INF, 189)
      obj
1      cap_per_hour(h0)
1      cap_total(h0)
1      cap_total(h1)

```

```

1      cap_total(h2)
1      cap_total(h4)
1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c1)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h4,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h0,c2)

```

      (.LO, .L, .UP, .M = 0, 2, +INF, 189)
-189  obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

```

x(h1,c1)

```

      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)

```

```

-1      shortage(h1,c1)
-1      shortage(h2,c1)
-1      shortage(h3,c1)
-1      shortage(h5,c1)
-1      shortage(h6,c1)
-1      shortage(h7,c1)
 1      min_coverage(h1)
 1      min_coverage(h2)
 1      min_coverage(h3)
 1      min_coverage(h5)
 1      min_coverage(h6)
 1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1    shortage(h0,c1)
 1    shortage_cap(h0,c1)
 1    shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1    shortage(h0,c2)
 1    shortage_cap(h0,c2)
 1    shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 8, +INF, 30)
-30   obj
-1    shortage(h1,c1)
 1    shortage_cap(h1,c1)
 1    shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [           NA,           NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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 General Algebraic Modeling System
 Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS t t14

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.016 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
 L O O P S t t14

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 General Algebraic Modeling System
 Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	173

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 8 Integer Solution
 **** OBJECTIVE VALUE 184707.0000

RESOURCE USAGE, LIMIT 1.656 10000000000.000
 ITERATION COUNT, LIMIT 14147 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (102): integer optimal, tolerance.
 --- Cplex Time: 1.66sec (det. 222.26 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).

--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.

--- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances

MIP Solution: 184707.000000 (14147 iterations, 6403 nodes)

Final Solve: 184707.000000 (0 iterations)

Best possible: 184690.499996

Absolute gap: 16.500004

Relative gap: 0.000089

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	2.0000	100.0000	.
h1	-INF	4.0000	100.0000	.
h2	-INF	19.0000	100.0000	.
h3	-INF	13.0000	100.0000	.
h4	-INF	25.0000	100.0000	.
h5	-INF	36.0000	100.0000	.
h6	-INF	28.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	65.0000	100.0000	.
h9	-INF	71.0000	100.0000	.
h10	-INF	65.0000	100.0000	.
h11	-INF	43.0000	100.0000	.
h12	-INF	89.0000	100.0000	.
h13	-INF	40.0000	100.0000	.
h14	-INF	56.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	65.0000	100.0000	.
h17	-INF	83.0000	100.0000	.
h18	-INF	87.0000	100.0000	.
h19	-INF	49.0000	100.0000	.
h20	-INF	33.0000	100.0000	.
h21	-INF	30.0000	100.0000	.
h22	-INF	21.0000	100.0000	.

h23	-INF	.	100.0000	.
-----	------	---	----------	---

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	192.0000	420.0000	.
h1	-INF	118.0000	420.0000	.
h2	-INF	109.0000	420.0000	.
h3	-INF	87.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	114.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	343.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	408.0000	420.0000	.
h14	-INF	386.0000	420.0000	.
h15	-INF	375.0000	420.0000	.
h16	-INF	418.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	415.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	319.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-122.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-57.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-37.0000	-30.0000	.
h5 .c2	-INF	-43.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-58.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.

h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-213.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-248.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-166.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-145.0000	-130.0000	.
h16.c1	-INF	-258.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	10.0000	10.0000	.
h1 .c2	-INF	7.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	1.0000	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.

h6 .c1	-INF	4.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	5.0000	10.0000	.
h18.c2	-INF	5.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	5.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	1.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	50.0000	52.0000	.
c2	-INF	28.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	192.0000	+INF	.
h1	80.0000	118.0000	+INF	.

h2	80.0000	109.0000	+INF	.
h3	80.0000	87.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	114.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	343.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	408.0000	+INF	.
h14	80.0000	386.0000	+INF	.
h15	80.0000	375.0000	+INF	.
h16	80.0000	418.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	415.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	319.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	184707.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	2.0000	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	4.0000	+INF	189.0000
h2 .c1	.	19.0000	+INF	189.0000
h2 .c2	.	.	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	13.0000	+INF	180.0000
h4 .c1	.	3.0000	+INF	171.0000
h4 .c2	.	22.0000	+INF	171.0000
h5 .c1	.	34.0000	+INF	162.0000
h5 .c2	.	2.0000	+INF	162.0000
h6 .c1	.	.	+INF	153.0000
h6 .c2	.	28.0000	+INF	153.0000
h7 .c1	.	57.0000	+INF	144.0000
h7 .c2	.	43.0000	+INF	144.0000

h8 .c1	.	61.0000	+INF	135.0000
h8 .c2	.	4.0000	+INF	135.0000
h9 .c1	.	45.0000	+INF	126.0000
h9 .c2	.	26.0000	+INF	126.0000
h10.c1	.	37.0000	+INF	126.0000
h10.c2	.	28.0000	+INF	126.0000
h11.c1	.	40.0000	+INF	135.0000
h11.c2	.	3.0000	+INF	135.0000
h12.c1	.	45.0000	+INF	144.0000
h12.c2	.	44.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	40.0000	+INF	153.0000
h14.c1	.	32.0000	+INF	162.0000
h14.c2	.	24.0000	+INF	162.0000
h15.c1	.	76.0000	+INF	171.0000
h15.c2	.	24.0000	+INF	171.0000
h16.c1	.	28.0000	+INF	180.0000
h16.c2	.	37.0000	+INF	180.0000
h17.c1	.	61.0000	+INF	189.0000
h17.c2	.	22.0000	+INF	189.0000
h18.c1	.	59.0000	+INF	189.0000
h18.c2	.	28.0000	+INF	189.0000
h19.c1	.	22.0000	+INF	189.0000
h19.c2	.	27.0000	+INF	189.0000
h20.c1	.	33.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000
h21.c1	.	30.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	8.0000	+INF	189.0000
h22.c2	.	13.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	10.0000	+INF	30.0000
h1 .c2	.	7.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	1.0000	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	4.0000	+INF	30.0000

h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	5.0000	+INF	30.0000
h18.c2	.	5.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	5.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	1.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -

$162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -$
 $144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -$
 $126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -$
 $144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -$
 $171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -$
 $189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -$
 $189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -$
 $189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -$
 $30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -$
 $30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -$
 $60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -$
 $60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -$
 $45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -$
 $60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -$
 $45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -$
 $30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -$
 $45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = 0)$

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 2)$

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 4)$

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 19)$

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;$
 (LHS = 192)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS$
 = 118)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS$
 = 109)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. - x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) -
x(h23,c1) - L(h0,c1) =L= -120 ; (LHS = -122)

shortage(h0,c2).. - x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) -
x(h23,c2) - L(h0,c2) =L= -70 ; (LHS = -70)

shortage(h1,c1).. - x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) -
x(h23,c1) - L(h1,c1) =L= -95 ; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. L(h0,c1) =L= 10 ; (LHS = 0)

shortage_cap(h0,c2).. L(h0,c2) =L= 10 ; (LHS = 0)

shortage_cap(h1,c1).. L(h1,c1) =L= 10 ; (LHS = 10)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 56 ; (LHS = 50)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS = 28)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 192)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 118)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 109)

REMAINING 21 ENTRIES SKIPPED

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Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

z

1 (.LO, .L, .UP, .M = -INF, 184707, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)

(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

x(h0,c2)

(.LO, .L, .UP, .M = 0, 2, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)

```

-1 shortage(h0,c2)
-1 shortage(h1,c2)
-1 shortage(h2,c2)
-1 shortage(h4,c2)
-1 shortage(h5,c2)
-1 shortage(h6,c2)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

```

x(h1,c1)

```

(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h1)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h3)
1 cap_total(h5)
1 cap_total(h6)
1 cap_total(h7)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h3,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
-1 shortage(h7,c1)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h3)
1 min_coverage(h5)
1 min_coverage(h6)
1 min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

(.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30 obj
-1 shortage(h0,c1)
1 shortage_cap(h0,c1)
1 shortage_total(c1)

```

L(h0,c2)

```

(.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45 obj

```

-1 shortage(h0,c2)
1 shortage_cap(h0,c2)
1 shortage_total(c2)

L(h1,c1)

(.LO, .L, .UP, .M = 0, 10, +INF, 30)
-30 obj
-1 shortage(h1,c1)
1 shortage_cap(h1,c1)
1 shortage_total(c1)

REMAINING 45 ENTRIES SKIPPED

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Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

RHS [min, max] : [1.000E+01, 4.200E+02] - Zero values observed as well
Bound [min, max] : [NA, NA] - Zero values observed as well
Matrix [min, max] : [1.000E+00, 1.890E+02]

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General Algebraic Modeling System

Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS

t t15

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.000 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI

L O O P S t t15

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General Algebraic Modeling System

Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	173

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 8 Integer Solution
 **** OBJECTIVE VALUE 184680.0000

RESOURCE USAGE, LIMIT 0.203 10000000000.000
 ITERATION COUNT, LIMIT 174 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (102): integer optimal, tolerance.
 --- Cplex Time: 0.20sec (det. 23.36 ticks)
 --- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
 --- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances
 MIP Solution: 184680.000000 (174 iterations, 0 nodes)
 Final Solve: 184680.000000 (0 iterations)
 Best possible: 184666.424370
 Absolute gap: 13.575630
 Relative gap: 0.000074

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

LOWER	LEVEL	UPPER	MARGINAL
-------	-------	-------	----------

h0	-INF	1.0000	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	30.0000	100.0000	.
h3	-INF	15.0000	100.0000	.
h4	-INF	26.0000	100.0000	.
h5	-INF	38.0000	100.0000	.
h6	-INF	17.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	63.0000	100.0000	.
h9	-INF	68.0000	100.0000	.
h10	-INF	78.0000	100.0000	.
h11	-INF	41.0000	100.0000	.
h12	-INF	91.0000	100.0000	.
h13	-INF	47.0000	100.0000	.
h14	-INF	40.0000	100.0000	.
h15	-INF	99.0000	100.0000	.
h16	-INF	67.0000	100.0000	.
h17	-INF	75.0000	100.0000	.
h18	-INF	100.0000	100.0000	.
h19	-INF	52.0000	100.0000	.
h20	-INF	32.0000	100.0000	.
h21	-INF	35.0000	100.0000	.
h22	-INF	8.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	193.0000	420.0000	.
h1	-INF	120.0000	420.0000	.
h2	-INF	106.0000	420.0000	.
h3	-INF	88.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	112.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	251.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	342.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	410.0000	420.0000	.
h14	-INF	387.0000	420.0000	.
h15	-INF	373.0000	420.0000	.
h16	-INF	416.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	413.0000	420.0000	.
h20	-INF	390.0000	420.0000	.

h21	-INF	360.0000	420.0000	.
h22	-INF	317.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-123.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-58.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-35.0000	-30.0000	.
h5 .c2	-INF	-45.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-57.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-141.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-212.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-250.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-167.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-143.0000	-130.0000	.
h16.c1	-INF	-256.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.

h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	10.0000	10.0000	.
h1 .c2	-INF	5.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	4.0000	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	5.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	8.0000	10.0000	.
h18.c2	-INF	2.0000	10.0000	.
h19.c1	-INF	.	10.0000	.

h19.c2	-INF	7.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	3.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	56.0000	56.0000	.
c2	-INF	28.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	193.0000	+INF	.
h1	80.0000	120.0000	+INF	.
h2	80.0000	106.0000	+INF	.
h3	80.0000	88.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	112.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	251.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	342.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	410.0000	+INF	.
h14	80.0000	387.0000	+INF	.
h15	80.0000	373.0000	+INF	.
h16	80.0000	416.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	413.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	317.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	184680.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	189.0000
h0 .c2	.	1.0000	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	20.0000	+INF	189.0000
h2 .c2	.	10.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	15.0000	+INF	180.0000
h4 .c1	.	2.0000	+INF	171.0000
h4 .c2	.	24.0000	+INF	171.0000
h5 .c1	.	33.0000	+INF	162.0000
h5 .c2	.	5.0000	+INF	162.0000
h6 .c1	.	.	+INF	153.0000
h6 .c2	.	17.0000	+INF	153.0000
h7 .c1	.	57.0000	+INF	144.0000
h7 .c2	.	43.0000	+INF	144.0000
h8 .c1	.	62.0000	+INF	135.0000
h8 .c2	.	1.0000	+INF	135.0000
h9 .c1	.	46.0000	+INF	126.0000
h9 .c2	.	22.0000	+INF	126.0000
h10.c1	.	37.0000	+INF	126.0000
h10.c2	.	41.0000	+INF	126.0000
h11.c1	.	39.0000	+INF	135.0000
h11.c2	.	2.0000	+INF	135.0000
h12.c1	.	45.0000	+INF	144.0000
h12.c2	.	46.0000	+INF	144.0000
h13.c1	.	1.0000	+INF	153.0000
h13.c2	.	46.0000	+INF	153.0000
h14.c1	.	29.0000	+INF	162.0000
h14.c2	.	11.0000	+INF	162.0000
h15.c1	.	78.0000	+INF	171.0000
h15.c2	.	21.0000	+INF	171.0000
h16.c1	.	28.0000	+INF	180.0000
h16.c2	.	39.0000	+INF	180.0000
h17.c1	.	59.0000	+INF	189.0000
h17.c2	.	16.0000	+INF	189.0000
h18.c1	.	60.0000	+INF	189.0000
h18.c2	.	40.0000	+INF	189.0000
h19.c1	.	23.0000	+INF	189.0000
h19.c2	.	29.0000	+INF	189.0000
h20.c1	.	32.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000
h21.c1	.	30.0000	+INF	189.0000

h21.c2	.	5.0000	+INF	189.0000
h22.c1	.	8.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	10.0000	+INF	30.0000
h1 .c2	.	5.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	4.0000	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	5.0000	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	8.0000	+INF	30.0000
h18.c2	.	2.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	7.0000	+INF	45.0000

h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	3.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

GAMS 52.5.0 9bc771bf Jan 28, 2026 WEX-WEI x86 64bit/MS Windows - 02/15/26

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = 0)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
 capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 1)

cap_per_hour(h1).. x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 0)

cap_per_hour(h2).. x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 30)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420$;
(LHS = 193)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420$; (LHS = 120)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420$; (LHS = 106)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) =L= -120$; (LHS = -123)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) =L= -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) =L= -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. $L(h0,c1) =L= 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) =L= 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) =L= 10$; (LHS = 10)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. $L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) + L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +$

$L(h_{12},c_1) + L(h_{13},c_1) + L(h_{14},c_1) + L(h_{15},c_1) + L(h_{16},c_1) + L(h_{17},c_1) + L(h_{18},c_1) + L(h_{19},c_1) + L(h_{20},c_1) + L(h_{21},c_1) + L(h_{22},c_1) + L(h_{23},c_1) =L= 60 ; (LHS = 56)$

shortage_total(c2).. $L(h_0,c_2) + L(h_1,c_2) + L(h_2,c_2) + L(h_3,c_2) + L(h_4,c_2) + L(h_5,c_2) + L(h_6,c_2) + L(h_7,c_2) + L(h_8,c_2) + L(h_9,c_2) + L(h_{10},c_2) + L(h_{11},c_2) + L(h_{12},c_2) + L(h_{13},c_2) + L(h_{14},c_2) + L(h_{15},c_2) + L(h_{16},c_2) + L(h_{17},c_2) + L(h_{18},c_2) + L(h_{19},c_2) + L(h_{20},c_2) + L(h_{21},c_2) + L(h_{22},c_2) + L(h_{23},c_2) =L= 30 ; (LHS = 28)$

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. $x(h_0,c_1) + x(h_0,c_2) + x(h_{18},c_1) + x(h_{18},c_2) + x(h_{19},c_1) + x(h_{19},c_2) + x(h_{20},c_1) + x(h_{20},c_2) + x(h_{22},c_1) + x(h_{22},c_2) + x(h_{23},c_1) + x(h_{23},c_2) =G= 80 ; (LHS = 193)$

min_coverage(h1).. $x(h_0,c_1) + x(h_0,c_2) + x(h_1,c_1) + x(h_1,c_2) + x(h_{19},c_1) + x(h_{19},c_2) + x(h_{20},c_1) + x(h_{20},c_2) + x(h_{21},c_1) + x(h_{21},c_2) + x(h_{23},c_1) + x(h_{23},c_2) =G= 80 ; (LHS = 120)$

min_coverage(h2).. $x(h_0,c_1) + x(h_0,c_2) + x(h_1,c_1) + x(h_1,c_2) + x(h_2,c_1) + x(h_2,c_2) + x(h_{20},c_1) + x(h_{20},c_2) + x(h_{21},c_1) + x(h_{21},c_2) + x(h_{22},c_1) + x(h_{22},c_2) =G= 80 ; (LHS = 106)$

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 184680, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
-189 (.LO, .L, .UP, .M = 0, 0, +INF, 189)
obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)

```

-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

```

x(h0,c2)

```

(.LO, .L, .UP, .M = 0, 1, +INF, 189)
-189 obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c2)
-1 shortage(h1,c2)
-1 shortage(h2,c2)
-1 shortage(h4,c2)
-1 shortage(h5,c2)
-1 shortage(h6,c2)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

```

x(h1,c1)

```

(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h1)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h3)
1 cap_total(h5)
1 cap_total(h6)
1 cap_total(h7)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h3,c1)
-1 shortage(h5,c1)

```

```

-1      shortage(h6,c1)
-1      shortage(h7,c1)
 1      min_coverage(h1)
 1      min_coverage(h2)
 1      min_coverage(h3)
 1      min_coverage(h5)
 1      min_coverage(h6)
 1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
 1     shortage_cap(h0,c1)
 1     shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1     shortage(h0,c2)
 1     shortage_cap(h0,c2)
 1     shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 10, +INF, 30)
-30   obj
-1     shortage(h1,c1)
 1     shortage_cap(h1,c1)
 1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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General Algebraic Modeling System

Model Statistics SOLVE Telecom_MIP Using MIP From line 173

MODEL STATISTICS

```

GENERATION TIME      =      0.015 SECONDS      4 MB  52.5.0 9bc771bf WEX-WEI
      L O O P S      t t16

```

General Algebraic Modeling System
Solution Report SOLVE Telecom MIP Using MIP From line 173

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	173

RESOURCE USAGE, LIMIT	1.172 10000000000.000
-----------------------	-----------------------

```
*** This solver runs with a demo license. No commercial use.
```

--- GMO memory 0.53 Mb (peak 0.53 Mb)

```

--- Dictionary memory 0.00 Mb

```

```
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
```

```

--- Starting Cplex

```

```
--- MIP status (102): integer optimal, tolerance.
```

```
--- Cplex Time: 1.17sec (det. 161.64 ticks)
```

```
--- Returning a primal only solution to GAMS (marginals all set to NA).
```

```
--- Fixing integer variables and solving final LP...
```

```
--- Fixed MIP status (1): optimal.
```


--- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances

MIP Solution: 184662.000000 (12436 iterations, 5323 nodes)

Final Solve: 184662.000000 (0 iterations)

Best possible: 184644.257143

Absolute gap: 17.742857

Relative gap: 0.000096

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	6.0000	100.0000	.
h1	-INF	.	100.0000	.
h2	-INF	17.0000	100.0000	.
h3	-INF	15.0000	100.0000	.
h4	-INF	20.0000	100.0000	.
h5	-INF	39.0000	100.0000	.
h6	-INF	27.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	71.0000	100.0000	.
h9	-INF	65.0000	100.0000	.
h10	-INF	68.0000	100.0000	.
h11	-INF	40.0000	100.0000	.
h12	-INF	84.0000	100.0000	.
h13	-INF	48.0000	100.0000	.
h14	-INF	54.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	73.0000	100.0000	.
h17	-INF	75.0000	100.0000	.
h18	-INF	86.0000	100.0000	.
h19	-INF	52.0000	100.0000	.
h20	-INF	25.0000	100.0000	.
h21	-INF	35.0000	100.0000	.
h22	-INF	22.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	191.0000	420.0000	.
h1	-INF	118.0000	420.0000	.
h2	-INF	105.0000	420.0000	.
h3	-INF	89.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	109.0000	420.0000	.
h7	-INF	198.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	339.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	408.0000	420.0000	.
h14	-INF	390.0000	420.0000	.
h15	-INF	375.0000	420.0000	.
h16	-INF	419.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	415.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	316.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-121.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-59.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-37.0000	-30.0000	.
h5 .c2	-INF	-43.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-54.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.

h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-209.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-248.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-170.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-145.0000	-130.0000	.
h16.c1	-INF	-259.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	10.0000	10.0000	.
h1 .c2	-INF	7.0000	10.0000	.
h2 .c1	-INF	.	10.0000	.
h2 .c2	-INF	5.0000	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	5.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	2.0000	10.0000	.
h7 .c2	-INF	.	10.0000	.

h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	9.0000	10.0000	.
h18.c2	-INF	1.0000	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	5.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	4.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	60.0000	60.0000	.
c2	-INF	28.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	191.0000	+INF	.
h1	80.0000	118.0000	+INF	.
h2	80.0000	105.0000	+INF	.
h3	80.0000	89.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.

h6	80.0000	109.0000	+INF	.	
h7	80.0000	198.0000	+INF	.	
h8	80.0000	250.0000	+INF	.	
h9	80.0000	310.0000	+INF	.	
h10	80.0000	290.0000	+INF	.	
h11	80.0000	339.0000	+INF	.	
h12	80.0000	390.0000	+INF	.	
h13	80.0000	408.0000	+INF	.	
h14	80.0000	390.0000	+INF	.	
h15	80.0000	375.0000	+INF	.	
h16	80.0000	419.0000	+INF	.	
h17	80.0000	420.0000	+INF	.	
h18	80.0000	420.0000	+INF	.	
h19	80.0000	415.0000	+INF	.	
h20	80.0000	390.0000	+INF	.	
h21	80.0000	360.0000	+INF	.	
h22	80.0000	316.0000	+INF	.	
h23	80.0000	270.0000	+INF	.	
		LOWER	LEVEL	UPPER	MARGINAL
----	VAR z	-INF	184662.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	6.0000	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	.	+INF	189.0000
h2 .c1	.	17.0000	+INF	189.0000
h2 .c2	.	.	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	15.0000	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	20.0000	+INF	171.0000
h5 .c1	.	31.0000	+INF	162.0000
h5 .c2	.	8.0000	+INF	162.0000
h6 .c1	.	1.0000	+INF	153.0000
h6 .c2	.	26.0000	+INF	153.0000
h7 .c1	.	59.0000	+INF	144.0000
h7 .c2	.	41.0000	+INF	144.0000
h8 .c1	.	63.0000	+INF	135.0000
h8 .c2	.	8.0000	+INF	135.0000
h9 .c1	.	47.0000	+INF	126.0000
h9 .c2	.	18.0000	+INF	126.0000

h10.c1	.	38.0000	+INF	126.0000
h10.c2	.	30.0000	+INF	126.0000
h11.c1	.	33.0000	+INF	135.0000
h11.c2	.	7.0000	+INF	135.0000
h12.c1	.	46.0000	+INF	144.0000
h12.c2	.	38.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	48.0000	+INF	153.0000
h14.c1	.	26.0000	+INF	162.0000
h14.c2	.	28.0000	+INF	162.0000
h15.c1	.	86.0000	+INF	171.0000
h15.c2	.	14.0000	+INF	171.0000
h16.c1	.	30.0000	+INF	180.0000
h16.c2	.	43.0000	+INF	180.0000
h17.c1	.	55.0000	+INF	189.0000
h17.c2	.	20.0000	+INF	189.0000
h18.c1	.	64.0000	+INF	189.0000
h18.c2	.	22.0000	+INF	189.0000
h19.c1	.	19.0000	+INF	189.0000
h19.c2	.	33.0000	+INF	189.0000
h20.c1	.	25.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000
h21.c1	.	35.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	7.0000	+INF	189.0000
h22.c2	.	15.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	10.0000	+INF	30.0000
h1 .c2	.	7.0000	+INF	45.0000
h2 .c1	.	.	+INF	30.0000
h2 .c2	.	5.0000	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	5.0000	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	2.0000	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000

h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	9.0000	+INF	30.0000
h18.c2	.	1.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	5.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	4.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -

$171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -$
 $189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -$
 $189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -$
 $189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -$
 $30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -$
 $30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -$
 $60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -$
 $60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -$
 $45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -$
 $60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -$
 $45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -$
 $30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -$
 $45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -5.82076609134674E-11,$
 $INFES = 5.82076609134674E-11 ****)$

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 6)$

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 0)$

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 17)$

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;$
 $(LHS = 191.000000000035)$

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS$
 $= 118.000000000018)$

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS$
 $= 104.999999999965)$

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $- x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) -$
 $x(h23,c1) - L(h0,c1) =L= -120 ; (LHS = -121)$

shortage(h0,c2).. - x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) -
x(h23,c2) - L(h0,c2) =L= -70 ; (LHS = -70.000000000035)

shortage(h1,c1).. - x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) -
x(h23,c1) - L(h1,c1) =L= -95 ; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. L(h0,c1) =L= 10 ; (LHS = 0)

shortage_cap(h0,c2).. L(h0,c2) =L= 10 ; (LHS = 0)

shortage_cap(h1,c1).. L(h1,c1) =L= 10 ; (LHS = 10)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 64 ; (LHS = 60)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS =
28.0000000001576)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 191.000000000035)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 118.000000000018)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 104.999999999965)

REMAINING 21 ENTRIES SKIPPED

---- z total cost (objective)

```

z
      (.LO, .L, .UP, .M = -INF, 184662.000000009, +INF, 0)
1      obj

```

---- x operators starting at hour for company

```

x(h0,c1)
      (.LO, .L, .UP, .M = 0, 6, +INF, 189)
-189  obj
      1  cap_per_hour(h0)
      1  cap_total(h0)
      1  cap_total(h1)
      1  cap_total(h2)
      1  cap_total(h4)
      1  cap_total(h5)
      1  cap_total(h6)
      -1  shortage(h0,c1)
      -1  shortage(h1,c1)
      -1  shortage(h2,c1)
      -1  shortage(h4,c1)
      -1  shortage(h5,c1)
      -1  shortage(h6,c1)
      1  min_coverage(h0)
      1  min_coverage(h1)
      1  min_coverage(h2)
      1  min_coverage(h4)
      1  min_coverage(h5)
      1  min_coverage(h6)

```

```

x(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1  cap_per_hour(h0)
      1  cap_total(h0)
      1  cap_total(h1)
      1  cap_total(h2)
      1  cap_total(h4)
      1  cap_total(h5)
      1  cap_total(h6)
      -1  shortage(h0,c2)
      -1  shortage(h1,c2)

```

-1	shortage(h2,c2)
-1	shortage(h4,c2)
-1	shortage(h5,c2)
-1	shortage(h6,c2)
1	min_coverage(h0)
1	min_coverage(h1)
1	min_coverage(h2)
1	min_coverage(h4)
1	min_coverage(h5)
1	min_coverage(h6)

x(h1,c1)

	(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189	obj
1	cap_per_hour(h1)
1	cap_total(h1)
1	cap_total(h2)
1	cap_total(h3)
1	cap_total(h5)
1	cap_total(h6)
1	cap_total(h7)
-1	shortage(h1,c1)
-1	shortage(h2,c1)
-1	shortage(h3,c1)
-1	shortage(h5,c1)
-1	shortage(h6,c1)
-1	shortage(h7,c1)
1	min_coverage(h1)
1	min_coverage(h2)
1	min_coverage(h3)
1	min_coverage(h5)
1	min_coverage(h6)
1	min_coverage(h7)

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

	(.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30	obj
-1	shortage(h0,c1)
1	shortage_cap(h0,c1)
1	shortage_total(c1)

L(h0,c2)

	(.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45	obj
-1	shortage(h0,c2)
1	shortage_cap(h0,c2)

```

1 shortage_total(c2)

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 10, +INF, 30)
-30 obj
-1 shortage(h1,c1)
1 shortage_cap(h1,c1)
1 shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

GAMS 52.5.0 9bc771bf Jan 28, 2026 WEX-WEI x86 64bit/MS Windows - 02/15/26

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

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General Algebraic Modeling System

Model Statistics SOLVE Telecom_MIP Using MIP From line 173

```

LOOPS                                     t   t17

```

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

```

GENERATION TIME      =      0.016 SECONDS      4 MB  52.5.0 9bc771bf WEX-WEI
      L O O P S      t t17

```

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General Algebraic Modeling System

Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

```

MODEL  Telecom_MIP      OBJECTIVE  z

```

TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	173

**** SOLVER STATUS 1 Normal Completion
 **** MODEL STATUS 8 Integer Solution
 **** OBJECTIVE VALUE 184635.0000

RESOURCE USAGE, LIMIT 0.391 10000000000.000
 ITERATION COUNT, LIMIT 883 2147483647
 *** This solver runs with a demo license. No commercial use.
 --- GMO setup time: 0.00s
 --- GMO memory 0.53 Mb (peak 0.53 Mb)
 --- Dictionary memory 0.00 Mb
 --- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
 --- Starting Cplex

--- MIP status (102): integer optimal, tolerance.
 --- Cplex Time: 0.39sec (det. 25.51 ticks)
 --- Returning a primal only solution to GAMS (marginals all set to NA).
 --- Fixing integer variables and solving final LP...
 --- Fixed MIP status (1): optimal.
 --- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances
 MIP Solution: 184635.000000 (883 iterations, 330 nodes)
 Final Solve: 184635.000000 (0 iterations)
 Best possible: 184618.500000
 Absolute gap: 16.500000
 Relative gap: 0.000089

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	1.0000	100.0000	.
h1	-INF	4.0000	100.0000	.

h2	-INF	19.0000	100.0000	.
h3	-INF	17.0000	100.0000	.
h4	-INF	23.0000	100.0000	.
h5	-INF	35.0000	100.0000	.
h6	-INF	25.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	66.0000	100.0000	.
h9	-INF	69.0000	100.0000	.
h10	-INF	72.0000	100.0000	.
h11	-INF	38.0000	100.0000	.
h12	-INF	89.0000	100.0000	.
h13	-INF	45.0000	100.0000	.
h14	-INF	51.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	70.0000	100.0000	.
h17	-INF	78.0000	100.0000	.
h18	-INF	87.0000	100.0000	.
h19	-INF	54.0000	100.0000	.
h20	-INF	28.0000	100.0000	.
h21	-INF	31.0000	100.0000	.
h22	-INF	20.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	190.0000	420.0000	.
h1	-INF	118.0000	420.0000	.
h2	-INF	103.0000	420.0000	.
h3	-INF	91.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	107.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	339.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	407.0000	420.0000	.
h14	-INF	392.0000	420.0000	.
h15	-INF	375.0000	420.0000	.
h16	-INF	420.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	415.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	361.0000	420.0000	.
h22	-INF	314.0000	420.0000	.

h23	-INF	270.0000	420.0000	.
-----	------	----------	----------	---

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-120.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-61.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-38.0000	-30.0000	.
h5 .c2	-INF	-42.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-52.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-209.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-247.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-172.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-145.0000	-130.0000	.
h16.c1	-INF	-260.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-111.0000	-110.0000	.

h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	10.0000	10.0000	.
h1 .c2	-INF	7.0000	10.0000	.
h2 .c1	-INF	1.0000	10.0000	.
h2 .c2	-INF	6.0000	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	5.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	10.0000	10.0000	.
h18.c2	-INF	.	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	5.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.

h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	6.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	62.0000	64.0000	.
c2	-INF	28.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	190.0000	+INF	.
h1	80.0000	118.0000	+INF	.
h2	80.0000	103.0000	+INF	.
h3	80.0000	91.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	107.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	339.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	407.0000	+INF	.
h14	80.0000	392.0000	+INF	.
h15	80.0000	375.0000	+INF	.
h16	80.0000	420.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	415.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	361.0000	+INF	.
h22	80.0000	314.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	184635.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	1.0000	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	3.0000	+INF	189.0000
h1 .c2	.	1.0000	+INF	189.0000
h2 .c1	.	17.0000	+INF	189.0000
h2 .c2	.	2.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	17.0000	+INF	180.0000
h4 .c1	.	2.0000	+INF	171.0000
h4 .c2	.	21.0000	+INF	171.0000
h5 .c1	.	32.0000	+INF	162.0000
h5 .c2	.	3.0000	+INF	162.0000
h6 .c1	.	.	+INF	153.0000
h6 .c2	.	25.0000	+INF	153.0000
h7 .c1	.	58.0000	+INF	144.0000
h7 .c2	.	42.0000	+INF	144.0000
h8 .c1	.	63.0000	+INF	135.0000
h8 .c2	.	3.0000	+INF	135.0000
h9 .c1	.	45.0000	+INF	126.0000
h9 .c2	.	24.0000	+INF	126.0000
h10.c1	.	38.0000	+INF	126.0000
h10.c2	.	34.0000	+INF	126.0000
h11.c1	.	36.0000	+INF	135.0000
h11.c2	.	2.0000	+INF	135.0000
h12.c1	.	45.0000	+INF	144.0000
h12.c2	.	44.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	45.0000	+INF	153.0000
h14.c1	.	29.0000	+INF	162.0000
h14.c2	.	22.0000	+INF	162.0000
h15.c1	.	82.0000	+INF	171.0000
h15.c2	.	18.0000	+INF	171.0000
h16.c1	.	30.0000	+INF	180.0000
h16.c2	.	40.0000	+INF	180.0000
h17.c1	.	57.0000	+INF	189.0000
h17.c2	.	21.0000	+INF	189.0000
h18.c1	.	59.0000	+INF	189.0000
h18.c2	.	28.0000	+INF	189.0000
h19.c1	.	23.0000	+INF	189.0000
h19.c2	.	31.0000	+INF	189.0000
h20.c1	.	27.0000	+INF	189.0000
h20.c2	.	1.0000	+INF	189.0000
h21.c1	.	31.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	10.0000	+INF	189.0000

h22.c2	.	10.0000	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	10.0000	+INF	30.0000
h1 .c2	.	7.0000	+INF	45.0000
h2 .c1	.	1.0000	+INF	30.0000
h2 .c2	.	6.0000	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	5.0000	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	10.0000	+INF	30.0000
h18.c2	.	.	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	5.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000

h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	6.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -5.82076609134674E-11,
 INFES = 5.82076609134674E-11 ****)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
 capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 1)

cap_per_hour(h1).. x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 4.0000000000492)

cap_per_hour(h2).. x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 18.9999999999859)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;
(LHS = 190.000000000014)

cap_total(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS = 118.000000000063)

cap_total(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS = 103.000000000042)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. - x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) =L= -120 ; (LHS = -120.000000000042)

shortage(h0,c2).. - x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) =L= -70 ; (LHS = -69.999999999719, INFES = 2.81090706266696E-11 ****)

shortage(h1,c1).. - x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) =L= -95 ; (LHS = -95.0000000000211)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. L(h0,c1) =L= 10 ; (LHS = 0)

shortage_cap(h0,c2).. L(h0,c2) =L= 10 ; (LHS = 0)

shortage_cap(h1,c1).. L(h1,c1) =L= 10 ; (LHS = 10)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) + L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +

L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 68 ; (LHS =
61.999999999578)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS =
27.9999999998524)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 190.000000000014)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 118.000000000063)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 103.000000000042)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System

Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 184634.999999999, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
-189 (.LO, .L, .UP, .M = 0, 1, +INF, 189)
obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)

```

1      cap_total(h5)
1      cap_total(h6)
-1     shortage(h0,c1)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h4,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h0,c2)

```

      (.L0, .L, .UP, .M = 0, 0, +INF, 189)
-189  obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

```

x(h1,c1)

```

      (.L0, .L, .UP, .M = 0, 2.99999999997892, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)

```

```

-1      shortage(h3,c1)
-1      shortage(h5,c1)
-1      shortage(h6,c1)
-1      shortage(h7,c1)
 1      min_coverage(h1)
 1      min_coverage(h2)
 1      min_coverage(h3)
 1      min_coverage(h5)
 1      min_coverage(h6)
 1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
 1     shortage_cap(h0,c1)
 1     shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1     shortage(h0,c2)
 1     shortage_cap(h0,c2)
 1     shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 10, +INF, 30)
-30   obj
-1     shortage(h1,c1)
 1     shortage_cap(h1,c1)
 1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

```

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Range Statistics      SOLVE Telecom_MIP Using MIP From line 173

```

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

```

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```


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General Algebraic Modeling System

Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS t t18

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.000 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t18

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General Algebraic Modeling System

Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	173

**** SOLVER STATUS 1 Normal Completion
**** MODEL STATUS 8 Integer Solution
**** OBJECTIVE VALUE 184617.0000

RESOURCE USAGE, LIMIT 0.094 10000000000.000
ITERATION COUNT, LIMIT 85 2147483647

*** This solver runs with a demo license. No commercial use.

--- GMO setup time: 0.00s

--- GMO memory 0.53 Mb (peak 0.53 Mb)

--- Dictionary memory 0.00 Mb

--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)

--- Starting Cplex

--- MIP status (102): integer optimal, tolerance.

--- Cplex Time: 0.08sec (det. 10.16 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).

--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
 --- Cplex Time: 0.00sec (det. 0.14 ticks)

Solution satisfies tolerances

MIP Solution: 184617.000000 (85 iterations, 0 nodes)
 Final Solve: 184617.000000 (0 iterations)

Best possible: 184599.000000
 Absolute gap: 18.000000
 Relative gap: 0.000097

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

---- EQU cap_per_hour sum of operators starting at hour across companies <= capHour

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	2.0000	100.0000	.
h1	-INF	4.0000	100.0000	.
h2	-INF	29.0000	100.0000	.
h3	-INF	18.0000	100.0000	.
h4	-INF	24.0000	100.0000	.
h5	-INF	33.0000	100.0000	.
h6	-INF	16.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	63.0000	100.0000	.
h9	-INF	72.0000	100.0000	.
h10	-INF	82.0000	100.0000	.
h11	-INF	38.0000	100.0000	.
h12	-INF	91.0000	100.0000	.
h13	-INF	44.0000	100.0000	.
h14	-INF	38.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	70.0000	100.0000	.
h17	-INF	77.0000	100.0000	.
h18	-INF	100.0000	100.0000	.
h19	-INF	55.0000	100.0000	.
h20	-INF	27.0000	100.0000	.
h21	-INF	31.0000	100.0000	.
h22	-INF	7.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	191.0000	420.0000	.
h1	-INF	119.0000	420.0000	.
h2	-INF	100.0000	420.0000	.
h3	-INF	89.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	81.0000	420.0000	.
h6	-INF	108.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	341.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	408.0000	420.0000	.
h14	-INF	390.0000	420.0000	.
h15	-INF	374.0000	420.0000	.
h16	-INF	419.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	414.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	312.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-121.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-59.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-37.0000	-30.0000	.
h5 .c2	-INF	-44.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-54.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.

h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-211.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-248.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-170.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-144.0000	-130.0000	.
h16.c1	-INF	-259.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	10.0000	10.0000	.
h1 .c2	-INF	6.0000	10.0000	.
h2 .c1	-INF	4.0000	10.0000	.
h2 .c2	-INF	6.0000	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	6.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.

h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	10.0000	10.0000	.
h18.c2	-INF	.	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	6.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	8.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	68.0000	68.0000	.
c2	-INF	28.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	191.0000	+INF	.
h1	80.0000	119.0000	+INF	.
h2	80.0000	100.0000	+INF	.
h3	80.0000	89.0000	+INF	.

h4	80.0000	80.0000	+INF	.
h5	80.0000	81.0000	+INF	.
h6	80.0000	108.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	341.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	408.0000	+INF	.
h14	80.0000	390.0000	+INF	.
h15	80.0000	374.0000	+INF	.
h16	80.0000	419.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	414.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	312.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	184617.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	2.0000	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	4.0000	+INF	189.0000
h2 .c1	.	19.0000	+INF	189.0000
h2 .c2	.	10.0000	+INF	189.0000
h3 .c1	.	2.0000	+INF	180.0000
h3 .c2	.	16.0000	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	24.0000	+INF	171.0000
h5 .c1	.	33.0000	+INF	162.0000
h5 .c2	.	.	+INF	162.0000
h6 .c1	.	.	+INF	153.0000
h6 .c2	.	16.0000	+INF	153.0000
h7 .c1	.	56.0000	+INF	144.0000
h7 .c2	.	44.0000	+INF	144.0000
h8 .c1	.	63.0000	+INF	135.0000
h8 .c2	.	.	+INF	135.0000

h9 .c1	.	46.0000	+INF	126.0000
h9 .c2	.	26.0000	+INF	126.0000
h10.c1	.	38.0000	+INF	126.0000
h10.c2	.	44.0000	+INF	126.0000
h11.c1	.	38.0000	+INF	135.0000
h11.c2	.	-2.132955E-8	+INF	135.0000
h12.c1	.	45.0000	+INF	144.0000
h12.c2	.	46.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	44.0000	+INF	153.0000
h14.c1	.	28.0000	+INF	162.0000
h14.c2	.	10.0000	+INF	162.0000
h15.c1	.	80.0000	+INF	171.0000
h15.c2	.	20.0000	+INF	171.0000
h16.c1	.	30.0000	+INF	180.0000
h16.c2	.	40.0000	+INF	180.0000
h17.c1	.	57.0000	+INF	189.0000
h17.c2	.	20.0000	+INF	189.0000
h18.c1	.	60.0000	+INF	189.0000
h18.c2	.	40.0000	+INF	189.0000
h19.c1	.	25.0000	+INF	189.0000
h19.c2	.	30.0000	+INF	189.0000
h20.c1	.	27.0000	+INF	189.0000
h20.c2	.	-1.065248E-8	+INF	189.0000
h21.c1	.	31.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	7.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	10.0000	+INF	30.0000
h1 .c2	.	6.0000	+INF	45.0000
h2 .c1	.	4.0000	+INF	30.0000
h2 .c2	.	6.0000	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	6.0000	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000

h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	10.0000	+INF	30.0000
h18.c2	.	.	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	6.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	8.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -

$126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -$
 $144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -$
 $171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -$
 $189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -$
 $189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -$
 $189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -$
 $30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -$
 $30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -$
 $60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -$
 $60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -$
 $45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -$
 $60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -$
 $45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -$
 $30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -$
 $45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = -4.89281956106424E-6,$
 $INFES = 4.89281956106424E-6 ****)$

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 1.99999998578031)$

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 4)$

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 29.0000000106525)$

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;$
 $(LHS = 191.000000010652)$

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS$
 $= 119.000000003555)$

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS$
 $= 100.000000010652)$

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $- x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) -$

$x(h_{23},c_1) - L(h_0,c_1) = -120$; (LHS = -121)

$shortage(h_0,c_2) \dots - x(h_0,c_2) - x(h_{18},c_2) - x(h_{19},c_2) - x(h_{20},c_2) - x(h_{22},c_2) - x(h_{23},c_2) - L(h_0,c_2) = -70$; (LHS = -70.0000000106525)

$shortage(h_1,c_1) \dots - x(h_0,c_1) - x(h_1,c_1) - x(h_{19},c_1) - x(h_{20},c_1) - x(h_{21},c_1) - x(h_{23},c_1) - L(h_1,c_1) = -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

$shortage_cap(h_0,c_1) \dots L(h_0,c_1) = 10$; (LHS = 0)

$shortage_cap(h_0,c_2) \dots L(h_0,c_2) = 10$; (LHS = 0)

$shortage_cap(h_1,c_1) \dots L(h_1,c_1) = 10$; (LHS = 10)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

$shortage_total(c_1) \dots L(h_0,c_1) + L(h_1,c_1) + L(h_2,c_1) + L(h_3,c_1) + L(h_4,c_1) + L(h_5,c_1) + L(h_6,c_1) + L(h_7,c_1) + L(h_8,c_1) + L(h_9,c_1) + L(h_{10},c_1) + L(h_{11},c_1) + L(h_{12},c_1) + L(h_{13},c_1) + L(h_{14},c_1) + L(h_{15},c_1) + L(h_{16},c_1) + L(h_{17},c_1) + L(h_{18},c_1) + L(h_{19},c_1) + L(h_{20},c_1) + L(h_{21},c_1) + L(h_{22},c_1) + L(h_{23},c_1) = 72$; (LHS = 68)

$shortage_total(c_2) \dots L(h_0,c_2) + L(h_1,c_2) + L(h_2,c_2) + L(h_3,c_2) + L(h_4,c_2) + L(h_5,c_2) + L(h_6,c_2) + L(h_7,c_2) + L(h_8,c_2) + L(h_9,c_2) + L(h_{10},c_2) + L(h_{11},c_2) + L(h_{12},c_2) + L(h_{13},c_2) + L(h_{14},c_2) + L(h_{15},c_2) + L(h_{16},c_2) + L(h_{17},c_2) + L(h_{18},c_2) + L(h_{19},c_2) + L(h_{20},c_2) + L(h_{21},c_2) + L(h_{22},c_2) + L(h_{23},c_2) = 30$; (LHS = 27.999999999631)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

$min_coverage(h_0) \dots x(h_0,c_1) + x(h_0,c_2) + x(h_{18},c_1) + x(h_{18},c_2) + x(h_{19},c_1) + x(h_{19},c_2) + x(h_{20},c_1) + x(h_{20},c_2) + x(h_{22},c_1) + x(h_{22},c_2) + x(h_{23},c_1) + x(h_{23},c_2) = 80$; (LHS = 191.000000010652)

$min_coverage(h_1) \dots x(h_0,c_1) + x(h_0,c_2) + x(h_1,c_1) + x(h_1,c_2) + x(h_{19},c_1) + x(h_{19},c_2) + x(h_{20},c_1) + x(h_{20},c_2) + x(h_{21},c_1) + x(h_{21},c_2) + x(h_{23},c_1) + x(h_{23},c_2) = 80$; (LHS = 119.00000003555)

$min_coverage(h_2) \dots x(h_0,c_1) + x(h_0,c_2) + x(h_1,c_1) + x(h_1,c_2) + x(h_2,c_1) + x(h_2,c_2) + x(h_{20},c_1) + x(h_{20},c_2) + x(h_{21},c_1) + x(h_{21},c_2) + x(h_{22},c_1) + x(h_{22},c_2) = 80$; (LHS = 100.000000010652)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System
Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 184617.000000064, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
-189 (.LO, .L, .UP, .M = 0, 1.9999999857803, +INF, 189)
obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

x(h0,c2)
-189 (.LO, .L, .UP, .M = 0, 0, +INF, 189)
obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)

```

-1 shortage(h0,c2)
-1 shortage(h1,c2)
-1 shortage(h2,c2)
-1 shortage(h4,c2)
-1 shortage(h5,c2)
-1 shortage(h6,c2)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

```

x(h1,c1)

```

(.LO, .L, .UP, .M = 0, 0, +INF, 189)
-189 obj
1 cap_per_hour(h1)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h3)
1 cap_total(h5)
1 cap_total(h6)
1 cap_total(h7)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h3,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
-1 shortage(h7,c1)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h3)
1 min_coverage(h5)
1 min_coverage(h6)
1 min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

(.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30 obj
-1 shortage(h0,c1)
1 shortage_cap(h0,c1)
1 shortage_total(c1)

```

L(h0,c2)

```

(.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45 obj

```

```

-1 shortage(h0,c2)
1 shortage_cap(h0,c2)
1 shortage_total(c2)

```

L(h1,c1)

```

(.LO, .L, .UP, .M = 0, 10, +INF, 30)
-30 obj
-1 shortage(h1,c1)
1 shortage_cap(h1,c1)
1 shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix    [min, max] : [ 1.000E+00, 1.890E+02]

```

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General Algebraic Modeling System

Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS

t t19

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97 2 projected
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

```

GENERATION TIME      =      0.016 SECONDS      4 MB  52.5.0 9bc771bf WEX-WEI
      L O O P S      t t19

```

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General Algebraic Modeling System

Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	173

```

**** SOLVER STATUS      1 Normal Completion
**** MODEL STATUS      8 Integer Solution
**** OBJECTIVE VALUE    184599.0000

```

```

RESOURCE USAGE, LIMIT      0.312 10000000000.000
ITERATION COUNT, LIMIT    2582  2147483647
*** This solver runs with a demo license. No commercial use.
--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

```

```

--- MIP status (102): integer optimal, tolerance.
--- Cplex Time: 0.31sec (det. 48.79 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
--- Fixing integer variables and solving final LP...

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.00sec (det. 0.14 ticks)

```

```

Solution satisfies tolerances
MIP Solution:      184599.000000    (2582 iterations, 883 nodes)
Final Solve:      184599.000000    (0 iterations)

Best possible:      184581.000000
Absolute gap:       18.000000
Relative gap:       0.000098

```

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

```

---- EQU cap_per_hour  sum of operators starting at hour across companies <= capHour

          LOWER          LEVEL          UPPER          MARGINAL

```

h0	-INF	6.0000	100.0000	.
h1	-INF	3.0000	100.0000	.
h2	-INF	27.0000	100.0000	.
h3	-INF	17.0000	100.0000	.
h4	-INF	22.0000	100.0000	.
h5	-INF	32.0000	100.0000	.
h6	-INF	19.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	65.0000	100.0000	.
h9	-INF	74.0000	100.0000	.
h10	-INF	78.0000	100.0000	.
h11	-INF	39.0000	100.0000	.
h12	-INF	89.0000	100.0000	.
h13	-INF	42.0000	100.0000	.
h14	-INF	41.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	72.0000	100.0000	.
h17	-INF	78.0000	100.0000	.
h18	-INF	98.0000	100.0000	.
h19	-INF	55.0000	100.0000	.
h20	-INF	24.0000	100.0000	.
h21	-INF	31.0000	100.0000	.
h22	-INF	8.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	191.0000	420.0000	.
h1	-INF	119.0000	420.0000	.
h2	-INF	99.0000	420.0000	.
h3	-INF	86.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	109.0000	420.0000	.
h7	-INF	198.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	342.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	409.0000	420.0000	.
h14	-INF	389.0000	420.0000	.
h15	-INF	374.0000	420.0000	.
h16	-INF	419.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	414.0000	420.0000	.
h20	-INF	390.0000	420.0000	.

h21	-INF	360.0000	420.0000	.
h22	-INF	311.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-121.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-56.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-36.0000	-30.0000	.
h5 .c2	-INF	-44.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-55.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-212.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-249.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-169.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-144.0000	-130.0000	.
h16.c1	-INF	-259.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.

h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	10.0000	10.0000	.
h1 .c2	-INF	6.0000	10.0000	.
h2 .c1	-INF	6.0000	10.0000	.
h2 .c2	-INF	5.0000	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	6.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	2.0000	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	9.0000	10.0000	.
h18.c2	-INF	1.0000	10.0000	.
h19.c1	-INF	.	10.0000	.

h19.c2	-INF	6.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	9.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	72.0000	72.0000	.
c2	-INF	28.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	191.0000	+INF	.
h1	80.0000	119.0000	+INF	.
h2	80.0000	99.0000	+INF	.
h3	80.0000	86.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	109.0000	+INF	.
h7	80.0000	198.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	342.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	409.0000	+INF	.
h14	80.0000	389.0000	+INF	.
h15	80.0000	374.0000	+INF	.
h16	80.0000	419.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	414.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	311.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	184599.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	4.0000	+INF	189.0000
h0 .c2	.	2.0000	+INF	189.0000
h1 .c1	.	1.0000	+INF	189.0000
h1 .c2	.	2.0000	+INF	189.0000
h2 .c1	.	16.0000	+INF	189.0000
h2 .c2	.	11.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	17.0000	+INF	180.0000
h4 .c1	.	2.0000	+INF	171.0000
h4 .c2	.	20.0000	+INF	171.0000
h5 .c1	.	29.0000	+INF	162.0000
h5 .c2	.	3.0000	+INF	162.0000
h6 .c1	.	2.0000	+INF	153.0000
h6 .c2	.	17.0000	+INF	153.0000
h7 .c1	.	60.0000	+INF	144.0000
h7 .c2	.	40.0000	+INF	144.0000
h8 .c1	.	60.0000	+INF	135.0000
h8 .c2	.	5.0000	+INF	135.0000
h9 .c1	.	49.0000	+INF	126.0000
h9 .c2	.	25.0000	+INF	126.0000
h10.c1	.	38.0000	+INF	126.0000
h10.c2	.	40.0000	+INF	126.0000
h11.c1	.	34.0000	+INF	135.0000
h11.c2	.	5.0000	+INF	135.0000
h12.c1	.	46.0000	+INF	144.0000
h12.c2	.	43.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	42.0000	+INF	153.0000
h14.c1	.	27.0000	+INF	162.0000
h14.c2	.	14.0000	+INF	162.0000
h15.c1	.	82.0000	+INF	171.0000
h15.c2	.	18.0000	+INF	171.0000
h16.c1	.	32.0000	+INF	180.0000
h16.c2	.	40.0000	+INF	180.0000
h17.c1	.	56.0000	+INF	189.0000
h17.c2	.	22.0000	+INF	189.0000
h18.c1	.	60.0000	+INF	189.0000
h18.c2	.	38.0000	+INF	189.0000
h19.c1	.	25.0000	+INF	189.0000
h19.c2	.	30.0000	+INF	189.0000
h20.c1	.	24.0000	+INF	189.0000
h20.c2	.	.	+INF	189.0000
h21.c1	.	31.0000	+INF	189.0000

h21.c2	.	.	+INF	189.0000
h22.c1	.	8.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	10.0000	+INF	30.0000
h1 .c2	.	6.0000	+INF	45.0000
h2 .c1	.	6.0000	+INF	30.0000
h2 .c2	.	5.0000	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	6.0000	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	2.0000	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	9.0000	+INF	30.0000
h18.c2	.	1.0000	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	6.0000	+INF	45.0000

h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	9.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1) -
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -
 144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -
 171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -
 189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -
 189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -
 189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -
 30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -
 30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -
 60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -
 60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -
 45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -
 60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -
 45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -
 30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -
 45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = 0)

---- cap_per_hour =L= sum of operators starting at hour across companies <=
 capHour

cap_per_hour(h0).. x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 5.99999999991127)

cap_per_hour(h1).. x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 3.00000000008872)

cap_per_hour(h2).. x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 27.0000000000444)

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420$;
(LHS = 191)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420$; (LHS = 119)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420$; (LHS = 99.0000000000443)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $-x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) - x(h23,c1) - L(h0,c1) =L= -120$; (LHS = -121)

shortage(h0,c2).. $-x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) - x(h23,c2) - L(h0,c2) =L= -70$; (LHS = -70)

shortage(h1,c1).. $-x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) - x(h23,c1) - L(h1,c1) =L= -95$; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. $L(h0,c1) =L= 10$; (LHS = 0)

shortage_cap(h0,c2).. $L(h0,c2) =L= 10$; (LHS = 0)

shortage_cap(h1,c1).. $L(h1,c1) =L= 10$; (LHS = 10)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. $L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) + L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +$

L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 76 ; (LHS = 72)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS =
28.0000000001331)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 191)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 119)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 99.0000000000444)

REMAINING 21 ENTRIES SKIPPED

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Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 184599.000000015, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
-189 (.LO, .L, .UP, .M = 0, 4, +INF, 189)
obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)

```

1      cap_total(h6)
-1     shortage(h0,c1)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h4,c1)
-1     shortage(h5,c1)
-1     shortage(h6,c1)
1      min_coverage(h0)
1      min_coverage(h1)
1      min_coverage(h2)
1      min_coverage(h4)
1      min_coverage(h5)
1      min_coverage(h6)

```

x(h0,c2)

```

      (.L0, .L, .UP, .M = 0, 1.99999999991127, +INF, 189)
-189  obj
      1      cap_per_hour(h0)
      1      cap_total(h0)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h4)
      1      cap_total(h5)
      1      cap_total(h6)
-1     shortage(h0,c2)
-1     shortage(h1,c2)
-1     shortage(h2,c2)
-1     shortage(h4,c2)
-1     shortage(h5,c2)
-1     shortage(h6,c2)
      1      min_coverage(h0)
      1      min_coverage(h1)
      1      min_coverage(h2)
      1      min_coverage(h4)
      1      min_coverage(h5)
      1      min_coverage(h6)

```

x(h1,c1)

```

      (.L0, .L, .UP, .M = 0, 1, +INF, 189)
-189  obj
      1      cap_per_hour(h1)
      1      cap_total(h1)
      1      cap_total(h2)
      1      cap_total(h3)
      1      cap_total(h5)
      1      cap_total(h6)
      1      cap_total(h7)
-1     shortage(h1,c1)
-1     shortage(h2,c1)
-1     shortage(h3,c1)

```



```

-1      shortage(h5,c1)
-1      shortage(h6,c1)
-1      shortage(h7,c1)
 1      min_coverage(h1)
 1      min_coverage(h2)
 1      min_coverage(h3)
 1      min_coverage(h5)
 1      min_coverage(h6)
 1      min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

```

L(h0,c1)
      (.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30   obj
-1     shortage(h0,c1)
 1     shortage_cap(h0,c1)
 1     shortage_total(c1)

```

```

L(h0,c2)
      (.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45   obj
-1     shortage(h0,c2)
 1     shortage_cap(h0,c2)
 1     shortage_total(c2)

```

```

L(h1,c1)
      (.LO, .L, .UP, .M = 0, 10, +INF, 30)
-30   obj
-1     shortage(h1,c1)
 1     shortage_cap(h1,c1)
 1     shortage_total(c1)

```

REMAINING 45 ENTRIES SKIPPED

```

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General Algebraic Modeling System
Range Statistics    SOLVE Telecom_MIP Using MIP From line 173

```

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

```

RHS      [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound    [min, max] : [          NA,          NA] - Zero values observed as well
Matrix   [min, max] : [ 1.000E+00, 1.890E+02]

```

```

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```

General Algebraic Modeling System
Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS t t20

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.000 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t20

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General Algebraic Modeling System
Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	173

**** SOLVER STATUS 1 Normal Completion
**** MODEL STATUS 1 Optimal
**** OBJECTIVE VALUE 184572.0000

RESOURCE USAGE, LIMIT 0.140 10000000000.000
ITERATION COUNT, LIMIT 82 2147483647
*** This solver runs with a demo license. No commercial use.
--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

--- MIP status (101): integer optimal solution.
--- Cplex Time: 0.14sec (det. 9.91 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
--- Fixing integer variables and solving final LP...

```

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.00sec (det. 0.14 ticks)

```

Proven optimal solution

```

MIP Solution:      184572.000000      (82 iterations, 0 nodes)
Final Solve:      184572.000000      (0 iterations)

```

```

Best possible:      184572.000000
Absolute gap:        0.000000
Relative gap:        0.000000

```

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

```

---- EQU cap_per_hour  sum of operators starting at hour across companies <= capHour

```

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	3.0000	100.0000	.
h1	-INF	8.0000	100.0000	.
h2	-INF	24.0000	100.0000	.
h3	-INF	20.0000	100.0000	.
h4	-INF	21.0000	100.0000	.
h5	-INF	28.0000	100.0000	.
h6	-INF	21.0000	100.0000	.
h7	-INF	99.0000	100.0000	.
h8	-INF	65.0000	100.0000	.
h9	-INF	77.0000	100.0000	.
h10	-INF	78.0000	100.0000	.
h11	-INF	38.0000	100.0000	.
h12	-INF	89.0000	100.0000	.
h13	-INF	39.0000	100.0000	.
h14	-INF	44.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	71.0000	100.0000	.
h17	-INF	83.0000	100.0000	.
h18	-INF	94.0000	100.0000	.
h19	-INF	56.0000	100.0000	.
h20	-INF	25.0000	100.0000	.
h21	-INF	25.0000	100.0000	.
h22	-INF	12.0000	100.0000	.
h23	-INF	.	100.0000	.

```

---- EQU cap_total  total operators active in window across companies <= capTotal

```

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	190.0000	420.0000	.
h1	-INF	117.0000	420.0000	.
h2	-INF	97.0000	420.0000	.
h3	-INF	89.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	105.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	341.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	407.0000	420.0000	.
h14	-INF	392.0000	420.0000	.
h15	-INF	376.0000	420.0000	.
h16	-INF	420.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	416.0000	420.0000	.
h20	-INF	390.0000	420.0000	.
h21	-INF	360.0000	420.0000	.
h22	-INF	310.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-120.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-59.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-38.0000	-30.0000	.
h5 .c2	-INF	-42.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-52.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.

h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-211.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-247.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-172.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-146.0000	-130.0000	.
h16.c1	-INF	-260.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.
h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	10.0000	10.0000	.
h1 .c2	-INF	8.0000	10.0000	.
h2 .c1	-INF	7.0000	10.0000	.
h2 .c2	-INF	6.0000	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	7.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.

h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	10.0000	10.0000	.
h18.c2	-INF	.	10.0000	.
h19.c1	-INF	.	10.0000	.
h19.c2	-INF	4.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	10.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	74.0000	76.0000	.
c2	-INF	28.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	190.0000	+INF	.
h1	80.0000	117.0000	+INF	.
h2	80.0000	97.0000	+INF	.
h3	80.0000	89.0000	+INF	.
h4	80.0000	80.0000	+INF	.

h5	80.0000	80.0000	+INF	.	
h6	80.0000	105.0000	+INF	.	
h7	80.0000	200.0000	+INF	.	
h8	80.0000	250.0000	+INF	.	
h9	80.0000	310.0000	+INF	.	
h10	80.0000	290.0000	+INF	.	
h11	80.0000	341.0000	+INF	.	
h12	80.0000	390.0000	+INF	.	
h13	80.0000	407.0000	+INF	.	
h14	80.0000	392.0000	+INF	.	
h15	80.0000	376.0000	+INF	.	
h16	80.0000	420.0000	+INF	.	
h17	80.0000	420.0000	+INF	.	
h18	80.0000	420.0000	+INF	.	
h19	80.0000	416.0000	+INF	.	
h20	80.0000	390.0000	+INF	.	
h21	80.0000	360.0000	+INF	.	
h22	80.0000	310.0000	+INF	.	
h23	80.0000	270.0000	+INF	.	
		LOWER	LEVEL	UPPER	MARGINAL
----	VAR z	-INF	184572.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	3.0000	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	7.0000	+INF	189.0000
h1 .c2	.	1.0000	+INF	189.0000
h2 .c1	.	15.0000	+INF	189.0000
h2 .c2	.	9.0000	+INF	189.0000
h3 .c1	.	.	+INF	180.0000
h3 .c2	.	20.0000	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	21.0000	+INF	171.0000
h5 .c1	.	28.0000	+INF	162.0000
h5 .c2	.	.	+INF	162.0000
h6 .c1	.	.	+INF	153.0000
h6 .c2	.	21.0000	+INF	153.0000
h7 .c1	.	60.0000	+INF	144.0000
h7 .c2	.	39.0000	+INF	144.0000
h8 .c1	.	65.0000	+INF	135.0000
h8 .c2	.	.	+INF	135.0000
h9 .c1	.	47.0000	+INF	126.0000

h9 .c2	.	30.0000	+INF	126.0000
h10.c1	.	40.0000	+INF	126.0000
h10.c2	.	38.0000	+INF	126.0000
h11.c1	.	36.0000	+INF	135.0000
h11.c2	.	2.0000	+INF	135.0000
h12.c1	.	39.0000	+INF	144.0000
h12.c2	.	50.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	39.0000	+INF	153.0000
h14.c1	.	29.0000	+INF	162.0000
h14.c2	.	15.0000	+INF	162.0000
h15.c1	.	78.0000	+INF	171.0000
h15.c2	.	22.0000	+INF	171.0000
h16.c1	.	38.0000	+INF	180.0000
h16.c2	.	33.0000	+INF	180.0000
h17.c1	.	59.0000	+INF	189.0000
h17.c2	.	24.0000	+INF	189.0000
h18.c1	.	55.0000	+INF	189.0000
h18.c2	.	39.0000	+INF	189.0000
h19.c1	.	29.0000	+INF	189.0000
h19.c2	.	27.0000	+INF	189.0000
h20.c1	.	21.0000	+INF	189.0000
h20.c2	.	4.0000	+INF	189.0000
h21.c1	.	25.0000	+INF	189.0000
h21.c2	.	.	+INF	189.0000
h22.c1	.	12.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	10.0000	+INF	30.0000
h1 .c2	.	8.0000	+INF	45.0000
h2 .c1	.	7.0000	+INF	30.0000
h2 .c2	.	6.0000	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	7.0000	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	.	+INF	45.0000

h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	10.0000	+INF	30.0000
h18.c2	.	.	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	4.0000	+INF	45.0000
h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	10.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

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General Algebraic Modeling System

Equation Listing SOLVE Telecom_MIP Using MIP From line 173

---- obj =E= objective definition

obj.. z - 189*x(h0,c1) - 189*x(h0,c2) - 189*x(h1,c1) - 189*x(h1,c2) - 189*x(h2,c1)
 - 189*x(h2,c2) - 180*x(h3,c1) - 180*x(h3,c2) - 171*x(h4,c1) - 171*x(h4,c2) -
 162*x(h5,c1) - 162*x(h5,c2) - 153*x(h6,c1) - 153*x(h6,c2) - 144*x(h7,c1) -
 144*x(h7,c2) - 135*x(h8,c1) - 135*x(h8,c2) - 126*x(h9,c1) - 126*x(h9,c2) -
 126*x(h10,c1) - 126*x(h10,c2) - 135*x(h11,c1) - 135*x(h11,c2) - 144*x(h12,c1) -

$144*x(h12,c2) - 153*x(h13,c1) - 153*x(h13,c2) - 162*x(h14,c1) - 162*x(h14,c2) -$
 $171*x(h15,c1) - 171*x(h15,c2) - 180*x(h16,c1) - 180*x(h16,c2) - 189*x(h17,c1) -$
 $189*x(h17,c2) - 189*x(h18,c1) - 189*x(h18,c2) - 189*x(h19,c1) - 189*x(h19,c2) -$
 $189*x(h20,c1) - 189*x(h20,c2) - 189*x(h21,c1) - 189*x(h21,c2) - 189*x(h22,c1) -$
 $189*x(h22,c2) - 189*x(h23,c1) - 189*x(h23,c2) - 30*L(h0,c1) - 45*L(h0,c2) -$
 $30*L(h1,c1) - 45*L(h1,c2) - 30*L(h2,c1) - 45*L(h2,c2) - 30*L(h3,c1) - 45*L(h3,c2) -$
 $30*L(h4,c1) - 45*L(h4,c2) - 30*L(h5,c1) - 45*L(h5,c2) - 30*L(h6,c1) - 45*L(h6,c2) -$
 $60*L(h7,c1) - 45*L(h7,c2) - 60*L(h8,c1) - 45*L(h8,c2) - 60*L(h9,c1) - 45*L(h9,c2) -$
 $60*L(h10,c1) - 45*L(h10,c2) - 60*L(h11,c1) - 45*L(h11,c2) - 60*L(h12,c1) -$
 $45*L(h12,c2) - 60*L(h13,c1) - 45*L(h13,c2) - 60*L(h14,c1) - 45*L(h14,c2) -$
 $60*L(h15,c1) - 45*L(h15,c2) - 60*L(h16,c1) - 45*L(h16,c2) - 60*L(h17,c1) -$
 $45*L(h17,c2) - 30*L(h18,c1) - 45*L(h18,c2) - 30*L(h19,c1) - 45*L(h19,c2) -$
 $30*L(h20,c1) - 45*L(h20,c2) - 30*L(h21,c1) - 45*L(h21,c2) - 30*L(h22,c1) -$
 $45*L(h22,c2) - 30*L(h23,c1) - 45*L(h23,c2) =E= 0 ; (LHS = 0)$

---- cap_per_hour =L= sum of operators starting at hour across companies <= capHour

cap_per_hour(h0).. $x(h0,c1) + x(h0,c2) =L= 100 ; (LHS = 2.99999999868109)$

cap_per_hour(h1).. $x(h1,c1) + x(h1,c2) =L= 100 ; (LHS = 8.0000000001257)$

cap_per_hour(h2).. $x(h2,c1) + x(h2,c2) =L= 100 ; (LHS = 23.9999999985264)$

REMAINING 21 ENTRIES SKIPPED

---- cap_total =L= total operators active in window across companies <= capTotal

cap_total(h0).. $x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2) =L= 420 ;$
 (LHS = 189.999999998399)

cap_total(h1).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) + x(h19,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2) =L= 420 ; (LHS$
 = 116.999999997413)

cap_total(h2).. $x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2) +$
 $x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =L= 420 ; (LHS$
 = 97.000000000051)

REMAINING 21 ENTRIES SKIPPED

---- shortage =L= demand - covered <= L

shortage(h0,c1).. $- x(h0,c1) - x(h18,c1) - x(h19,c1) - x(h20,c1) - x(h22,c1) -$
 $x(h23,c1) - L(h0,c1) =L= -120 ; (LHS = -119.999999998399, INFES =$
 1.60119384418067E-9 ****)

shortage(h0,c2).. - x(h0,c2) - x(h18,c2) - x(h19,c2) - x(h20,c2) - x(h22,c2) -
x(h23,c2) - L(h0,c2) =L= -70 ; (LHS = -70)

shortage(h1,c1).. - x(h0,c1) - x(h1,c1) - x(h19,c1) - x(h20,c1) - x(h21,c1) -
x(h23,c1) - L(h1,c1) =L= -95 ; (LHS = -95)

REMAINING 45 ENTRIES SKIPPED

---- shortage_cap =L= L <= maxShort

shortage_cap(h0,c1).. L(h0,c1) =L= 10 ; (LHS = 0)

shortage_cap(h0,c2).. L(h0,c2) =L= 10 ; (LHS = 0)

shortage_cap(h1,c1).. L(h1,c1) =L= 10 ; (LHS = 10)

REMAINING 45 ENTRIES SKIPPED

---- shortage_total =L= sum_h L(h,j) <= totShort(j)

shortage_total(c1).. L(h0,c1) + L(h1,c1) + L(h2,c1) + L(h3,c1) + L(h4,c1) +
L(h5,c1) + L(h6,c1) + L(h7,c1) + L(h8,c1) + L(h9,c1) + L(h10,c1) + L(h11,c1) +
L(h12,c1) + L(h13,c1) + L(h14,c1) + L(h15,c1) + L(h16,c1) + L(h17,c1) + L(h18,c1) +
L(h19,c1) + L(h20,c1) + L(h21,c1) + L(h22,c1) + L(h23,c1) =L= 80 ; (LHS =
73.9999999999999)

shortage_total(c2).. L(h0,c2) + L(h1,c2) + L(h2,c2) + L(h3,c2) + L(h4,c2) +
L(h5,c2) + L(h6,c2) + L(h7,c2) + L(h8,c2) + L(h9,c2) + L(h10,c2) + L(h11,c2) +
L(h12,c2) + L(h13,c2) + L(h14,c2) + L(h15,c2) + L(h16,c2) + L(h17,c2) + L(h18,c2) +
L(h19,c2) + L(h20,c2) + L(h21,c2) + L(h22,c2) + L(h23,c2) =L= 30 ; (LHS =
27.99999999998487)

---- min_coverage =G= minimum coverage in each window across companies >= minCov

min_coverage(h0).. x(h0,c1) + x(h0,c2) + x(h18,c1) + x(h18,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h22,c1) + x(h22,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 189.999999998399)

min_coverage(h1).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h19,c1) +
x(h19,c2) + x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h23,c1) + x(h23,c2)
=G= 80 ; (LHS = 116.999999997413)

min_coverage(h2).. x(h0,c1) + x(h0,c2) + x(h1,c1) + x(h1,c2) + x(h2,c1) + x(h2,c2)
+ x(h20,c1) + x(h20,c2) + x(h21,c1) + x(h21,c2) + x(h22,c1) + x(h22,c2) =G= 80 ;
(LHS = 97.0000000000051)

REMAINING 21 ENTRIES SKIPPED

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General Algebraic Modeling System
Column Listing SOLVE Telecom_MIP Using MIP From line 173

---- z total cost (objective)

z
1 (.LO, .L, .UP, .M = -INF, 184572.000000003, +INF, 0)
obj

---- x operators starting at hour for company

x(h0,c1)
-189 (.LO, .L, .UP, .M = 0, 2.99999999868109, +INF, 189)
obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)
-1 shortage(h0,c1)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h4,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

x(h0,c2)
-189 (.LO, .L, .UP, .M = 0, 0, +INF, 189)
obj
1 cap_per_hour(h0)
1 cap_total(h0)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h4)
1 cap_total(h5)
1 cap_total(h6)

```

-1 shortage(h0,c2)
-1 shortage(h1,c2)
-1 shortage(h2,c2)
-1 shortage(h4,c2)
-1 shortage(h5,c2)
-1 shortage(h6,c2)
1 min_coverage(h0)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h4)
1 min_coverage(h5)
1 min_coverage(h6)

```

x(h1,c1)

```

(.LO, .L, .UP, .M = 0, 7.00000000159984, +INF, 189)
-189 obj
1 cap_per_hour(h1)
1 cap_total(h1)
1 cap_total(h2)
1 cap_total(h3)
1 cap_total(h5)
1 cap_total(h6)
1 cap_total(h7)
-1 shortage(h1,c1)
-1 shortage(h2,c1)
-1 shortage(h3,c1)
-1 shortage(h5,c1)
-1 shortage(h6,c1)
-1 shortage(h7,c1)
1 min_coverage(h1)
1 min_coverage(h2)
1 min_coverage(h3)
1 min_coverage(h5)
1 min_coverage(h6)
1 min_coverage(h7)

```

REMAINING 45 ENTRIES SKIPPED

---- L shortage at hour for company

L(h0,c1)

```

(.LO, .L, .UP, .M = 0, 0, +INF, 30)
-30 obj
-1 shortage(h0,c1)
1 shortage_cap(h0,c1)
1 shortage_total(c1)

```

L(h0,c2)

```

(.LO, .L, .UP, .M = 0, 0, +INF, 45)
-45 obj

```

-1 shortage(h0,c2)
1 shortage_cap(h0,c2)
1 shortage_total(c2)

L(h1,c1)

(.LO, .L, .UP, .M = 0, 10, +INF, 30)
-30 obj
-1 shortage(h1,c1)
1 shortage_cap(h1,c1)
1 shortage_total(c1)

REMAINING 45 ENTRIES SKIPPED

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General Algebraic Modeling System

Range Statistics SOLVE Telecom_MIP Using MIP From line 173

RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

RHS [min, max] : [1.000E+01, 4.200E+02] - Zero values observed as well
Bound [min, max] : [NA, NA] - Zero values observed as well
Matrix [min, max] : [1.000E+00, 1.890E+02]

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General Algebraic Modeling System

Model Statistics SOLVE Telecom_MIP Using MIP From line 173

LOOPS

t t21

MODEL STATISTICS

BLOCKS OF EQUATIONS	7	SINGLE EQUATIONS	171
BLOCKS OF VARIABLES	3	SINGLE VARIABLES	97
NON ZERO ELEMENTS	1,153	DISCRETE VARIABLES	96

GENERATION TIME = 0.047 SECONDS 4 MB 52.5.0 9bc771bf WEX-WEI
L O O P S t t21

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General Algebraic Modeling System

Solution Report SOLVE Telecom_MIP Using MIP From line 173

S O L V E S U M M A R Y

MODEL	Telecom_MIP	OBJECTIVE	z
TYPE	MIP	DIRECTION	MINIMIZE
SOLVER	CPLEX	FROM LINE	173

```

**** SOLVER STATUS      1 Normal Completion
**** MODEL STATUS      1 Optimal
**** OBJECTIVE VALUE    184545.0000

```

```

RESOURCE USAGE, LIMIT      0.125 10000000000.000
ITERATION COUNT, LIMIT     74    2147483647
*** This solver runs with a demo license. No commercial use.
--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex

```

```

--- MIP status (101): integer optimal solution.
--- Cplex Time: 0.12sec (det. 7.08 ticks)

--- Returning a primal only solution to GAMS (marginals all set to NA).
--- Fixing integer variables and solving final LP...

```

```

--- Fixed MIP status (1): optimal.
--- Cplex Time: 0.00sec (det. 0.14 ticks)

```

```

Proven optimal solution
MIP Solution:      184545.000000    (74 iterations, 0 nodes)
Final Solve:      184545.000000    (0 iterations)

Best possible:      184545.000000
Absolute gap:       0.000000
Relative gap:       0.000000

```

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU obj	.	.	.	1.0000

obj objective definition

```

---- EQU cap_per_hour  sum of operators starting at hour across companies <= capHour

          LOWER          LEVEL          UPPER          MARGINAL

```

h0	-INF	3.0000	100.0000	.
h1	-INF	4.0000	100.0000	.
h2	-INF	24.0000	100.0000	.
h3	-INF	22.0000	100.0000	.
h4	-INF	24.0000	100.0000	.
h5	-INF	27.0000	100.0000	.
h6	-INF	23.0000	100.0000	.
h7	-INF	100.0000	100.0000	.
h8	-INF	57.0000	100.0000	.
h9	-INF	80.0000	100.0000	.
h10	-INF	79.0000	100.0000	.
h11	-INF	36.0000	100.0000	.
h12	-INF	95.0000	100.0000	.
h13	-INF	42.0000	100.0000	.
h14	-INF	36.0000	100.0000	.
h15	-INF	100.0000	100.0000	.
h16	-INF	69.0000	100.0000	.
h17	-INF	78.0000	100.0000	.
h18	-INF	100.0000	100.0000	.
h19	-INF	57.0000	100.0000	.
h20	-INF	28.0000	100.0000	.
h21	-INF	28.0000	100.0000	.
h22	-INF	7.0000	100.0000	.
h23	-INF	.	100.0000	.

---- EQU cap_total total operators active in window across companies <= capTotal

	LOWER	LEVEL	UPPER	MARGINAL
h0	-INF	195.0000	420.0000	.
h1	-INF	120.0000	420.0000	.
h2	-INF	94.0000	420.0000	.
h3	-INF	85.0000	420.0000	.
h4	-INF	80.0000	420.0000	.
h5	-INF	80.0000	420.0000	.
h6	-INF	105.0000	420.0000	.
h7	-INF	200.0000	420.0000	.
h8	-INF	250.0000	420.0000	.
h9	-INF	310.0000	420.0000	.
h10	-INF	290.0000	420.0000	.
h11	-INF	345.0000	420.0000	.
h12	-INF	390.0000	420.0000	.
h13	-INF	410.0000	420.0000	.
h14	-INF	389.0000	420.0000	.
h15	-INF	373.0000	420.0000	.
h16	-INF	415.0000	420.0000	.
h17	-INF	420.0000	420.0000	.
h18	-INF	420.0000	420.0000	.
h19	-INF	413.0000	420.0000	.
h20	-INF	390.0000	420.0000	.

h21	-INF	360.0000	420.0000	.
h22	-INF	310.0000	420.0000	.
h23	-INF	270.0000	420.0000	.

---- EQU shortage demand - covered <= L

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-125.0000	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	.
h1 .c2	-INF	-40.0000	-40.0000	.
h2 .c1	-INF	-90.0000	-90.0000	.
h2 .c2	-INF	-20.0000	-20.0000	.
h3 .c1	-INF	-55.0000	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	.
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	.
h5 .c1	-INF	-35.0000	-30.0000	.
h5 .c2	-INF	-45.0000	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	.
h6 .c2	-INF	-55.0000	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	.
h7 .c2	-INF	-90.0000	-90.0000	.
h8 .c1	-INF	-140.0000	-140.0000	.
h8 .c2	-INF	-110.0000	-110.0000	.
h9 .c1	-INF	-200.0000	-200.0000	.
h9 .c2	-INF	-110.0000	-110.0000	.
h10.c1	-INF	-180.0000	-180.0000	.
h10.c2	-INF	-110.0000	-110.0000	.
h11.c1	-INF	-215.0000	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	.
h12.c1	-INF	-240.0000	-240.0000	.
h12.c2	-INF	-150.0000	-150.0000	.
h13.c1	-INF	-250.0000	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	.
h14.c1	-INF	-220.0000	-220.0000	.
h14.c2	-INF	-169.0000	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	.
h15.c2	-INF	-143.0000	-130.0000	.
h16.c1	-INF	-255.0000	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	.
h17.c1	-INF	-260.0000	-260.0000	.
h17.c2	-INF	-180.0000	-180.0000	.
h18.c1	-INF	-230.0000	-230.0000	.
h18.c2	-INF	-200.0000	-200.0000	.
h19.c1	-INF	-250.0000	-250.0000	.
h19.c2	-INF	-170.0000	-170.0000	.
h20.c1	-INF	-260.0000	-260.0000	.
h20.c2	-INF	-140.0000	-140.0000	.

h21.c1	-INF	-250.0000	-250.0000	.
h21.c2	-INF	-110.0000	-110.0000	.
h22.c1	-INF	-220.0000	-220.0000	.
h22.c2	-INF	-100.0000	-100.0000	.
h23.c1	-INF	-190.0000	-190.0000	.
h23.c2	-INF	-90.0000	-90.0000	.

---- EQU shortage_cap L <= maxShort

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	.	10.0000	.
h0 .c2	-INF	.	10.0000	.
h1 .c1	-INF	10.0000	10.0000	.
h1 .c2	-INF	5.0000	10.0000	.
h2 .c1	-INF	10.0000	10.0000	.
h2 .c2	-INF	6.0000	10.0000	.
h3 .c1	-INF	.	10.0000	.
h3 .c2	-INF	.	10.0000	.
h4 .c1	-INF	.	10.0000	.
h4 .c2	-INF	.	10.0000	.
h5 .c1	-INF	.	10.0000	.
h5 .c2	-INF	.	10.0000	.
h6 .c1	-INF	10.0000	10.0000	.
h6 .c2	-INF	.	10.0000	.
h7 .c1	-INF	.	10.0000	.
h7 .c2	-INF	.	10.0000	.
h8 .c1	-INF	.	10.0000	.
h8 .c2	-INF	.	10.0000	.
h9 .c1	-INF	.	10.0000	.
h9 .c2	-INF	.	10.0000	.
h10.c1	-INF	.	10.0000	.
h10.c2	-INF	.	10.0000	.
h11.c1	-INF	.	10.0000	.
h11.c2	-INF	.	10.0000	.
h12.c1	-INF	.	10.0000	.
h12.c2	-INF	.	10.0000	.
h13.c1	-INF	.	10.0000	.
h13.c2	-INF	.	10.0000	.
h14.c1	-INF	.	10.0000	.
h14.c2	-INF	.	10.0000	.
h15.c1	-INF	.	10.0000	.
h15.c2	-INF	.	10.0000	.
h16.c1	-INF	.	10.0000	.
h16.c2	-INF	.	10.0000	.
h17.c1	-INF	10.0000	10.0000	.
h17.c2	-INF	10.0000	10.0000	.
h18.c1	-INF	10.0000	10.0000	.
h18.c2	-INF	.	10.0000	.
h19.c1	-INF	.	10.0000	.

h19.c2	-INF	7.0000	10.0000	.
h20.c1	-INF	10.0000	10.0000	.
h20.c2	-INF	.	10.0000	.
h21.c1	-INF	.	10.0000	.
h21.c2	-INF	.	10.0000	.
h22.c1	-INF	10.0000	10.0000	.
h22.c2	-INF	.	10.0000	.
h23.c1	-INF	10.0000	10.0000	.
h23.c2	-INF	.	10.0000	.

---- EQU shortage_total sum_h L(h,j) <= totShort(j)

	LOWER	LEVEL	UPPER	MARGINAL
c1	-INF	80.0000	80.0000	.
c2	-INF	28.0000	30.0000	.

---- EQU min_coverage minimum coverage in each window across companies >= minCov

	LOWER	LEVEL	UPPER	MARGINAL
h0	80.0000	195.0000	+INF	.
h1	80.0000	120.0000	+INF	.
h2	80.0000	94.0000	+INF	.
h3	80.0000	85.0000	+INF	.
h4	80.0000	80.0000	+INF	.
h5	80.0000	80.0000	+INF	.
h6	80.0000	105.0000	+INF	.
h7	80.0000	200.0000	+INF	.
h8	80.0000	250.0000	+INF	.
h9	80.0000	310.0000	+INF	.
h10	80.0000	290.0000	+INF	.
h11	80.0000	345.0000	+INF	.
h12	80.0000	390.0000	+INF	.
h13	80.0000	410.0000	+INF	.
h14	80.0000	389.0000	+INF	.
h15	80.0000	373.0000	+INF	.
h16	80.0000	415.0000	+INF	.
h17	80.0000	420.0000	+INF	.
h18	80.0000	420.0000	+INF	.
h19	80.0000	413.0000	+INF	.
h20	80.0000	390.0000	+INF	.
h21	80.0000	360.0000	+INF	.
h22	80.0000	310.0000	+INF	.
h23	80.0000	270.0000	+INF	.

	LOWER	LEVEL	UPPER	MARGINAL
---- VAR z	-INF	184545.0000	+INF	.

z total cost (objective)

---- VAR x operators starting at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	3.0000	+INF	189.0000
h0 .c2	.	.	+INF	189.0000
h1 .c1	.	.	+INF	189.0000
h1 .c2	.	4.0000	+INF	189.0000
h2 .c1	.	15.0000	+INF	189.0000
h2 .c2	.	9.0000	+INF	189.0000
h3 .c1	.	5.0000	+INF	180.0000
h3 .c2	.	17.0000	+INF	180.0000
h4 .c1	.	.	+INF	171.0000
h4 .c2	.	24.0000	+INF	171.0000
h5 .c1	.	27.0000	+INF	162.0000
h5 .c2	.	.	+INF	162.0000
h6 .c1	.	5.0000	+INF	153.0000
h6 .c2	.	18.0000	+INF	153.0000
h7 .c1	.	58.0000	+INF	144.0000
h7 .c2	.	42.0000	+INF	144.0000
h8 .c1	.	57.0000	+INF	135.0000
h8 .c2	.	.	+INF	135.0000
h9 .c1	.	53.0000	+INF	126.0000
h9 .c2	.	27.0000	+INF	126.0000
h10.c1	.	38.0000	+INF	126.0000
h10.c2	.	41.0000	+INF	126.0000
h11.c1	.	34.0000	+INF	135.0000
h11.c2	.	2.0000	+INF	135.0000
h12.c1	.	48.0000	+INF	144.0000
h12.c2	.	47.0000	+INF	144.0000
h13.c1	.	.	+INF	153.0000
h13.c2	.	42.0000	+INF	153.0000
h14.c1	.	24.0000	+INF	162.0000
h14.c2	.	12.0000	+INF	162.0000
h15.c1	.	81.0000	+INF	171.0000
h15.c2	.	19.0000	+INF	171.0000
h16.c1	.	30.0000	+INF	180.0000
h16.c2	.	39.0000	+INF	180.0000
h17.c1	.	57.0000	+INF	189.0000
h17.c2	.	21.0000	+INF	189.0000
h18.c1	.	61.0000	+INF	189.0000
h18.c2	.	39.0000	+INF	189.0000
h19.c1	.	27.0000	+INF	189.0000
h19.c2	.	30.0000	+INF	189.0000
h20.c1	.	27.0000	+INF	189.0000
h20.c2	.	1.0000	+INF	189.0000
h21.c1	.	28.0000	+INF	189.0000

h21.c2	.	.	+INF	189.0000
h22.c1	.	7.0000	+INF	189.0000
h22.c2	.	.	+INF	189.0000
h23.c1	.	.	+INF	189.0000
h23.c2	.	.	+INF	189.0000

---- VAR L shortage at hour for company

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	30.0000
h0 .c2	.	.	+INF	45.0000
h1 .c1	.	10.0000	+INF	30.0000
h1 .c2	.	5.0000	+INF	45.0000
h2 .c1	.	10.0000	+INF	30.0000
h2 .c2	.	6.0000	+INF	45.0000
h3 .c1	.	.	+INF	30.0000
h3 .c2	.	.	+INF	45.0000
h4 .c1	.	.	+INF	30.0000
h4 .c2	.	.	+INF	45.0000
h5 .c1	.	.	+INF	30.0000
h5 .c2	.	.	+INF	45.0000
h6 .c1	.	10.0000	+INF	30.0000
h6 .c2	.	.	+INF	45.0000
h7 .c1	.	.	+INF	60.0000
h7 .c2	.	.	+INF	45.0000
h8 .c1	.	.	+INF	60.0000
h8 .c2	.	.	+INF	45.0000
h9 .c1	.	.	+INF	60.0000
h9 .c2	.	.	+INF	45.0000
h10.c1	.	.	+INF	60.0000
h10.c2	.	.	+INF	45.0000
h11.c1	.	.	+INF	60.0000
h11.c2	.	.	+INF	45.0000
h12.c1	.	.	+INF	60.0000
h12.c2	.	.	+INF	45.0000
h13.c1	.	.	+INF	60.0000
h13.c2	.	.	+INF	45.0000
h14.c1	.	.	+INF	60.0000
h14.c2	.	.	+INF	45.0000
h15.c1	.	.	+INF	60.0000
h15.c2	.	.	+INF	45.0000
h16.c1	.	.	+INF	60.0000
h16.c2	.	.	+INF	45.0000
h17.c1	.	10.0000	+INF	60.0000
h17.c2	.	10.0000	+INF	45.0000
h18.c1	.	10.0000	+INF	30.0000
h18.c2	.	.	+INF	45.0000
h19.c1	.	.	+INF	30.0000
h19.c2	.	7.0000	+INF	45.0000

h20.c1	.	10.0000	+INF	30.0000
h20.c2	.	.	+INF	45.0000
h21.c1	.	.	+INF	30.0000
h21.c2	.	.	+INF	45.0000
h22.c1	.	10.0000	+INF	30.0000
h22.c2	.	.	+INF	45.0000
h23.c1	.	10.0000	+INF	30.0000
h23.c2	.	.	+INF	45.0000

**** REPORT SUMMARY : 0 NONOPT
 0 INFEASIBLE
 0 UNBOUNDED

GAMS 52.5.0 9bc771bf Jan 28, 2026 WEX-WEI x86 64bit/MS Windows - 02/15/26
 15:53:30 Page 212
 General Algebraic Modeling System
 Execution

---- 195 PARAMETER coefVal trial values for objective coefficient cost('h8')

t1 67.500,	t2 74.250,	t3 81.000,	t4 87.750,	t5 94.500,	t6
101.250,	t7 108.000,	t8 114.750,	t9 121.500,	t10 128.250,	t11
135.000,	t12 141.750,	t13 148.500,	t14 155.250,	t15 162.000,	t16
168.750,	t17 175.500,	t18 182.250,	t19 189.000,	t20 195.750,	t21
202.500					

---- 195 PARAMETER objCoef optimal objective for coefficient sweep

t1 178479.000,	t2 179154.000,	t3 179829.000,	t4 180504.000,	t5
181179.000,	t6 181854.000,	t7 182529.000,	t8 183192.000,	t9
183804.000,	t10 184367.250,	t11 184848.000,	t12 185205.000,	t13
185556.000,	t14 185893.500,	t15 186225.000,	t16 186531.000,	t17
186729.000,	t18 186931.500,	t19 187134.000,	t20 187287.750,	t21
187425.000				

---- 195 PARAMETER rhsVal trial values for RHS capHour('h0')

t2 4.000,	t3 8.000,	t4 12.000,	t5 16.000,	t6 20.000,	t7
24.000,	t8 28.000,	t9 32.000,	t10 36.000,	t11 40.000,	t12 44.000,
t13 48.000,	t14 52.000,	t15 56.000,	t16 60.000,	t17 64.000,	t18
68.000,	t19 72.000,	t20 76.000,	t21 80.000		

---- 195 PARAMETER objRhs optimal objective for RHS sweep

t1 187425.000,	t2 187425.000,	t3 187425.000,	t4 185592.000,	t5
185487.000,	t6 185355.000,	t7 185214.000,	t8 185127.000,	t9

184977.000,	t10 184908.000,	t11 184848.000,	t12 184794.000,	t13
184749.000,	t14 184707.000,	t15 184680.000,	t16 184662.000,	t17
184635.000,	t18 184617.000,	t19 184599.000,	t20 184572.000,	t21
184545.000				

**** REPORT FILE SUMMARY

f1 C:\Users\m\OneDrive\Desktop\C238~1\NEWFOL~1\sweep_coef.csv
 f2 C:\Users\m\OneDrive\Desktop\C238~1\NEWFOL~1\sweep_rhs.csv

EXECUTION TIME = 23.313 SECONDS 6 MB 52.5.0 9bc771bf WEX-WEI

USER: GAMS Demo, for EULA and demo limitations see G251112+0003Cc-GEN
 <https://www.gams.com/latest/docs/UG%5FLicense.html> DC0000

**** FILE SUMMARY

Input C:\Users\m\OneDrive\Desktop\C238~1\NEWFOL~1\telecom_part3.gms
 Output C:\Users\m\OneDrive\Desktop\C238~1\NEWFOL~1\telecom_part3.lst